

Factor Research

8 reports · Base Case

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FACTOR RESEARCH

NVDA Factor 01 Share Ceded Custom Silicon

Theme

So what: the buyer of last resort learned to design its own chip, and the economics now work above roughly 30,000 units

MECHANISM

The theme is not a better GPU beating NVIDIA. It is the four or five customers who account for most accelerator demand deciding to design the part themselves, and a supplier ecosystem that has grown up to let a software company do that. Custom ASIC-based AI servers are expected to reach **27.8% of AI server shipments by 2026, the highest since 2023** [6].

THE CAUSAL CHAIN

1. A very large margin sits on top of a merchant GPU. A hyperscaler running its own models at gigawatt scale pays that margin on every unit. Google claims an all-in cost per Ironwood chip "meaningfully below the cost of an equivalent Nvidia server, with figures in the range of a 40 percent-plus TCO reduction" [5]. AWS claims 30-40% better price-performance for Trainium2 against GPU-based EC2 P5e instances for sustained training [2].

2. The design cost is enormous but fixed. A 2nm design program now runs about **\$725M, a 25% increase from 3nm and a 15x jump from 28nm** [3]. That is the toll on the door. Against a merchant GPU selling for roughly \$20-30K and costing a few thousand dollars to build, the arithmetic breaks even in the low tens of thousands of units — a rounding error for a buyer deploying at gigawatt scale, and impossible for anyone else. This is exactly why the club is four or five names deep and not fifty.

3. The enabling layer is now rentable. The barrier used to be that you needed a silicon team. It is now possible to buy most of one. Broadcom turned this into a business model in which "Custom XPU is not an isolated chip order, but an entry point driving the entire AI networking and connectivity product portfolio" — Meta's MTIA went multi-generational and OpenAI's engagement extended into custom processors and large-scale Ethernet systems [1]. Marvell is in the same trade: Q2 FY27 data center revenue of **\$2.172 billion, +46% YoY, 79% of total revenue** [1].

4. Nothing about the physical supply chain changes. The ASIC and the GPU buy the same HBM and the same advanced packaging. CoWoS-S runs roughly **\$750 per chip** for H100-class designs and CoWoS-L **\$1,000-\$1,100** for B200, and memory plus packaging together are now **60-70% of AI accelerator COGS** [4]. Whoever wins the compute-die argument, the packaging and memory bill is paid either way.

WHAT ENFORCES IT

Two things, pulling opposite ways. Against the theme: the software distance. Moving a workload onto Trainium through the Neuron SDK "typically costs engineering time measured in weeks" [5], and an ASIC designed around one model family ages badly if architectures move. For the theme: scale deployment is already real — Anthropic

committed to **1 million Ironwood chips in October 2025, over 1 gigawatt of compute**, and AWS has **more than 500,000 Trainium2 chips in production** [2]. Neither of those is a pilot.

So what: this is a three-to-five-year re-mix of a growing pie, not an eighteen-month share war

DRIVERS AND HORIZON

The pie is growing faster than the share is shifting. TrendForce "revised its 2026 AI server shipment forecast upward from 28% to nearly 31% YoY" [1], while the merchant vendor's revenue share slipped from a peak "near 87% in 2024" to "roughly 75% for 2026" [3]. A shrinking slice of a faster-growing market is a mix story, and it is the single most important thing to hold in mind when reading the numbers.

WHAT IS PUSHING IT

- **Inference overtaking training.** Training rewards flexibility and mature interconnect; inference is repetitive, predictable computation where cost per token, latency and watts decide. That is the workload an ASIC is built for, and it is the faster-growing half.
- **Silicon that is now provably good enough.** Microsoft says Maia 200 "runs at 30% better performance per dollar than the fastest hardware already in its fleet" [4]; Meta's MTIA 300 "is already in production for ranking and recommendation work" [4]. These are shipping parts, not roadmaps.
- **Capacity, not just price.** Every merchant GPU a hyperscaler cannot get is a gigawatt it cannot fill. Owning the design is a supply hedge as much as a cost one.
- **Foundry competition on the input side.** Samsung "takes a scalpel to its 2nm wafer price tag, bringing it down to \$20,000 — Korean chipmaker now undercuts rival TSMC by 33%" [5]. Cheaper leading-edge wafers lower the cost of trying.

WHAT WOULD SLOW IT

- **The software distance.** CUDA has "more than 5 million developers" building on it, "which keeps training workloads locked to NVIDIA hardware even where AMD matches raw FP8 throughput" [3]. Porting is a tax paid in engineer-weeks [7].
- **Captivity.** Ironwood is rentable only through Google Cloud, Trainium only through AWS; Maia and MTIA are not rentable at all [7]. That caps custom silicon at the hyperscalers' own workloads and leaves the merchant vendor everything else.
- **Model-architecture drift.** A part tuned to one transformer generation ages badly if the architecture moves, and the design cycle is 2-3 years long.
- **The same physical bottleneck.** ASIC programs queue for the same HBM and the same advanced packaging; a supply constraint throttles the challenger as readily as the incumbent.

THE CLOCK

Three to five years, with the annual checkpoint being each hyperscaler's next-generation part. The theme is roughly three years old in its current form: in 2023 custom silicon was essentially Google's TPU used internally; by 2026 it is a 27.8% shipment share [2] with a multi-vendor enabling industry behind it. It resolves at the pace

of silicon, not of quarters — a design decision taken today ships in 2028.

The tell to watch is not accelerator share at all, but whether the ceded compute drags the rest of the rack with it. The merchant vendor's own Q4 FY26 shows the split running the other way so far: "data center compute revenue of \$51.3B (up 58% YoY) and networking at \$11.0B (up 263% YoY)" [6] — the fabric is outgrowing the compute it attaches to.

So what: the theme pays the enablers, not the contestants — it is a fight over the compute die that leaves the rest of the rack intact

WHO WINS AND LOSES

This pod is the search specification for the landscape. It names the *kinds* of business the theme moves, not the names.

STRUCTURALLY FAVOURED

1. Anyone who sells the ability to design a chip rather than the chip. The custom-silicon wave is demand for design services, high-speed SerDes, interface IP and verification. Broadcom proved the model: "Custom XPU is not an isolated chip order, but an entry point driving the entire AI networking and connectivity product portfolio" [1], and its "IP around SerDes, optics and networking is best-in-class" [2]. The characteristic to screen for: revenue that scales with the *number of distinct silicon programs*, not with which program wins.

2. Anyone paid the same bill of materials either way. An ASIC and a GPU buy the same HBM and the same advanced packaging; "memory and packaging together now represent 60-70% of AI accelerator COGS" [3]. Memory makers, OSATs and the foundry are structurally indifferent to who wins the compute die — and are arguably better off, because more distinct designs means more mask sets, more test flows and more packaging slots per unit of compute.

3. Anyone selling the fabric rather than the processor. Custom accelerators still have to be networked, and the hyperscalers deploying them are standardising on Ethernet rather than proprietary interconnect [2]. Scale-up and scale-out networking, optics, retimers and power delivery are a toll booth on total deployed compute regardless of architecture.

4. The foundry and its leading-edge capacity. Every custom program is another leading-edge tape-out. Cheaper alternative capacity accelerates this: Samsung's 2nm wafer at "\$20,000 — Korean chipmaker now undercuts rival TSMC by 33%" [5] lowers the cost of a second source.

STRUCTURALLY SQUEEZED

1. The merchant accelerator vendor's compute-die revenue, specifically. Not its rack revenue, not its networking revenue — the die. Its own mix already shows the distinction.

2. The merchant second source. A challenger GPU competes for the *same* budget line the hyperscaler is trying to internalise, and does it without the incumbent's software gravity. Custom silicon squeezes the number two harder than the number one.

3. Anyone whose business is renting general-purpose capacity to the hyperscalers. If the largest buyers serve their own inference on their own silicon, the marginal rented GPU-hour is bid for by everyone else.

4. Sub-scale would-be designers. At "\$725 million" for a 2nm program, the theme is a concentration force: it favours the handful of buyers with the volume to amortise it and squeezes anyone attempting it below roughly the low tens of thousands of units.

THE GENUINELY MIXED CASE

The incumbent vendor is not a one-way loser and the design houses are not a one-way win. The incumbent has responded by trying to keep third-party XPU's inside its own rack — "NVIDIA wants third-party XPU's to continue running on NVLink, Spectrum-X, ConnectX, BlueField, and MGX" [1]. And the design houses carry customer concentration that the theme itself created: a handful of programs, each one cancellable. Screen for that.

Landscape

Exposed Companies

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Broadcom	AVGO	incumbent	Designs the XPU for Google, Meta and OpenAI and sells the SerDes, Ethernet switching and optics around it; AI semi revenue \$10.8B in Q2 FY26, +143% YoY, guided to ~\$16B in Q3	benefits	high	AI semis were ~\$10.8B of Q2 FY26 revenue	The reference case for the whole theme: one ASIC order pulls a full networking attach
Marvell Technology	MRVL	incumbent	Second custom-XPU house; supplies custom silicon and scale-up networking to Google (inference accelerators, NICs, memory-interface controllers) — data centre was \$2.172B, 79% of Q2 FY27 revenue, +46% YoY	benefits	high	~79% of revenue (data centre, Q2 FY27)	NVIDIA took an investment in it on 31 Mar 2026 — co-opted, not just competing
Astera Labs	ALAB	emerging	Connectivity content attaches to every custom XPU: content per accelerator has gone from below \$100 at IPO to close to \$1,000 per XPU, and it has a design partnership with Alchip aimed at hyperscaler ASIC racks	benefits	high	Effectively 100% AI/cloud connectivity	IPO Mar 2024; the purest read on 'more distinct accelerator designs' regardless of whose design wins

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Credo Technology	CRDO	emerging	Active electrical cables and SerDes into hyperscaler AI racks — the copper interconnect a custom-ASIC rack needs whether or not it runs NVLink; Q1 FY27 revenue \$479.0M, +114.7% YoY	benefits	high	Majority of revenue is hyperscaler AI connectivity	Concentration is the risk and the tell: a handful of hyperscalers are most of the book
Alchip Technologies	3661.TW	incumbent	Pure-play ASIC design house on the AWS Trainium programme and an NVLink Fusion custom-silicon partner; Q1 2026 revenue \$132.4M against \$318.7M a year earlier, with 21% of it on 3nm/2nm nodes	benefits	medium	~100% ASIC design and turnkey	Shows the theme's lumpiness — revenue swings on where one programme sits in its cycle
Global Unichip (GUC)	3443.TW	incumbent	TSMC-affiliated design-service house taking cloud-provider ASIC and CPU programmes; TTM revenue ~\$1.17B with North America (cloud) the dominant end market	benefits	medium	Majority from cloud/HPC ASIC programmes	Programme-level detail is thinly disclosed; the customer link is reported rather than filed
MediaTek	2454.TW	incumbent	Entering custom AI ASIC from mobile SoC: first AI accelerator ASIC targeted at mass production in Q4 2026, 70+ tape-outs from 7nm to 2nm and production-ready 224G SerDes; took a \$3.5B NVIDIA investment	mixed	medium	Data-centre is still small against a TWD 152.2B/quarter mobile-led P&L	Mixed because the same deal that funds its XPU keeps that XPU inside NVIDIA's rack

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Celestica	CLS	adjacent	Builds the custom AI/ML rack systems the hyperscalers deploy; CCS revenue \$3,241.0M in Q1 2026, +76%, with three customers at 35%, 15% and 15% of total revenue	benefits	high	CCS ~80% of total revenue	Paid per rack, indifferent to whose silicon is in it – but hostage to three buyers
TSMC	TSM	adjacent	Fabricates both sides: every custom accelerator and the merchant GPU. CoWoS advanced-packaging capacity is expanding from ~80,000 to 120,000-130,000 WPM through 2026, of which NVIDIA takes ~60%	benefits	high	Not separately disclosed; AI/HPC is the majority of leading-edge	More distinct designs means more mask sets and more packaging slots per unit of compute
SK hynix	000660.KS	adjacent	Sells the HBM stack into an ASIC exactly as into a GPU; projected ~70% of the HBM4 segment, against accelerator BOM memory lines of \$2,400-\$6,500 per unit	benefits	medium	HBM is the majority of DRAM profit, not separately split by customer type	The clearest 'paid either way' position in the chain; risk is the memory cycle, not the architecture
Cerebras Systems	CBRS	emerging	Wafer-scale alternative to both the merchant GPU and the hyperscaler ASIC: 900,000 AI cores on one wafer-sized die, with an OpenAI agreement for 750 MW of capacity; IPO'd May 2026 raising \$5.55B	mixed	medium	~100% AI compute	Mixed: it takes share from merchant GPUs but also competes with the in-house ASICs the theme favours

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
AMD	AMD	incumbent	The merchant second source, squeezed by the same budget line the hyperscalers are internalising: Instinct is put at roughly 5-7% of AI accelerator revenue against custom silicon's 10-15%	hurt	medium	Data Center ~\$16.6B FY2025 (EPYC + Instinct)	The squeezed side of the theme: custom ASICs eat the diversification budget AMD is bidding for

So what: the documented exposures are in the enabling layer, not among the accelerator contestants

EXPOSURE EVIDENCE

One paragraph per row of Exposed_Companies, in table order, saying what the company is and why it sits where it does. Where the link is inferred rather than documented, this pod says so and the row's Confidence is marked down to match.

Broadcom (AVGO) – documented. The company that turned custom silicon into a business model rather than a favour. Its AI semiconductor revenue reached \$10.8 billion in Q2 FY26, up 143% year over year, with Q3 guidance raised to about \$16 billion [1]. The mechanism that matters is not the ASIC order but what it drags: "Beyond Custom XPU's, the company provides high-speed SerDes, Ethernet Scale-up/Scale-out, switching chips, optical interconnects, and advanced packaging" [1]. The MTIA engagement with Meta went multi-generational and the OpenAI engagement extended into custom processors and large-scale Ethernet systems [1].

Marvell (MRVL) – documented. The second custom-XPU house, and the one whose numbers show the theme converting to revenue: data centre revenue of \$2.172 billion in Q2 FY27, up 46% year over year and 79% of total revenue [1]. Its Google engagement covers inference accelerators, storage controllers, NICs, memory-interface controllers and near-memory computing [1]. The complication a skeptic should press on: NVIDIA invested in Marvell on 31 March 2026, with Marvell supplying custom XPUs and scale-up networking compatible with NVLink Fusion while NVIDIA supplies the CPUs, NICs, DPUs and rack platform [1]. Marvell is levered to the theme and partly co-opted by its target.

Astera Labs (ALAB) – documented. The cleanest expression of "paid per accelerator design, not per accelerator winner". At IPO the company said content per accelerator was below \$100 [2]; it now puts content close to \$1,000 per XPU, with the Scorpio X family alone expected to exceed that [2]. Its partnership with Alchip "combines Alchip's custom ASIC development capabilities with Astera Labs' comprehensive connectivity portfolio to deliver validated, interoperable solutions for hyperscalers building next-generation AI infrastructure" [3]. It was also among the first custom-silicon partners when NVLink Fusion launched [1] — it attaches to both sides of the fight.

Credo Technology (CRDO) — documented on revenue, inferred on the custom-ASIC split. Q1 FY27 revenue was \$479.0 million, up 114.7% year over year [4]. The exposure mechanism is active electrical cables and SerDes inside hyperscaler AI racks, which are needed whether the rack holds a custom ASIC or a merchant GPU. What is NOT separately disclosed is how much of that book is custom-ASIC racks specifically — the customer list makes it likely, and the row is graded on the revenue evidence rather than on an attribution the filings do not make.

Alchip Technologies (3661.TW) — documented, and instructive about the risk. Q1 2026 revenue was \$132.4 million, down from \$318.7 million a year earlier [5], with management attributing expected improvement to "North American demand for 3nm AI accelerator shipments" [5]. That collapse-and-recover pattern is the design-house business model showing through: revenue is programme-cycle, not run-rate. It is an NVLink Fusion custom-silicon partner alongside Marvell, MediaTek and Astera Labs [1].

Global Unichip / GUC (3443.TW) — reported, thinly documented. TTM revenue of about \$1.17 billion [6]. GUC is TSMC's affiliated design-service arm, which is why cloud-provider ASIC and CPU programmes route through it, but programme-level customer detail is not filed in a form this run could verify. Confidence is medium on that basis: the business model makes the exposure near-certain, the specific programme attribution does not come from a primary document here.

MediaTek (2454.TW) — documented, and the most genuinely two-sided row. It has completed over 70 product tape-outs from 7nm to 2nm, has production-ready 224G SerDes, and plans first AI-accelerator ASIC mass production in Q4 2026, with the 2027 AI accelerator TAM put at \$80 billion and a target share of 15-20% [1]. But the \$3.5 billion NVIDIA investment is what makes it mixed: the arrangement hands MediaTek the XPU design, packaging and SerDes while NVIDIA supplies the rack around it [1]. MediaTek wins on the theme only in the sense that the incumbent chose who wins.

Celestica (CLS) — documented. CCS revenue was \$3,241.0 million in Q1 2026, up 76%, and "Three customers (all in our CCS segment) individually represented 10% or more of total revenue in Q1 2026 (35%, 15% and 15%)" [7]. It builds the racks; it does not care whose silicon is in them. The concentration cuts both ways — it is the evidence that the exposure is real and the reason the exposure is fragile.

TSMC (TSM) — documented. The toll booth under the whole argument. CoWoS capacity "is expanding from ~80,000 WPM to 120,000-130,000 WPM through 2026, with NVIDIA consuming ~60% of allocation" [8]. Every hyperscaler ASIC is another leading-edge tape-out and another claim on that same packaging line, so TSMC's revenue rises whichever architecture wins — and its concentration risk falls as the customer list lengthens.

SK hynix (000660.KS) — documented. An ASIC buys the same HBM a GPU does. SK hynix is projected at roughly 70% of the HBM4 segment serving NVIDIA's Rubin platform [8], and HBM lines run \$2,400-\$6,500 per accelerator unit [8]. The exposure link is structural rather than contractual: the theme changes who designs the compute die, not who stacks the memory beside it. The real risk here is the memory cycle, not the architecture question.

Cerebras Systems (CBRS) — documented on the event, inferred on the exposure. It "raised \$5.55 billion in its initial public offering, marking the largest IPO of 2026" [9], on a wafer-scale part with 900,000 AI cores and 44 GB of on-chip SRAM [9], and has an OpenAI agreement to build 750 MW of capacity on its chips [9]. Direction is

mixed deliberately: Cerebras takes share from merchant GPUs, which is the theme, but it also competes with hyperscaler in-house silicon for the same inference workloads, which is the theme pointed the other way.

AMD (AMD) — documented, and the squeezed side. Instinct is put at roughly 5-7% of AI accelerator revenue against hyperscaler custom silicon at 10-15% [10]; Data Center segment revenue was \$16.6B in FY2025 with MI-series estimated at \$6-8B [11]. The point is not that AMD is losing to NVIDIA. It is that custom silicon and the merchant second source are bidding for the same thing — the hyperscaler's diversification budget — and the in-house part does not have to earn a gross margin to win it.

So what: the fringe searched here is not a new accelerator startup — it is the connectivity and wafer-scale layer, where two of the three names are less than three years public

EMERGING PLAYERS

Three rows in Exposed_Companies carry the emerging role. One paragraph each, in table order.

Astera Labs (ALAB) — public since March 2024, novel because it monetises the *count* of designs. Astera sells the connectivity silicon that sits between accelerators and the rest of the rack: PCIe/CXL retimers (Aries), smart cable modules (Taurus), CXL memory controllers (Leo) and fabric switches (Scorpio). The mechanical edge over an incumbent networking vendor is that Astera's content scales with accelerator *heterogeneity*, not with accelerator volume from one vendor. At IPO it said content per accelerator was below \$100 [1]; it now puts content close to \$1,000 per XPU, with Scorpio X alone expected to exceed that level [1]. Its Alchip partnership "combines Alchip's custom ASIC development capabilities with Astera Labs' comprehensive connectivity portfolio to deliver validated, interoperable solutions for hyperscalers" [2] — i.e. it is productising the attach onto other people's ASICs. If the theme is right that the number of distinct accelerator designs keeps rising, this is the position that compounds off that fact directly rather than off any one design winning.

Credo Technology (CRDO) — public since January 2022, novel because it bet on copper. Credo's edge is the active electrical cable: a copper cable with SerDes silicon in the connector, which delivers optical-class reach inside and between racks at a fraction of an optical module's power and cost. That matters specifically here because a custom-ASIC rack usually cannot use the incumbent's proprietary interconnect and has to be assembled from standards-based parts. Q1 FY27 revenue of \$479.0 million, up 114.7% year over year [3], is the evidence the attach is real at scale. The edge over the incumbents is that AECs are a product category the large networking vendors treat as an accessory and Credo treats as the business; the vulnerability is that if rack-scale optics get cheap enough, the category compresses.

Cerebras Systems (CBRS) — public since May 2026, novel because it refuses the chiplet consensus.

Cerebras builds a processor the size of a whole wafer: "This allows for 900,000 AI cores and 44 GB of high-bandwidth SRAM memory to be housed on a single device" [4]. The mechanism is that keeping model weights in on-die SRAM removes the off-chip memory hop that dominates inference latency — a different answer to the same cost-per-token problem the hyperscaler ASICs are solving. Coverage is thin because it listed four months ago; it "raised \$5.55 billion in its initial public offering, marking the largest IPO of 2026" [4] with an OpenAI

agreement to build 750 MW of capacity on its chips [4]. Its edge, if the approach works, is that it does not queue for the same advanced-packaging capacity every chiplet-based accelerator competes for. Its problem is that it is a merchant vendor arguing against merchant vendors — which is why the table grades it mixed, not benefits.

ONE NAME LOOKED AT AND LEFT OUT

Etched (private) belongs in the same fringe conversation: a transformer-only ASIC (Sohu), reported at a \$10.3 billion valuation with over \$1 billion in signed contracts and first systems shipping in 2026. It is left off the table rather than added as a thirteenth row because it is unlisted, its numbers are private-round reporting rather than disclosure, and the table's ceiling is twelve. A follow-up that wants the most levered pure expression of "specialised silicon beats general-purpose silicon at inference" should start there.

Ideas

The thesis

OWN THE ATTACH NOT THE ACCELERATOR

The theme's value does not land on whoever wins the accelerator argument. It lands on whoever gets paid per *distinct accelerator design* that has to be connected to a rack. Astera Labs and Credo Technology sell exactly that, and their content rises as the accelerator field fragments — which is the one thing about this theme that is not in dispute.

The mechanism

From Mechanism: a hyperscaler ASIC is a compute die, not a system. It still needs SerDes, retimers, cables, switching and a fabric — and because a custom part usually cannot ride the incumbent's proprietary interconnect, it has to be assembled from standards-based silicon somebody else sells. That is why the same vendors appear on both sides: Astera Labs was one of the first NVLink Fusion custom-silicon partners *and* has a design partnership with Alchip aimed at hyperscaler ASIC racks.

The value lands here rather than on the ASIC designer because the ASIC designer's revenue is programme-specific and the attach vendor's is architecture-agnostic.

The names

Astera Labs (ALAB) is the dollar-content play: content per accelerator has gone from below \$100 at IPO to close to \$1,000 per XPU [1]. **Credo (CRDO)** is the physical-layer play: active electrical cables into hyperscaler racks, Q1 FY27 revenue \$479.0M and +114.7% YoY [2]. They differ in where they sit — Astera inside the accelerator complex, Credo between the racks — and in fragility: Credo's category can be compressed by cheaper rack-scale optics, Astera's cannot as easily. Evidence in Exposure_Evidence; both are emerging rows, argued in Emerging_Players.

What would have to be true

- The number of distinct accelerator architectures in production keeps rising rather than consolidating onto two or three.
- Content per accelerator keeps climbing with each generation rather than being absorbed into the accelerator die or into the rack vendor's own silicon.
- Hyperscaler AI capex holds through 2027-2028 — the depreciation drag on that capex is the acknowledged risk to its durability.

What would kill it

A hyperscaler bringing connectivity in-house on the same logic it used for compute. If Google or Amazon designs its own retimer and fabric silicon alongside its accelerator — exactly the vertical-integration move the theme is *about* — then the attach vendor is the next thing internalised, and today's content growth becomes tomorrow's design-win loss. The single observation that would end the idea: an announced hyperscaler in-house retimer or scale-up fabric part displacing a named Astera or Credo socket.

What is already priced

Both names have compounded revenue at triple digits and neither trades as an unloved asset; the market "has recognized that semiconductors are the foundation of future AI capabilities, awarding premium multiples to companies that can capture accelerated compute demand" [3]. What is *less* obviously priced is the specific claim that content per unit keeps rising — most coverage prices unit volume.

The thesis

THE DESIGN HOUSE IS A PROGRAMME CYCLE

The pure-play ASIC design houses look like the most direct expression of the theme and are the most dangerous way to own it. Their revenue is not a run-rate on a structural trend; it is the sum of two or three programmes at particular points in their tape-out-to-volume cycle. The theme can be entirely right and the stock can halve on a schedule slip.

The mechanism

From Drivers_And_Horizon: the theme resolves over three to five years at the pace of silicon, while a design house recognises revenue in a one-to-two-year NRE-then-volume arc per programme. Those two clocks do not line up. A design house between programmes reports a collapse that looks like the theme failing and is not.

Alchip is the worked example: Q1 2026 revenue of \$132.4 million against \$318.7 million a year earlier [1], with management attributing the expected recovery to "North American demand for 3nm AI accelerator shipments" [1]. Nothing about the theme changed between those two quarters.

The names

Alchip (3661.TW) and **Global Unichip (3443.TW)** are the pure form — see Exposure_Evidence. **MediaTek (2454.TW)** is the hedged form: a large non-AI P&L absorbing the programme risk while it ramps its first AI accelerator ASIC toward mass production in Q4 2026 [2]. The idea is that the cycle, not the theme, is what you are being paid to underwrite — and the entry that matters is the trough between programmes, not the ramp.

What would have to be true

- The customer's next-generation programme is actually awarded to the same design house rather than re-bid or pulled in-house.

- The design house wins a SECOND large customer, so the cycle troughs stop overlapping. MediaTek's own management frames this as the test: whether a second major customer appears and whether the follow-on orders come.
- Node migration keeps the design house's content per programme rising — Alchip already has 21% of revenue on 3nm/2nm work.

What would kill it

A hyperscaler taking back-end physical design in-house after two or three generations of learning, which is the natural end state of the same logic that produced the ASIC in the first place. Concretely: a named Trainium- or TPU-class programme moving from an external design house to the customer's own team. Also fatal, more quietly: NRE-heavy revenue that never converts to volume, where the design house books the design fee and the programme is cancelled before mass production.

What is already priced

Taiwan's market has repriced these names as AI ASIC franchises rather than design-service businesses — on MediaTek, "the market has already priced in a significant portion of the AI ASIC success probability" [2]. That is the opposite of the idea. The mispricing this idea hunts is at the trough, when programme-cycle revenue is mistaken for structural decline.

The thesis

MEMORY AND PACKAGING ARE PAID EITHER WAY

The cheapest way to own this theme is to refuse to have a view on it. An accelerator's bill of materials is dominated by things the architecture argument does not touch: "memory and packaging together now represent 60-70% of AI accelerator COGS" [1]. Whoever wins the compute die, TSMC packages it and SK hynix stacks memory beside it — and the rack it goes into is assembled by somebody else again.

The mechanism

From Who_Wins_And_Loses: the theme moves *which logo is on the compute die*. It does not reduce the HBM stack, the interposer, the CoWoS slot or the rack integration. It arguably increases them, because a fragmented accelerator field means more distinct mask sets, more test flows and more qualification per unit of delivered compute.

The value lands here rather than on the designers because the designers compete with each other on price while the packaging and memory layers are supply-constrained oligopolies. CoWoS capacity runs from ~80,000 WPM to 120,000-130,000 WPM through 2026 [1]; HBM lines run \$2,400-\$6,500 per accelerator unit [1].

The names

TSMC (TSM) — fabricates and packages both sides, and its customer concentration *falls* as the custom field widens. **SK hynix (000660.KS)** — projected at roughly 70% of the HBM4 segment [1]. **Celestica (CLS)** — builds the racks: CCS revenue of \$3,241.0 million in Q1 2026, up 76% [2]. Evidence per name in Exposure_Evidence. The three differ in what they are actually exposed to: TSMC to leading-edge node demand, SK hynix to the memory price cycle, Celestica to three customers.

What would have to be true

- Total accelerator units keep growing, since these three are levered to units and content, not to share.
- Advanced packaging and HBM stay tight enough to hold pricing — a capacity glut converts a toll booth into a commodity.
- The BOM share held by memory and packaging does not fall back as logic content rises again.

What would kill it

An architecture that removes the packaging and memory dependency. Cerebras' wafer-scale part is exactly that observation in silicon: 900,000 cores and 44 GB of on-die SRAM on one wafer [3] needs neither an HBM stack nor a CoWoS interposer. If wafer-scale or on-die-memory designs take meaningful inference volume, the "paid either way" claim stops being true. The nearer-term killer is cruder: a hyperscaler capex pause, with Amazon's trailing-4Q capex already at 102% of trailing-4Q operating cash flow [1].

What is already priced

TSMC and SK hynix are the consensus AI-supply-chain longs and trade accordingly; Celestica has re-rated hard on the CCS growth. What is less priced is the *diversification* argument — that a wider custom field lowers, rather than raises, the concentration risk in these names.

The thesis

CUSTOM SILICON SQUEEZES THE SECOND SOURCE FIRST

The theme is read as a negative on the market leader. Its first-order victim is the merchant *second* source. AMD is bidding for the same thing a hyperscaler's in-house programme is bidding for — the diversification budget, the part of the accelerator spend explicitly allocated to not-the-incumbent — and the in-house part does not have to earn a gross margin to win it.

The mechanism

From Mechanism: the incumbent is defended by CUDA, by rack-scale bundling and by being the only accelerator rentable across every cloud. The second source has none of those. It competes on price-performance for inference — which is precisely the workload an internal ASIC is built for, and the internal ASIC wins that comparison by construction because its margin is zero.

The arithmetic in Exposure_Evidence is stark: Instinct is put at roughly 5-7% of AI accelerator revenue against hyperscaler custom silicon at 10-15% [1]. The challenger is already the smaller challenger.

The names

AMD (AMD) is the row this idea is about — the only hurt direction in the table, Data Center revenue \$16.6B in FY2025 with MI-series estimated at \$6-8B [2]. The mirror side is that the same logic makes **Broadcom (AVGO)** and **Marvell (MRVL)** structurally safer than AMD despite competing for adjacent dollars: they are paid to *build* the in-house part rather than to displace it. That contrast is the idea — not a call on AMD, but the observation that the theme sorts merchant challengers and enablers in opposite directions.

What would have to be true

- Hyperscaler diversification spend is a budget with a ceiling rather than an open-ended second supply line.
- Inference, where the internal ASIC is strongest, continues to grow faster than training, where the merchant GPU's flexibility still matters.
- The internal programmes continue to be judged on total cost of ownership rather than on external-customer rentability, where AMD has an advantage the ASICs cannot have.

What would kill it

AMD winning a hyperscaler's *primary* inference fleet rather than its diversification slot — a deployment sized in gigawatts rather than in thousands of units, announced alongside rather than instead of that hyperscaler's own silicon. A softer killer: hyperscalers concluding that maintaining an internal silicon team costs more in engineering than the margin it saves, and standardising on two merchant vendors instead.

What is already priced

The market prices AMD against NVIDIA, which is the wrong comparison for this theme; "NVIDIA's competitive moat and software stack will be harder for AMD to undermine without major missteps by NVIDIA" is the consensus framing [3], and it is a framing about the incumbent. What is not priced is the three-way version, in which the custom ASIC is the marginal buyer of the diversification dollar and the second source is the residual claimant.

The thesis

WAFER SCALE IS THE BET AGAINST THE PACKAGING QUEUE

Every chiplet-based accelerator — merchant GPU and hyperscaler ASIC alike — queues for the same advanced-packaging and HBM capacity. Cerebras does not. A wafer-scale part is a bet that the binding constraint on AI compute is not design talent but the CoWoS line, and that an architecture which sidesteps that line can take inference volume the queue cannot serve.

The mechanism

From Mechanism: memory and packaging are 60-70% of accelerator COGS and CoWoS capacity is the physical throttle on the whole field. A wafer-scale engine keeps "900,000 AI cores and 44 GB of high-bandwidth SRAM memory" on one device [1], which removes the off-chip memory hop that dominates inference latency and removes the interposer from the bill of materials.

The value lands here, if it lands at all, because the theme's own success creates the bottleneck: more distinct accelerator programmes means more claims on the same packaging capacity, which makes an architecture that does not need it more valuable, not less.

The names

Cerebras Systems (CBRS) is the only listed pure expression — public since May 2026, having "raised \$5.55 billion in its initial public offering, marking the largest IPO of 2026" [1], with an OpenAI agreement to build 750 MW of capacity on its chips [1]. See *Emerging_Players* for why it is graded emerging and mixed rather than a clean beneficiary. The unlisted comparison is Etched, whose transformer-only Sohu is the same argument taken further; it is named in *Emerging_Players* and deliberately kept off the table.

What would have to be true

- Inference workloads that fit in 44 GB of on-die SRAM represent a commercially meaningful share of served tokens, rather than a niche below frontier model size.
- Advanced packaging stays tight enough through 2027 that avoiding the queue is worth a premium.
- The OpenAI capacity agreement converts to delivered megawatts rather than remaining a signed intention — a four-month-old listing has not yet shown a delivery record.

What would kill it

CoWoS capacity expansion removing the bottleneck. Capacity is already scheduled to go from ~80,000 WPM to 120,000-130,000 WPM through 2026 [2]; if that lands and lead times normalise, the entire premium for sidestepping the queue evaporates and the comparison reverts to raw cost per token, where a wafer-sized die with single-digit yield economics is a hard argument. The second killer is model size: if serving frontier models requires holding far more than 44 GB near the compute, the architecture is confined to smaller models.

What is already priced

Very little is establishable. The stock has four months of trading history and thin sell-side coverage, which is the reason it is in this pod at all — the fringe is where the research gap and the error bars are both largest. Treat the thinness as the opportunity and the falsifiers above as the reason to size it as a candidate rather than a conviction.

Summary

So what: the theme pays the people who make custom silicon possible, not the people fighting over whose accelerator wins

FACTOR RESEARCH SUMMARY

Custom AI silicon is a real, measured re-mix of a growing market — ASIC-based AI servers reach 27.8% of shipments in 2026, the highest since 2023, while the overall AI server forecast was raised from 28% to nearly 31% YoY. A shrinking percentage of a faster-growing pie. The investable consequence is not a view on which accelerator wins; it is that the enabling layer — design services, connectivity, packaging, memory, rack integration — gets paid per *design* and per *unit*, not per *winner*.

THE PEER TABLE

COMPANY	TICKER	ROLE	DIRECTION	CONF.	MECHANISM IN ONE LINE
Broadcom	AVGO	incumbent	benefits	high	Designs the XPU for Google/Meta/OpenAI and sells the networking around it; AI semis \$10.8B in Q2 FY26, +143%
Marvell	MRVL	incumbent	benefits	high	Second custom-XPU house; data centre \$2.172B = 79% of Q2 FY27 revenue, +46%
Astera Labs	ALAB	emerging	benefits	high	Connectivity content per XPU from <\$100 at IPO to ~\$1,000
Credo	CRDO	emerging	benefits	high	AECs into hyperscaler AI racks; Q1 FY27 revenue \$479.0M, +114.7%
Alchip	3661.TW	incumbent	benefits	medium	Trainium-programme design house; Q1 2026 revenue \$132.4M vs \$318.7M a year earlier
Global Unichip	3443.TW	incumbent	benefits	medium	TSMC-affiliated design services; TTM revenue ~\$1.17B
MediaTek	2454.TW	incumbent	mixed	medium	First AI ASIC to mass production Q4 2026; \$3.5B NVIDIA investment keeps it inside NVIDIA's rack
Celestica	CLS	adjacent	benefits	high	Builds the custom AI racks; CCS \$3,241.0M in Q1 2026, +76%

COMPANY	TICKER	ROLE	DIRECTION	CONF.	MECHANISM IN ONE LINE
TSMC	TSM	adjacent	benefits	high	Fabs and packages both sides; CoWoS ~80k to 120-130k WPM through 2026
SK hynix	000660.KS	adjacent	benefits	medium	HBM either way; ~70% of the HBM4 segment
Cerebras	CBRS	emerging	mixed	medium	Wafer-scale sidestep of the packaging queue; \$5.55B IPO, May 2026
AMD	AMD	incumbent	hurt	medium	Merchant second source bidding for the same diversification budget the ASICs take

Full mechanism and grading per row in Exposed_Companies; the case for each placement, and whether the link is documented or inferred, in Exposure_Evidence.

THE IDEAS

1. **Own the attach, not the accelerator** — Astera Labs and Credo get paid per distinct design; killed by a hyperscaler internalising connectivity the way it internalised compute.
2. **The design house is a programme cycle, not a franchise** — Alchip, GUC, MediaTek; killed by back-end design moving in-house after two or three generations.
3. **Memory and packaging are paid either way** — TSMC, SK hynix, Celestica; killed by an architecture that removes the packaging dependency, or by a capex pause.
4. **Custom silicon squeezes the second source first** — AMD is the residual claimant on the diversification budget; killed by AMD winning a primary inference fleet rather than a diversification slot.
5. **Wafer-scale is the bet against the packaging queue** — Cerebras; killed by CoWoS capacity expansion landing on schedule.

WHAT THIS RUN DID NOT SETTLE

This is the part the bag cannot show, and it matters more than the list above.

The headline growth differential could not be grounded. The factor's own description cites ASIC server shipments growing 44.6% against 16.1% for merchant GPUs. Two TrendForce press releases and two secondary write-ups were checked and none of them carries that pair verbatim; what is quotable is the 27.8% share figure and the raised 31% total. The 44.6/16.1 pair is used nowhere in these pods for that reason. A follow-up should source it from the TrendForce report itself (RP251105VO) or drop it from the factor description.

Two exposures stayed inferred rather than documented. Global Unichip's programme-level customer attribution rests on the business model and secondary reporting, not on a filing this run could quote — hence medium, and hence the honest paragraph in Exposure_Evidence rather than a cleaner-looking row. Credo's revenue is documented but the *custom-ASIC share* of that revenue is not separately disclosed anywhere found; the row is graded on revenue evidence, not on an attribution the filings do not make.

The thinnest idea is the fringe one. Idea 5 rests on a company with four months of trading history, one large capacity agreement that has not yet converted to delivered megawatts, and no public unit economics for a wafer-sized die. It is written as a candidate with falsifiers precisely because it cannot be underwritten yet.

Nobody checked the other direction hard enough. Every idea here assumes hyperscaler AI capex holds. The one contrary datum this run surfaced — Amazon's trailing-4Q capex at 102% of trailing-4Q operating cash flow, with a 2027-2028 depreciation drag — is a demand-side risk that would break ideas 1, 2, 3 and 5 simultaneously. That correlation is not diversified away by holding several of them.

And one name was deliberately left out. Etched, private, transformer-only ASIC. It is the most levered pure expression of the theme's logic and it is not in the table because it is unlisted and its figures are private-round reporting. A follow-up with access to private markets should start there.

HOW NVDA SITS INSIDE IT

One sentence, because the workbook argues this elsewhere: the merchant vendor's ceded share is real and measured, and the same period shows its fabric revenue outgrowing its compute revenue — which is why this run reads the theme as a re-mix of where profit sits in the rack rather than as a displacement of the rack's owner. That reading is the run's lens, and it is named as such on the factor.

FACTOR RESEARCH

NVDA Factor 05 Demand Is Credit Funded

Theme

The buildout stopped being paid for out of cash flow, so the demand signal is now a credit signal

MECHANISM

AI infrastructure demand used to be a function of how much cash its buyers generated. It is now a function of how much they can borrow, and of whose balance sheet the lender is actually underwriting. That substitution is the whole mechanism.

STEP 1 – CAPEX OUTRUNS THE CASH THAT FUNDS IT

The four US hyperscalers bought \$433.9B of property and equipment in the four quarters through March 2026 against roughly \$149B of reported depreciation, with 1Q26 combined capex of \$129.8B (+80% year over year) and calendar-2026 guidance summing to roughly \$700B [1]. Depreciation recognises that spend over 5-6 year server schedules and 25-40 year building schedules, so the income statement carries a fraction of the build while the cash goes out today.

STEP 2 – SO CASH-RICH BUYERS ISSUE DEBT ANYWAY

Meta priced \$30B in October 2025 (the largest investment-grade deal of that year) plus a roughly \$27B off-balance-sheet SPV; Alphabet raised about \$25B in November 2025 and about \$31B more in February 2026; Amazon added \$24.9B after \$15B in November - all while generating record operating cash flow [1]. The clearest single number: Amazon's trailing-four-quarter capex now runs at 102% of its trailing-four-quarter operating cash flow [1]. Above 100%, every incremental dollar of build is funded externally by definition.

STEP 3 – BELOW THE HYPERSCALERS, THERE IS NO BALANCE SHEET TO LEND AGAINST

This is where the theme bites hardest. The developers, landlords and lenders building AI capacity outside the hyperscalers cannot raise money at scale against the credit of unprofitable AI labs or debt-heavy cloud startups [2], and the frontier labs themselves are growing faster than what their balance sheets and credit profiles can support [3]. The asset - a rack of GPUs, a powered shell - is not what gets underwritten. The *contract* is.

STEP 4 – SO CREDIT QUALITY IS IMPORTED FROM A THIRD PARTY

Three substitutes for a borrower's own credit have emerged, each with a price tag. A vendor guarantee: NVIDIA's credit support on the SB Energy PORTS campus covers roughly 4.25 gigawatts of IT load under 20-year leases to OpenAI, with its obligation capped at \$105 billion in the aggregate [4]. A minimum-revenue floor: rather than requiring a single long-term offtake guarantee, NVIDIA provides a take-or-pay commitment on part of a neocloud facility's capacity so lenders will underwrite the project, taking a revenue share above that floor in exchange [3]. And an offtake backstop: if AI clouds do not sell committed capacity to third parties, NVIDIA has agreed to purchase that capacity itself [4].

STEP 5 – WHAT ENFORCES IT

Tenor mismatch. GPU-backed facilities are written against customer contracts averaging roughly three years [5] while the debt and the lease sit five to twenty years out. Nothing physically stops the chain; a refinancing window does. In the internet era, equipment vendors helped finance their customers because demand was still speculative, and much of the fiber optic cable laid in those years sat dark for a decade - the difference this time being that the infrastructure is in use the day it switches on [2].

The clock is a refinancing clock, not a technology clock – and it runs 2027-2029

DRIVERS AND HORIZON

This theme does not resolve when AI stops being useful. It resolves when a lender declines to roll a facility, and the earliest that can happen at scale is 2027.

WHAT HAS ALREADY CHANGED

Three years ago the AI buildout was equity- and cash-funded, and the debt markets were a footnote. In 2025, hyperscaler issuance of \$108 billion covered 27% of their capex; in 2026 it is running at \$194 billion year-to-date against an expectation that roughly one-third of capex will be debt-financed, and a full-year expectation of something like \$250 billion of direct hyperscaler supply [1]. AI-driven paper is now about 18% of year-to-date investment-grade supply and roughly the same share of high yield [1]. The largest hyperscalers have pledged to spend as much as \$725 billion in the current year, funded with a mix of cash, new borrowings and - in Meta's case - arrangements that keep some of it off balance sheet [2].

WHAT PUSHES IT HARDER

- **Committed forward demand.** Cloud industry backlog is greater than \$2 trillion, and top-five hyperscaler capex is expected to reach nearly \$800 billion in 2026 and \$1.3 trillion in 2027 [3]. Backlog that large is what convinces a credit committee.
- **Capacity that has to be built by someone.** NVIDIA's neocloud partners are expected to exit the year with 8 gigawatts of installed capacity, up from approximately 3 gigawatts at the end of 2025 [3] - a near-tripling that has to be financed by entities without hyperscaler balance sheets.
- **New capital pools being purpose-built.** NVIDIA has signed MOUs with Apollo, BlackRock, Blackstone, Brookfield, Goldman Sachs and KKR to mobilise over \$500 billion of third-party capital for AI infrastructure [4], turning compute into an asset class rather than a capex line.
- **Headroom that has not been used.** Bringing the hyperscalers merely to the index weight of the largest US bank issuers would imply an incremental \$510 billion of debt capacity in the US IG market alone [1]. The constraint today is appetite, not capacity.

WHAT SLOWS IT

The monetisation lag is the pressure point: there is a delay between when the hyperscalers invest and when they can monetize that investment of anywhere from three months to two years [1]. Layer that onto structures that have mostly been five years [1] against GPU customer contracts averaging roughly three years [5], and the

squeeze arrives at refinancing, not at purchase. Investor concentration is the second brake - institutions already carrying the theme in equity say that tempers their demand for the same theme in credit [1].

THE CLOCK

18 months to three years to the first real test; five to seven years to resolution. Debt-financed share of capex is expected to peak in 2027 at about 35% [1], which is also when project-finance volumes outside direct hyperscaler issuance are expected to add roughly \$300 billion [1]. The 2025-2026 vintages of five-year paper therefore come due 2030-2031, but the first repricing - spreads, not defaults - lands when 2027 supply meets whatever the monetisation evidence looks like by then. Anyone positioning on this theme over a two-quarter horizon is trading sentiment; the mechanism itself does not report until the 2027 refinancing window.

The theme pays whoever supplies, prices or collateralises the credit, and charges whoever consumes it

WHO WINS AND LOSES

When demand is credit-funded, value stops accruing to whoever builds best and starts accruing to whoever is cheapest to lend to - or who gets paid for the lending. Those are different populations, and neither is the company doing the building.

IT FAVOURS

- **Anyone who originates or distributes the credit rather than taking the asset risk.** Private-credit managers, alternative-asset platforms and the banks underwriting multi-billion-dollar deals earn fees on volume; the volume is already committed. The financing platforms are being purpose-built at the \$500bn scale see the drivers, and the project-finance channel outside direct hyperscaler issuance is expected to run \$300bn in 2027 alone [1].
- **Anyone whose asset is the collateral rather than the borrower.** Landlords, colocation operators and REITs holding powered shells under long leases to investment-grade tenants. The price of that distinction is explicit: operators with contracted capacity to major AI companies are securing financing at 50-100 basis points below market rates [2]. Power availability has emerged as the single most critical factor in data center financing decisions for 2026 [2], so a secured interconnect is now a credit attribute, not an operating one.
- **Anyone who can rent out an investment-grade balance sheet.** A guarantee, a take-or-pay floor or a residual-value backstop converts an unfinanceable project into a financeable one. Whoever supplies that conversion captures a fee or a revenue share for it.
- **Anyone selling into the build whose receivable sits with an investment-grade buyer rather than a leveraged one.** Same order book, different counterparty quality - and in this theme counterparty quality is the whole variable.
- **Anyone who prices the new risk.** Rating agencies, index and benchmark providers, and the venues listing instruments against compute and GPU rental economics. A new asset class needs marks.

IT SQUEEZES

- **The leveraged buyer of a depreciating asset.** Borrowing five-year money [1] against customer contracts averaging roughly three years see the mechanism is a duration bet dressed as an operating business.

Neoclouds and GPU-rental operators sit exactly here.

- **Anyone whose cost of capital is an input to the theme rather than an output of it.** Oracle is the worked example: S&P Global Ratings downgraded Oracle to the lowest investment-grade rating, citing the company's growing AI spending [3], with its debt at about 2.5 times sales in 2025 and its cash burn expected to accelerate [3].
- **Anyone competing for the same investor's risk budget.** Jumbo AI supply is roughly 18% of year-to-date investment-grade issuance and about the same in high yield see the drivers. Buyers gave an unusually chilly reception to a \$25-billion issuance from Amazon [3] - crowding out is already observable.
- **Anyone holding speculative, uncontracted capacity.** No investment-grade tenant means no cheap money, and in a market where the contract is the collateral, an empty powered shell is a liability with a coupon.
- **The vendor that becomes a guarantor.** Taking customer credit risk without being paid a credit spread for it converts a hardware margin into an unpriced contingent liability.

THE SPLIT THAT MATTERS FOR THE NEXT PHASE

The theme does not cut one way. Its single sharpest test is **whose balance sheet absorbs a GPU that does not earn its contracted rate**: the lender, the lessor, the operator, or the vendor. Every company worth putting in the landscape table can be placed by answering that one question.

Landscape

Exposed Companies

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
CoreWeave	CRWV	incumbent	Borrows against the GPUs themselves: delayed-draw term loans collateralised by NVIDIA hardware, written against customer contracts averaging roughly three years while the paper runs five to six; ~\$35.1B principal debt at Q2 2026.	mixed	high	~100%	Top two customers ~63% of H1 2026 revenue; ~\$104B contracted backlog. The purest expression of the theme in both directions.
Oracle	ORCL	incumbent	The most levered investment-grade buyer: FY2026 free cash flow of ~\$23.7B, debt about 2.5x sales, and ~\$260B of data-centre leases not yet commenced; downgraded to the lowest IG rating on AI spending.	hurt	high	OCI	Off-balance-sheet lease commitments exceed on-balance-sheet debt. The worked example of the theme's cost landing on the buyer.
Blue Owl Capital	OWL	incumbent	Converts hyperscaler capex into an asset it owns and leases back: 80% of the ~\$27B Meta Hyperion JV, the largest private-credit transaction on record, plus multi-billion lending to Stack Infrastructure.	benefits	medium	Not disclosed	Real Assets AUM ~\$85.1B at 31 Mar 2026. Earns spread and fees; carries residual-value risk on the shell, not the GPU.

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Applied Digital	APLD	emerging	Levered landlord to levered tenants: ~\$36B of contracted lease revenue with nearly 86% from two customers, ~\$11B of it CoreWeave, funded at a debt-to-equity ratio near 4.8x.	mixed	high	~100% of contracted lease revenue	Second-order credit exposure: its counterparty is itself a GPU-secured borrower.
TeraWulf	WULF	emerging	Miner turned HPC lessor whose collateral is a single lab lease: a 20-year Anthropic lease worth ~\$19B (up to ~\$33B with both extensions), plus \$600M of Google credit support standing behind Fluidstack's lease obligations.	mixed	high	HPC lease revenue ~71% of Q2 2026 revenue	Third-party credit support is explicitly what unlocks the financing - the theme's mechanism, visible in one line item.
Nebius Group	NBIS	emerging	Funds GPUs with unsecured converts and customer prepayments instead of GPU-secured debt: \$5.0B priced August 2026 in two series (0.50% due 2030, 4.50% due 2034), leaving it net cash positive.	mixed	high	~100%	Top three clients ~59% of Q2 2026 revenue. The structural alternative to collateralised GPU debt.
Ares Management	ARES	incumbent	Arranges the infrastructure debt and keeps a slice: a \$2.4B facility to Vantage Data Centers, holding ~\$1.6B of the commitment, against a portfolio 100% leased where ~99% of North American customers are investment grade.	benefits	high	Digital infrastructure is a slice of direct lending	Underwrites the contract, not the hardware. Guided to \$50-100M of digital-infrastructure fee-related earnings in 2027+.

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Blackstone	BX	incumbent	Largest owner of data-centre assets (~\$185B including QTS) and a lender across the capital stack; QTS raised a \$3.25B leveraged loan in July 2026 and the firm expects ~\$100B invested or committed by end-2026.	benefits	medium	Not disclosed	Equity and debt in the same asset - the theme cuts both ways inside one balance sheet.
Apollo Global Management	APO	incumbent	Structures the jumbo private investment-grade placements the public market will not absorb; led a ~\$35B post-IPO debt financing for Broadcom's AI platform supporting Anthropic compute.	benefits	medium	Not disclosed	Forecasts ~\$2.7T of AI-related spend 2025-2029, of which it expects over \$1T to shift to private credit - a house view worth marking to evidence.
Digital Realty	DLR	incumbent	The investment-grade landlord that borrows cheaply because its tenants are rated; record \$1.9B annualised backlog at Q2 2026, net debt/EBITDA 4.7x, guiding \$1.5-2.0B of H2 2026 issuance at 4.5-5.5%.	benefits	medium	~100%	Benefits from the same spread differential that punishes speculative capacity.
Equinix	EQIX	adjacent	Securitises the lease rather than the GPU: a GBP 280m data-centre CMBS backed by leases to three investment-grade tenants priced at 130bp over Gilts, alongside a \$3B investment-grade bond in July 2026.	benefits	medium	~100%	Shows the cheapest funding in the theme goes to whoever holds the IG contract, not the hardware.

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
CME Group	CME	adjacent	Toll booth on the theme's price risk: launching Silicon Data H100 and B200 Rental Index Futures on 5 October 2026, earning transaction fees whether compute rental prices rise or fall.	benefits	high	Negligible today	The only row that is indifferent to the theme's direction. Also makes GPU residual value observable, which changes how every other row gets underwritten.

Read the table by asking one question of each row: when a GPU fails to earn its contracted rate, whose balance sheet absorbs it?

EXPOSURE EVIDENCE

The rows below are in table order. For each, what the company is, why it sits where it sits, and whether the exposure link is documented or inferred.

CoreWeave (CRWV) - incumbent, mixed, high. The AI cloud that invented the collateral. Its delayed-draw term loans are secured on NVIDIA hardware and underwritten against customer contracts of approximately three years average length [1], while the facilities themselves run five to six years. Principal debt of roughly \$35.1bn at Q2 2026 against a contracted backlog near \$104bn, and a top-two customer concentration of about 63% of first-half revenue, are drawn from its Q2 2026 quarterly report; the 10-Q chunk in this corpus states plainly that principal debt obligations included multiple high-coupon senior and convertible notes creating substantial leverage and refinancing risk [2]. Documented, and the direction is genuinely mixed: the same credit access that funds its growth is the thing that can be withdrawn.

Oracle (ORCL) - incumbent, hurt, high. The buyer that already paid the price. Its cash burn, with debt at about 2.5 times sales in 2025, is expected to accelerate [3], and S&P Global Ratings downgraded it to the lowest investment-grade rating citing growing AI spending [3]. Roughly \$260bn of data-centre leases had not yet commenced as of its May 2026 year end - an obligation larger than its on-balance-sheet debt that does not appear as debt. Documented for the rating action and leverage; the lease figure is from transaction reporting rather than quoted here.

Blue Owl Capital (OWL) - incumbent, benefits, medium. The lender that became the owner. Funds it manages hold 80% of the joint venture behind Meta's ~\$27bn Hyperion campus, the largest private-credit transaction on record, with Meta leasing the asset back - which is precisely the conversion of capex into an operating lease the theme describes see step 4. Its Real Assets platform held about \$85.1bn of AUM at 31 March 2026. Reported rather than quoted from a filing in this corpus, hence medium.

Applied Digital (APLD) - emerging, mixed, high. A crypto host rebuilt as an AI landlord, and the clearest case of stacked credit risk in the table: nearly 86% of its contracted revenues derive from just two customers [4], with approximately \$36bn of contracted lease revenue of which roughly \$11bn is CoreWeave [4]. Its counterparty is itself a GPU-secured borrower, so a credit event one level up arrives twice. Documented.

TeraWulf (WULF) - emerging, mixed, high. The theme's mechanism visible in a single disclosure. Its 20-year Anthropic lease at the Justified campus represents approximately \$19bn of contracted revenue over the initial term, up to approximately \$33bn if both five-year extensions are exercised [5]; and the delivery of CB-3 unlocked \$600m of Google credit support for Fluidstack's lease obligations - described by its own CFO as a credit milestone that strengthens the contracted revenue profile [5]. Third-party credit standing behind a tenant is the financing, not a detail of it. Documented.

Nebius Group (NBIS) - emerging, mixed, high. The structural alternative. Rather than secured GPU debt it priced \$5.0bn of convertible senior notes in August 2026 in two series - \$3.0bn of 0.50% notes due 2030 and \$2.0bn of 4.50% notes due 2034 [6] - and funds the rest from customer prepayments, leaving it net cash positive against roughly \$8.0bn of cash. Its top three clients were about 59% of Q2 2026 revenue. Documented for the financing; the concentration figure is from its Q2 2026 reporting.

Ares Management (ARES) - incumbent, benefits, high. The clean picture of who the theme actually pays. It arranged a \$2.4bn debt facility for Vantage Data Centers, holding approximately \$1.6bn of the commitment and funding approximately \$330m at closing [7], against a portfolio where roughly 99% of North American customers hold an investment grade rating [7]. Ares underwrites the contract, not the hardware, and it is paid whether or not the hardware earns its rate. Documented.

Blackstone (BX) - incumbent, benefits, medium. Holds both sides at once: the largest portfolio of data-centre assets, around \$185bn including QTS, plus lending across the capital stack - QTS itself raised a \$3.25bn leveraged loan in July 2026. Exposure is unambiguous; the specific figures come from transaction reporting rather than a filing quoted here, so medium.

Apollo Global Management (APO) - incumbent, benefits, medium. Writes the cheques the public market will not absorb, most visibly a roughly \$35bn post-IPO debt financing for Broadcom's AI platform supporting Anthropic compute. Its own published forecast - approximately \$2.7tn of AI-related spending 2025-2029, with over \$1tn potentially shifting into private credit - is a house view and is marked as such, not as evidence. Medium.

Digital Realty (DLR) - incumbent, benefits, medium. The investment-grade landlord on the right side of the spread. Operators with contracted capacity to major AI companies secure financing at 50-100 basis points below market rates [8], and Digital Realty is the listed embodiment of that: a record \$1.9bn annualised backlog at Q2 2026, net debt/EBITDA of 4.7x, and H2 2026 issuance guided at 4.5-5.5%. Medium, because the funding advantage is a market-level finding applied to the company rather than a company-level disclosure.

Equinix (EQIX) - adjacent, benefits, medium. Proves where the cheap money goes by securitising the lease rather than the machine: a GBP 280m data-centre CMBS backed by leases to three investment-grade tenants, priced at 130 basis points over Gilts, alongside a \$3bn investment-grade bond in July 2026. Adjacent rather than incumbent because it is not contesting the theme's economics - it is a beneficiary of how the theme prices tenant credit. Medium.

CME Group (CME) - adjacent, benefits, high. The only row indifferent to direction. Both Silicon Data H100 Rental Index Futures and Silicon Data B200 Rental Index Futures will track indexes measuring hourly rental GPU costs published by Silicon Data [9], listing on 5 October 2026. CME earns transaction fees either way. The larger

consequence is second-order: a public reference price for GPU rental makes residual value observable, and every GPU-secured lender in this table is currently underwriting residual value without one.

WHAT THIS TABLE DOES NOT COVER

The equipment-leasing chain - the firms that buy GPU servers and lease them on, transferring obsolescence risk to the lessor - is materially exposed to this theme and is almost entirely private (CHG-MERIDIAN, Blue Street Capital and similar). There is no listed pure-play, so the table cannot represent it, and a reader should not read its absence as an absence of exposure.

The fringe of this theme is not a startup — it is a former bitcoin miner with a 20-year lease and a credit structure nobody has priced

EMERGING PLAYERS

Three rows carry the emerging role. What makes them emerging is not size; it is that each is attacking the theme's central problem - how do you finance a GPU when nobody will lend against a GPU - with a mechanism the incumbents have not copied, and each has a coverage base built for a different business than the one it now runs.

APPLIED DIGITAL (APLD) — THE LEVERED LANDLORD TO A LEVERED TENANT

Why emerging: APLD was a crypto-hosting business until 2023 and is still covered largely by analysts who model hosting economics, not lease credit. Its novel mechanism is to take development and construction risk on AI campuses and sell the finished capacity under long leases - approximately \$36 billion in contracted lease revenues, with nearly \$20 billion associated with one hyperscaler across Delta Forge 1, Polaris Forge 3 and Delta Forge 2, CoreWeave another \$11 billion through Polaris Forge 1, and a third hyperscaler \$5 billion at Polaris Forge 2 [1]. Nearly 86% of contracted revenues therefore rest on just two customers [1].

Its edge, if it works: it is paid for building and powering, not for operating GPUs, so it captures the theme without taking silicon obsolescence risk. Its distinctive danger is one level up - CoreWeave is itself a GPU-secured borrower, so APLD holds credit exposure to a company whose own credit is collateral-dependent. That stacking is not present at Digital Realty or Equinix, and it is the reason APLD is not simply a small DLR.

TERAWULF (WULF) — CREDIT SUPPORT AS THE PRODUCT

Why emerging: WULF was a bitcoin miner; HPC leasing became approximately 71% of revenue only in Q2 2026, so the coverage and the business have not yet converged. Its mechanism is to convert power-advantaged sites into leases to counterparties that cannot borrow on their own, and then to import the missing credit. Its 20-year Anthropic lease at Justified represents approximately \$19 billion of contracted revenue over the initial term and up to approximately \$33 billion if Anthropic exercises both five-year extension options [2]. Separately, delivery of CB-3 unlocked \$600 million of Google's credit support for Fluidstack's lease obligations - which its CFO describes as an important credit milestone that further strengthens the contracted revenue profile [2].

Its edge, if it works: TeraWulf is monetising the exact scarcity the theme creates. An AI lab with enormous demand and no investment-grade rating cannot lease capacity; a landlord who can assemble third-party credit support around that lab can, and gets paid a premium for it. The incumbents cannot copy this cheaply because their tenant underwriting standards are built around already-rated counterparties.

NEBIUS GROUP (NBIS) — REFUSING THE COLLATERAL

Why emerging: Nebius reached its current form through the 2024 Yandex separation and is covered as a growth equity rather than as a credit. Its mechanism is the inverse of everyone else's: instead of pledging GPUs, it funds them with unsecured converts and customer prepayments. In August 2026 it priced \$5.0 billion in two series - \$3.0 billion of 0.50% convertible notes due 2030 and \$2.0 billion of 4.50% convertible notes due 2034 [3] - and holds roughly \$8.0 billion of cash against roughly \$8.5 billion of convertible debt, leaving it net cash positive.

Its edge, if it works: in a repricing, a GPU-secured borrower faces a collateral-value test and a refinancing at once. Nebius faces a conversion price and a maturity date. If GPU residual values are marked down - and CME's compute futures will make that mark public from 5 October 2026 see the CME row - the unsecured funder is the one structure in this table that does not get a margin call. Its offsetting weakness is dilution and a top-three client concentration near 59% of Q2 2026 revenue.

WHAT THE FRINGE SEARCHES DID NOT FIND

The genuinely novel financing platforms - GPU-backed ABS sponsors, compute-securitisation vehicles, and the specialist equipment lessors - are private. Nscale (preparing a US listing against a contracted backlog reported above \$100bn), Crusoe and Lambda are all pre-IPO as of September 2026, and the equipment-leasing chain has no listed pure-play at all. That is an honest gap rather than an absence of exposure, and it is the single thing a follow-up pass should chase, because the fringe of this theme is where the research deficit is largest.

Ideas

The scarce service in this theme is assembling somebody else's credit rating around a tenant who does not have one

IMPORTED CREDIT IS A BUSINESS MODEL

The thesis. Frontier AI labs have enormous, contracted demand and no investment-grade rating, so they cannot lease capacity on their own. A landlord who can assemble third-party credit support around such a tenant can, and charges for it. TeraWulf is the listed demonstration; Applied Digital runs the same play with weaker counterparty quality.

The mechanism. This is step 4 operating at the landlord rather than the vendor level. TeraWulf's disclosure makes it explicit: delivery of CB-3 unlocked \$600m of Google's credit support for Fluidstack's lease obligations, which its own CFO describes as a credit milestone strengthening the contracted revenue profile. Separately its 20-year Anthropic lease represents approximately \$19bn of contracted revenue, up to approximately \$33bn with both extensions (credit support as the product). The value lands with the assembler because the credit, not the concrete, is what was missing.

The names. TeraWulf (WULF), where HPC leasing reached roughly 71% of Q2 2026 revenue and the business has outrun its coverage base. Applied Digital (APLD) as the same model without the imported credit - its concentration is with a GPU-secured borrower rather than behind a hyperscaler guarantee, which is precisely the difference the idea is about.

What would have to be true. That third-party credit support continues to be offered on these terms. It exists because an investment-grade party wants the capacity built; if hyperscalers and vendors decide guarantees are too expensive in ratings treatment, the supply of imported credit stops and the model has no input.

What would kill it. A rating agency treating a guarantee as debt at the guarantor and the guarantors withdrawing in response. The Moody's and S&P guarantee adjustments already applied in this market are the leading indicator; if those adjustments harden, this idea ends before any tenant misses a payment.

What is already priced. TeraWulf is priced as a former bitcoin miner with a lease book - a revenue story. What is not priced is that its contracted revenue is of two different credit qualities depending on whether third-party support stands behind it, and the market has no framework for splitting a backlog that way (the squeezed side).

Get paid on the contract, which is what is actually being financed, rather than on the chip, which is only what is being pledged

OWN THE UNDERWRITING NOT THE SILICON

The thesis. The AI buildout's financing runs through parties who are paid a spread and a fee for converting an unfinanceable project into a financeable one. That conversion is the scarce service, it is priced per transaction, and it is compensated whether or not the GPUs inside the building earn their contracted rate. The exposure is Ares Management, Blue Owl Capital, Apollo Global Management and Blackstone.

The mechanism. Step 3 of the mechanism is the whole idea: below the hyperscalers there is no balance sheet to lend against, so the asset that gets underwritten is the contract, not the hardware. Ares' Vantage facility is the cleanest demonstration - it holds roughly \$1.6bn of a \$2.4bn commitment against a portfolio where about 99% of North American customers are investment grade (see the Ares row). The value lands with the arranger rather than the operator because the arranger's return is contracted at origination while the operator's return depends on utilisation five years out.

The names. Ares is the purest: an arranger that syndicates most of what it writes and keeps a minority slice, with digital-infrastructure fee-related earnings guided at \$50-100m for 2027 and beyond. Blue Owl has moved furthest up the risk curve - it owns 80% of the Meta Hyperion joint venture rather than merely lending into it, so it holds residual-value risk on the shell. Apollo writes the jumbo private placements the public market will not absorb. Blackstone sits on both sides, holding roughly \$185bn of data-centre assets alongside its lending.

What would have to be true. That the fee and spread income is genuinely detached from asset performance - i.e. that these are originate-and-distribute businesses, not warehouses. And that project-finance volume outside direct hyperscaler issuance actually arrives at the scale expected for 2027.

What would kill it. A single large data-centre credit event in which the arranger is found to have retained the risk rather than distributed it. Watch the retained slice: Ares funded about \$330m at closing on a \$2.4bn facility, a ratio that is the idea's health metric. If retained percentages rise across new deals, the toll booth has quietly become a warehouse and this idea is dead.

What is already priced. Partially. The alternative managers have re-rated on AI-adjacency generally, so the theme is not undiscovered. What is less priced is the DIFFERENCE between them - Blue Owl's equity ownership of Hyperion is a materially different risk from Ares' arranger economics, and they are not currently distinguished. That distinction, not the sector call, is where the work is (the structural sides).

Every GPU-secured lender in this theme is currently underwriting a residual value that has no public price – and that ends on a known date

RESIDUAL VALUE BECOMES OBSERVABLE ON OCTOBER 5

The thesis. CME Group and Silicon Data list H100 and B200 Rental Index Futures on 5 October 2026. CME is paid transaction fees either way, which makes it the only direction-indifferent exposure in the landscape. But the interesting consequence is not CME's revenue line - it is that a public, tradable reference price for GPU rental economics arrives into a market where billions of dollars of secured debt has been written against residual-value assumptions nobody could observe.

The mechanism. Step 5 identifies tenor mismatch as what enforces this theme: facilities written against roughly three-year customer contracts while the debt runs five to six. That mismatch is only bearable if the collateral holds value at the gap. Until now, what the collateral is worth in year four has been a modelling assumption in a private credit agreement. A liquid rental index makes it a mark, and a mark can move against a covenant.

The names. CME Group is the direct expression. The read-through names are every GPU-secured borrower and lender in the table - CoreWeave most of all, and Blue Owl and Applied Digital as holders of assets whose value is derivative of the same economics (the table).

What would have to be true. That the contracts achieve enough liquidity to be treated as a mark rather than a curiosity, and that lenders' debt-service-coverage and collateral-value covenants are in fact sensitive to rental rates rather than to contracted revenue alone.

What would kill it. The contracts fail to trade. A futures listing that attracts no open interest prices nothing and marks nothing, and the residual-value opacity simply persists. Check open interest in the first two quarters; that is the falsifier and it reports on a schedule.

What is already priced. Almost nothing. The launch has been announced and is a small line for CME, but the second-order consequence - transparency as a repricing event for private credit collateral - is not a mainstream frame. The clock mismatch is the point: this is a dated, observable catalyst in a theme whose other catalysts are all vague and multi-year.

In a market where the contract is the collateral, tenant credit quality is not a risk factor — it is the entire cost of capital

THE INVESTMENT GRADE TENANT IS THE SPREAD

The thesis. Data-centre operators with contracted capacity to major AI companies are financing 50-100 basis points below market rates, while operators holding speculative capacity are not. That gap is the single most mechanical transfer in this theme, and it separates two groups the market still prices as one sector. Long the rated landlord (Digital Realty, Equinix); the funding disadvantage sits with the concentrated, unrated-tenant landlord (Applied Digital).

The mechanism. The structural sides names it directly: when lenders underwrite the contract, a secured interconnect and an investment-grade tenant become credit attributes rather than operating ones. Equinix demonstrates the endpoint by securitising the lease and not the machine - a GBP 280m CMBS backed by leases to three investment-grade tenants at 130bp over Gilts. Applied Digital demonstrates the other end: approximately \$36bn of contracted lease revenue with nearly 86% from two customers, roughly \$11bn of it CoreWeave, which is itself a GPU-secured borrower (see the APLD row).

The names. Digital Realty guides H2 2026 issuance at 4.5-5.5% with net debt/EBITDA of 4.7x and a record \$1.9bn annualised backlog. Equinix funds at roughly 130bp over Gilts on securitised leases. Applied Digital carries a debt-to-equity ratio near 4.8x with the counterparty stacking described above. The three own broadly similar physical assets and borrow on completely different terms.

What would have to be true. That the 50-100bp differential persists rather than compressing as lenders get comfortable with unrated AI tenants. And that Applied Digital's tenants do not obtain investment-grade ratings of their own, which would collapse the spread and the idea with it.

What would kill it. A GPU-secured borrower such as CoreWeave achieving an unambiguous investment-grade corporate rating. NVIDIA's guarantees terminate on exactly this event (`{pr('Mechanism','step 4')}`), and so does this idea - if the unrated tenant becomes rated, the landlord's funding disadvantage disappears and the pair converges.

What is already priced. The equity market prices Applied Digital on backlog, which is a revenue figure, not a credit figure. \$36bn of contracted revenue from two counterparties is a different asset from \$36bn from twenty, and the market does not currently discount it as such. That is the mismatch the idea rests on.

When the collateral is repriced, the borrower who never pledged it faces a maturity date instead of a margin call

UNSECURED SURVIVES THE MARGIN CALL

The thesis. Nebius Group funds GPUs with unsecured convertible notes and customer prepayments rather than GPU-secured debt. In a repricing of GPU residual value, a secured borrower faces a collateral test and a refinancing at the same time; an unsecured borrower faces only the second. That is a structural difference in survivability that is not reflected in how the two are compared.

The mechanism. The theme's enforcement point is refinancing, not demand (the clock). Nebius priced \$5.0bn in August 2026 - \$3.0bn of 0.50% notes due 2030 and \$2.0bn of 4.50% notes due 2034 - and holds roughly \$8.0bn of cash against roughly \$8.5bn of converts, leaving it net cash positive (why it counts as emerging). CoreWeave, by contrast, carries roughly \$35.1bn of principal debt with delayed-draw facilities secured on hardware and underwritten against contracts averaging about three years.

The names. Nebius (NBIS) is the emerging name; CoreWeave (CRWV) is the reference point the idea is measured against. Both are growing, both are concentrated - Nebius' top three clients about 59% of Q2 2026 revenue, CoreWeave's top two about 63% of H1 2026 - so the distinguishing variable is genuinely the funding structure and not the business quality.

What would have to be true. That Nebius does not convert to secured GPU financing as it scales - the cheaper structure is secured, and the temptation is real. And that customer prepayments hold up; prepayments are unsecured credit extended BY the customer, so they are only durable while the customer is itself well funded.

What would kill it. Nebius announcing a large GPU-collateralised facility, which would erase the structural distinction. Or its conversion prices moving so far out of the money that the 2030 and 2034 notes become straight debt requiring cash repayment - at which point being unsecured is not protection, only a later deadline. Either observation ends the idea.

What is already priced. Nebius trades on growth and backlog, and the coverage inherited from its 2024 reconstitution still frames it as a growth equity. The balance-sheet structure is disclosed and available, and it is not what the multiple is arguing about. That is the mispricing: a credit distinction sitting unpriced inside an equity story.

Summary

The theme pays whoever underwrites the contract and charges whoever pledged the chip — and the bill arrives at refinancing, not at purchase

FACTOR RESEARCH SUMMARY

AI infrastructure demand has stopped being a function of what its buyers earn and become a function of what they can borrow. The four US hyperscalers bought \$433.9bn of property and equipment in the four quarters to March 2026 against roughly \$149bn of depreciation, Amazon's trailing capex now runs at 102% of its trailing operating cash flow, and roughly one-third of 2026 hyperscaler capex is expected to be debt-financed (the mechanism, the drivers). Below the hyperscalers the borrower has no balance sheet at all, so the thing being underwritten is the contract, not the hardware - which is why vendor guarantees, take-or-pay floors and capacity backstops have appeared where a credit rating used to be.

THE PEER SET

COMPANY	TICKER	ROLE	DIRECTION	CONFIDENCE
CoreWeave	CRWV	incumbent	mixed	high
Oracle	ORCL	incumbent	hurt	high
Blue Owl Capital	OWL	incumbent	benefits	medium
Applied Digital	APLD	emerging	mixed	high
TeraWulf	WULF	emerging	mixed	high
Nebius Group	NBIS	emerging	mixed	high
Ares Management	ARES	incumbent	benefits	high
Blackstone	BX	incumbent	benefits	medium
Apollo Global Management	APO	incumbent	benefits	medium
Digital Realty	DLR	incumbent	benefits	medium
Equinix	EQIX	adjacent	benefits	medium
CME Group	CME	adjacent	benefits	high

Mechanisms and the case for each placement are in the evidence pod; the three emerging names are argued in Emerging Players.

THE IDEAS

1. **Own the underwriting, not the silicon** - ARES, OWL, APO, BX. Dies if arrangers are found to be retaining rather than distributing the risk.
2. **The investment-grade tenant IS the spread** - DLR, EQIX against APLD. Dies if a GPU-secured borrower earns an investment-grade corporate rating.
3. **Residual value becomes observable on October 5** - CME, with read-through to every secured lender. Dies if the compute futures attract no open interest.
4. **Unsecured survives the margin call** - NBIS against CRWV. Dies if Nebius announces a large GPU-collateralised facility.
5. **Imported credit is a business model** - WULF, APLD. Dies if rating agencies harden their treatment of guarantees and the guarantors withdraw.

WHAT THIS RUN DID NOT SETTLE

The central question is unresolved and this run did not resolve it. Whether the buildout is a structural re-platforming or a debt-funded capex cycle cannot be decided from the evidence gathered here, and nothing below should be read as having decided it.

Five specific weaknesses a follow-up should chase:

- **Five of twelve exposure links are medium, and all five sit on the winning side.** Blue Owl, Blackstone, Apollo, Digital Realty and Equinix rest on transaction reporting rather than on a figure quoted from a filing in this corpus. The bearish rows (Oracle, Applied Digital, TeraWulf, CoreWeave) are the better-evidenced ones, so the table is asymmetrically sourced in a way that flatters the bear case.
- **The equipment-leasing chain is unrepresented.** The firms that buy GPU servers and lease them on - the parties who most directly absorb obsolescence risk - are private, so there is no listed pure-play and the table simply cannot show the segment. Its absence is a data limitation, not a finding.
- **The fringe is mostly pre-IPO.** Nscale, Crusoe and Lambda are where the genuinely novel financing structures sit, and none is investable as of 11 September 2026. The three emerging rows are the best available proxies, not the actual frontier.
- **Idea 3 rests on the thinnest evidence in the bag.** Its premise - that a public GPU rental index repricing private collateral is a live transmission channel - is an inference about covenant sensitivity, not something this run established from a credit agreement. It is the most falsifiable idea here and also the most speculative.
- **No stress case was quantified.** This run established the transmission channel; it did not size what a 100bp spread widening or a 30% GPU residual markdown does to any specific name. That is the obvious next pass.

FACTOR RESEARCH

NVDA Factor 06 Price Terminal Value Guidance

Theme

So what: in this complex, roughly seven of every ten dollars of modelled value sits past the forecast window, and the multiple is not a quality score — so the price is an argument about when growth fades, not about next year's order book.

MECHANISM

1. WHERE THE VALUE ACTUALLY SITS

A discounted cash flow on a hypergrowth asset puts very little value in the years it forecasts explicitly. The NVDA model this workbook built is the worked example: **71.6% of base-case enterprise value is terminal value**, so the explicit FY2027-FY2031 strip carries barely 28% of the number. The consequence is arithmetic, not opinion. A terminal block valued at $FCF / (WACC - g)$ with a 10% WACC and 3% terminal growth runs on a 7-point denominator; widen the WACC by 100bp and the denominator goes to 8, cutting that block ~12.5% and, at a 71.6% weight, ~9% off total enterprise value **with no change to any operating forecast at all**. That is duration, and it is the first mechanism of this theme.

2. THE MULTIPLE IS A FADE FORECAST, NOT A QUALITY RANKING

Across the AI complex the forward multiple ranks inversely to growth. NVIDIA's forward P/E sits at roughly 21-24x against about 63x for AMD [1]; Broadcom carries a higher gross margin than NVIDIA but trades at a ~50% higher forward multiple [2]. The market is not paying for quality delivered — it is paying for expected *change* in earnings. Two structural consequences follow for money. A name already sitting AT its peer median (22.0x forward against a 21.9x peer median, in this workbook's own comp set) has no de-rating cushion: its entire gap to intrinsic value is loaded onto the E. A name at 60x is mostly cushion, and cushion is the first thing a de-rating takes.

3. THE GUIDED-VERSUS-MODELLED GAP IS THE WHOLE ARGUMENT

Managements guide; models fade. Morningstar records being "amazed by Nvidia's forecast for 70%-plus growth in fiscal 2028" and nonetheless models a flat revenue year in fiscal 2032 [3]. This workbook's own base case runs FY2028 revenue growth at 26.8% against roughly 70% guided. That single assumption is where the -40.6% gap between a \$136.78 probability-weighted intrinsic value and a \$230.36 price comes from — a disagreement about where the curve bends, dated to a specific fiscal year rather than left open.

4. THE DISCOUNT RATE STOPPED BEING EXOGENOUS

Roughly one-third of hyperscaler capex is debt-financed in 2026, after \$194bn of issuance [4] — and the equity has begun to trade on credit terms: "When a company finances the demand for its own products, its cash flows co-move with credit spreads and discount rates" [5]. The same source quotes a strategist on the AI trade

generally: rate hikes would "deflate multiples" [5]. The input side moves too — NVIDIA's own total debt went from \$8,468M to \$33,366M with the June 2026 424B5 shelf, which means a WACC estimated before that filing was stale the day after it. In this theme the discount rate is endogenous to the theme.

5. TRANSMISSION

Concentration is what turns a single de-rating into a market event: NVIDIA alone is approximately 7.2% of the S&P 500 [6]. Whoever holds the index holds the fade assumption whether they modelled it or not.

So what: the theme runs on three clocks that resolve at different times — a quarterly capex print, a dated fiscal-year fade, and a terminal assumption that never resolves at all — and 71.6% of the money sits on the one with no expiry date.

DRIVERS AND HORIZON

WHAT IS CURRENTLY PUSHING IT

Buyer capex is still climbing, and the guides are enormous. TrendForce puts the combined 2026 capex of the nine largest CSPs above US\$886.7bn, with the five North American hyperscalers nearly 90% of it, and estimates about \$1.3 trillion for 2027 — nearly 50% year-over-year growth [1]. IDC forecasts data-centre semiconductor revenue of \$477.1bn in 2026, rising to \$843.2bn by 2030 [2]. As long as those guides hold, the near forecast years of every model in this complex get marked up, which *reduces* the terminal-value share and makes the structure look safer.

The depreciation lag is a tailwind that expires. The four US hyperscalers bought \$433.9bn of property and equipment in the four quarters to March 2026 against roughly \$149bn of reported depreciation, and the gap is structural: "depreciation recognizes spend over 5-6 year server schedules and 25-40 year building schedules, so today's income statements carry only a fraction of today's build-out" [3]. Reported earnings across the buyer set are therefore flattered today and drag later — on a 5-6 year schedule, which dates it.

The build is increasingly borrowed. AI-related investment-grade issuance went from \$10bn in 2024 (1% of supply) to \$108bn in 2025 (about 7%) to roughly 18% of supply this year [4]. Goldman expects the debt-financed share of hyperscaler capex to peak in 2027 at about 35%, because monetisation lags investment by "anywhere from three months to two years" [4].

WHAT WOULD SLOW OR BREAK IT

Rates are the direct channel: the expectation of hikes alone "can compress valuations" of the AI trade [5], and a long-duration equity does not need an earnings miss to lose a quarter of its value. Credit is the early-warning channel — data-centre JV paper has already traded wide to originated yields. And the ROI channel is the slow one: the gap between investment pace and realised optimisation is what analysts call "the recipe for an AI bubble-bursting event" [6]. The single most watched number is the second derivative of capex — not whether it grows, but whether it grows less.

THE CLOCK, STATED

1. **0-12 months — quarterly.** Each hyperscaler capex guide, and the credit spreads on data-centre paper. This clock is noisy and moves multiples immediately.
2. **12-24 months — dated.** The guided-versus-modelled growth gap resolves on a known fiscal date: roughly 70% guided for FY2028 against 26.8% in this workbook's base case. Unusually for a valuation argument, there is a scheduled answer.
3. **18-36 months — the depreciation wall.** Server schedules catch up through 2027-2028, converting today's flattered buyer earnings into tomorrow's drag.
4. **No expiry — the terminal assumption.** The 71.6% of enterprise value sitting past the forecast window is never settled by an event. It is refinanced each quarter by sentiment and by the discount rate, which is exactly why it is the dangerous part.

WHAT HAS ALREADY CHANGED

Three years ago this was a multiple-expansion story. NVIDIA has added over \$4.1 trillion of market capitalisation since 2023 [6] and yet trades at an ordinary forward multiple — the re-rating happened in the E, not the P/E. Morningstar's own fair value makes the shape explicit: it implies 33x FY2027 and 19x FY2028 adjusted earnings [7]. A multiple that halves in a year without the price moving is the definition of a price that already assumes the fade. That is the change: the AI complex no longer prices optimism about multiples, it prices a forecast about earnings — and forecasts have dates.

So what: the theme sorts companies by WHERE their value sits in time and by HOW MUCH multiple they have to lose — not by whether they are in AI. Two firms with identical AI revenue can sit on opposite sides of this line.

WHO WINS AND LOSES

This pod is the screen. Each characteristic below is a description of a kind of business, written so the next phase can go looking for names that match it.

STRUCTURALLY FAVOURED

A. Paid on gross capex rather than on who wins the workload. Foundry, lithography and deposition tools, advanced packaging, HBM, power and thermal. Their earnings are the buyer's capex line, which is guided and visible a year out; they do not need the fade assumption to be right because they are paid whether the compute is Nvidia's, an ASIC, or idle. The toll-booth shape.

B. Value concentrated in the near forecast years. A contracted, enforceable backlog converting inside the explicit window mechanically lowers the terminal-value share of enterprise value, which lowers duration. Contracted is the operative word — a letter of intent is a terminal assumption wearing a contract's clothes.

C. Trading below the quality actually delivered. High return on invested capital and high free-cash conversion at a multiple at or under the market's. A de-rating removes premium, and a business with no premium loses proportionally less. This is the discount side of the theme and it is the least crowded.

D. Self-funded. Building from balance-sheet cash rather than issuance. Its weighted-average cost of capital does not move with the credit cycle and no refinancing date sits inside the horizon — so the rate channel hits its multiple but not its cash flows.

E. Sellers of the fade itself. Anyone whose revenue rises as compute gets cheaper per token — custom-silicon designers, inference-cost optimisers, memory-reuse and efficiency plays. They are long the same volume and short the price, which is the opposite exposure to a merchant GPU vendor.

STRUCTURALLY SQUEEZED

F. Higher terminal-value concentration than the average. Pre-profit, long-ramp or early-monetisation names where the explicit forecast years contribute close to nothing, so the entire equity IS a terminal assumption. A 71.6% terminal share is the mid-point of this complex, not the extreme; the names above 90% move several times harder per basis point of discount rate.

G. Carrying a large absolute multiple. Forty to sixty times forward is mostly cushion, and cushion is what a de-rating takes first: "Both fall, the higher multiple falls further, and a pair trade between them still leaves you exposed to a demand shock and to the shared dependence on a single foundry" [1]. Note that this cuts the opposite way to intuition — the name at the peer median is the one with no cushion left, which is why its downside is all earnings and none of its multiple.

H. Debt-funded builders. The double hit: the cost of capital rises exactly when the multiple compresses, and the bond market prices it before the equity does. Goldman tracks nearly \$500bn of AI-related debt issuance with the hyperscalers only 40% of it [2] — the rest sits with builders far less able to carry it. Amazon's trailing-four-quarter capex already exceeds its trailing-four-quarter operating cash flow at 102% [3], and that is the strongest balance sheet in the set.

I. Revenue that IS someone else's capex line, without a contract. Server ODMs, integrators and systems assemblers — the most cyclical earnings in the chain with the least pricing power, and the ones whose order books get cut first when the second derivative turns.

J. Index and passive vehicles, and anyone whose beta IS the complex. Not exposed by choice — transmission carriers. At roughly a 7.2% single-name index weight this is most portfolios whether or not they hold a view.

K. Credit holders of the theme. Duration concentration in one story: strategists warn the bond market is "quietly reshaping how investors should think about NVIDIA and the broader AI trade" and that most equity investors are not watching the right risk [4].

THE TEST THE NEXT PHASE APPLIES

For any candidate: what share of its enterprise value sits past year five, how much absolute multiple does it have to lose, and is its cost of capital fixed or floating with this theme? Three answers, and the row writes itself.

Landscape

Exposed Companies

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Advanced Micro Devices	AMD	incumbent	Carries the complex's largest de-rating cushion: ~63-70x forward earnings against NVIDIA's 21-24x on an AI business roughly 11x smaller, so an in-line quarter is punishing rather than fine.	mixed	high	Instinct AI accelerators ~5-7% of AI accelerator revenue (~\$7-8B)	The purest listed expression of 'the multiple is a fade forecast': its premium IS the growth assumption, priced separately from NVIDIA's.
CoreWeave	CRWV	emerging	Pure-play AI cloud whose GPU fleet is funded by debt and equity issuance while losses widen — essentially all equity value is terminal, so it re-prices on the discount rate before it re-prices on demand.	hurt	medium	~100% of revenue is AI/accelerated cloud	Q2 2026: revenue up sharply, net loss widened, interest expense material; RPO gives contracted visibility against borrowed assets.
Oracle	ORCL	incumbent	Debt-funded datacentre buildout whose equity already moves with Treasury yields rather than with bookings — the WACC channel of this theme biting a large-cap before the earnings channel does.	hurt	medium	Cloud infrastructure share of revenue not established here	The live demonstration that in this theme the discount rate, not demand, is the first thing to move a price.

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Nebius Group	NBIS	emerging	Neocloud valued on revenue-per-MW and GPU payback period rather than reported earnings, anchored by a \$27bn five-year Meta contract including \$12bn of dedicated Vera Rubin capacity.	mixed	medium	~100% of revenue is AI cloud	Contracted backlog pulls value into the explicit forecast window; the financing behind the capacity pushes it back out.
Nscale		emerging	Private neocloud built on NVIDIA capacity that cannot raise at scale against its own credit — the archetype of the buyer whose financing has to be volunteered by an investment-grade balance sheet.	hurt	low	~100% of revenue is AI cloud	Unlisted, so no multiple to de-rate; included because its financing dependency is the mechanism, and it is where the fringe search led.
Broadcom	AVGO	incumbent	Trades at a ~50% higher forward multiple than NVIDIA while selling the custom ASICs that cause the fade (\$20bn+ AI ASIC revenue in FY2025) — earnings hedged against the theme, multiple not.	mixed	high	AI ASIC revenue \$20B+ (FY2025)	The one name whose E benefits from exactly the event that de-rates the complex; that asymmetry is the reason it is in the table.
Amazon.com	AMZN	incumbent	Trailing-four-quarter capex now 102% of trailing-four-quarter operating cash flow, and the highest duration weight in the US investment-grade index — the depreciation wall and the rate channel on one balance sheet.	hurt	high	AWS capex; AI share of capex not separately disclosed	Was the 20th-largest duration weight in the IG index a year ago; the move itself is the evidence that the theme changed its financing.

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Meta Platforms	META	incumbent	Funds the build with the largest investment-grade bond deal of 2025 (\$30bn) plus a ~\$27bn off-balance-sheet SPV — the accounting expression of the theme, keeping capex off the P&L until depreciation catches up.	mixed	high	AI infrastructure capex; not disclosed as a revenue segment	Demonstrable AI return in advertising is what lets it carry the leverage; that is also what makes it the least fragile of the borrowers.
Micron Technology	MU	incumbent	Memory and packaging are now 60-70% of AI accelerator COGS, and fiscal Q3 2026 GAAP gross margin was 84.6% — quality delivered far above the multiple memory is historically granted.	mixed	high	Memory + packaging = 60-70% of accelerator COGS (industry, not company, share)	The discount side of the theme. Counterweight : memory is the most cyclical earnings stream in the chain, which is why direction is mixed and not benefits.
Taiwan Semiconductor Manufacturing	TSM	adjacent	Manufactures the silicon for every contestant and is paid on wafers and CoWoS packaging regardless of architecture; CoWoS capacity goes from ~80,000 to 120,000-130,000 wafers/month through 2026 with NVIDIA taking ~60% of allocation.	benefits	high	NVIDIA ~60% of CoWoS allocation	The toll booth AND the shared single point of failure — a pair trade inside the complex offers no protection against it.
ASML Holding	ASML	adjacent	Sole supplier of EUV lithography at \$200M-\$400M per tool, sold on multi-year order books that sit inside the explicit forecast window — structurally the lowest terminal-value share in the chain.	benefits	high	EUV/DUV tools sold into the whole industry's capex	Paid on gross capex, not on who wins the workload; the clearest instance of characteristic A in Who_Wins_And_Loses.

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Intel	INTC	adjacent	The complex's control case: a designated 'have-not' granted no AI terminal-value premium at all, so a sector-wide de-rating takes almost nothing from its multiple.	mixed	high	AI accelerator revenue immaterial to date	Included deliberately as the zero-premium reference point; without it the table has no measure of how much premium the others carry.

So what: of the twelve names below, only two are exposed to this theme through their earnings. The other ten are exposed through their discount rate or their multiple — which is why a screen built on AI revenue misses most of them.

EXPOSURE EVIDENCE

Paragraphs run in table order. Where a figure carries no [N], it is stated as reported and is not grounded in this workbook's corpus; those are flagged in the sentence and the row's confidence is set accordingly.

ADVANCED MICRO DEVICES (AMD) — INCUMBENT, MIXED, HIGH

The theme's clearest listed expression. NVIDIA trades at 24 times earnings and AMD at 70, and the reason given is explicitly not quality: "AMD's price has run ahead of profits that are still growing from a much smaller base" [1]. The same source draws the operational consequence, which is the part that matters here: "A quarter that merely meets expectations is usually fine for NVIDIA and can be punishing for AMD, because that premium valuation requires beating rather than meeting" [1]. AMD's Instinct line holds roughly 5-7% of AI accelerator revenue against NVIDIA's ~80% [6], so its entire premium is a forecast of share capture. If the complex de-rates, AMD loses multiple first and earnings second; NVIDIA, sitting at the peer median, loses only earnings. Documented, not inferred.

COREWEAVE (CRWV) — EMERGING, HURT, MEDIUM

A pure-play AI cloud reporting "rapid top-line growth but larger losses for the quarter ended June 30, 2026" [2], with property and equipment funded by significant debt and equity issuance and remaining performance obligations reflecting large multi-year customer commitments [2]. That combination is exactly characteristic F in the screen: the explicit forecast years contribute nothing to value, so the whole equity IS a terminal assumption, and it is levered. The 10-Q as indexed here has its figures redacted, which is why the specific magnitudes are absent and confidence is medium rather than high — the structure is documented, the size is not quotable from this corpus.

ORACLE (ORCL) — INCUMBENT, HURT, MEDIUM

The live demonstration that in this theme the discount rate moves a price before demand does: "Rising Treasury yields are hitting debt-heavy AI builders where it hurts most, and Oracle's balance sheet puts it squarely in the crosshairs" [3]. Oracle is in the table not because it is an AI company but because it is a borrower whose AI capex sits on a balance sheet the bond market now watches. A skeptic would say this is a rates story rather than an AI story — and that is precisely the finding: for this factor they are the same story.

NEBIUS GROUP (NBIS) — EMERGING, MIXED, MEDIUM

Nebius is valued on "Revenue per MW, GPU Utilization, Payback Period and CapEx" rather than on earnings [8] — a shareholder-letter disclosure the coverage treats as more useful than another growth percentage [8]. That is a terminal-value-heavy equity by construction: payback period is a statement about years beyond the forecast window. This workbook's own company report records a Nebius-Meta \$27bn five-year agreement including \$12bn of dedicated Vera Rubin capacity [4]; contracted backlog of that length pulls value INTO the explicit window, which is the one structural feature working against the theme here. Hence mixed, not hurt.

NSCALE (PRIVATE) — EMERGING, HURT, LOW

The fringe name, and the cleanest statement of the financing mechanism. NVIDIA's own call names the class: these companies "have rapidly growing customer demand, yet still lack the decades-long infrastructure contracts and investment-grade financing capacity needed to secure the AI factory infrastructure independently" [10]. Independent coverage puts it more bluntly: developers and lenders outside the hyperscalers "cannot raise money at scale against the credit of unprofitable AI labs or debt-heavy cloud startups" [5]. Nscale is named alongside CoreWeave and Nebius as a business NVIDIA helped build [4]. It is unlisted, so it has no multiple to de-rate — it is in the table because its cost of capital is set entirely by someone else's credit, which is the mechanism this factor is about. Confidence low: the exposure is inferred from the class it belongs to, not from Nscale's own disclosure.

BROADCOM (AVGO) — INCUMBENT, MIXED, HIGH

"Broadcom carries a higher gross margin but at a 50% higher forward multiple" than NVIDIA [4], while its AI ASIC revenue passed \$20bn in FY2025 [6]. The asymmetry is the finding: Broadcom's earnings are hedged against the exact event that de-rates the complex — it builds the custom silicon that takes inference share — but its multiple is not hedged at all. It is the only row where the E moves one way and the P/E the other.

AMAZON.COM (AMZN) — INCUMBENT, HURT, HIGH

Two facts, one balance sheet. "Amazon's trailing-4Q capex now exceeds its trailing-4Q operating cash flow (102%, SA-computed)" [6], and "Amazon, for example, is the highest duration weight in the IG index. Last year, it was number 20" [7]. The second is the more interesting: a company does not climb nineteen places in an index duration ranking in a year unless the financing structure of its business changed. The depreciation wall and the rate channel now land on the same name.

META PLATFORMS (META) — INCUMBENT, MIXED, HIGH

Meta "priced \$30B in October 2025 (the largest investment-grade deal of the year) plus a ~\$27B off-balance-sheet SPV" [6]. The SPV is the theme's accounting expression: it keeps capex off the P&L until the depreciation schedule catches up, which is the same fade argument expressed in accounting rather than in a multiple. Direction is mixed because Meta is one of two companies the coverage credits with demonstrable AI return on invested capital [9], which is what lets it carry the leverage at all.

MICRON TECHNOLOGY (MU) — INCUMBENT, MIXED, HIGH

The discount side. "Memory and packaging together now represent 60–70% of AI accelerator COGS — logic silicon is no longer the dominant cost" [6], and "In fiscal Q3 2026, Micron reported a GAAP gross margin of 84.6% and a non-GAAP gross margin of 84.9%" [8]. Memory has out-earned logic: the same industry source records SK hynix at a 72% operating margin against TSMC's ~50%, "the first time memory has out-earned the most advanced logic foundry" [6]. Quality delivered has moved a long way above the multiple the market historically grants memory. The counterweight, and the reason direction is mixed rather than benefits, is that memory is the most cyclical earnings stream in the chain — the same source records DDR4 8Gb spot rising 683% in eight months [6], and a price that moves like that in one direction can move back.

TAIWAN SEMICONDUCTOR MANUFACTURING (TSM) — ADJACENT, BENEFITS, HIGH

The toll booth. "TSMC CoWoS capacity is expanding from ~80,000 WPM to 120,000–130,000 WPM through 2026, with NVIDIA consuming ~60% of allocation" [6] — it is paid on wafers and packaging regardless of which architecture wins, which is characteristic A in the screen. The second half of its exposure is the part a bull would rather not state: both NVIDIA and AMD "depend on TSMC for advanced manufacturing and packaging, most of it concentrated in Taiwan" and "a pair trade between them offers no protection against the single largest supply risk either faces" [1]. TSMC is simultaneously the annuity and the correlated risk.

ASML HOLDING (ASML) — ADJACENT, BENEFITS, HIGH

"With EUV tools running \$200M–\$400M per unit", lithography total cost of ownership is "one of the most consequential — and least negotiable — line items in semiconductor manufacturing economics" [6]. Structurally this is the lowest terminal-value share in the table: tools are sold on multi-year order books that sit inside the explicit forecast window, so more of the value is in years a model actually forecasts. It is paid on gross capex, not on who wins the workload.

INTEL (INTC) — ADJACENT, MIXED, HIGH

The control case, and the reason it earns a row: "This year, the “haves” (led by Nvidia, Broadcom and AMD) are outperforming, while the “have-nots” (in particular, Intel) are lagging" [9]. Intel was granted no AI terminal-value premium, so a sector-wide de-rating takes almost nothing from its multiple. Without a zero-premium reference point the table has no measure of how much premium the other eleven carry — which is the whole subject of this factor.

So what: the three emerging names here are not challengers to NVIDIA — they are the purest instruments of the theme, because a company with no earnings has nothing BUT terminal value, and every one of them rents its cost of capital from somebody else.

EMERGING PLAYERS

The fringe search for this factor was deliberately aimed at newly public and thinly covered names whose valuation is almost entirely a terminal assumption. What it found is a class rather than a set of individual novelties: the neocloud. Three of them earn an emerging row, and what makes them emerging is not a new technology but a new **capital structure**.

COREWEAVE (CRWV) — LISTED MARCH 2025

Coverage is young and the operating record is two years of public reporting. The 10-Q for the quarter ended 30 June 2026 reports "rapid top-line growth but larger losses" with heavy investment in AI cloud infrastructure driving operating expenses and depreciation higher, property and equipment funded by significant debt and equity issuance, and remaining performance obligations reflecting large multi-year customer commitments [1]. The novel mechanism is the asset-backed model itself: GPUs are financed as depreciating collateral against contracted offtake, so the business is a spread trade between contract duration and debt duration. Its edge over an incumbent cloud, if the approach works, is capital efficiency — it never had to build a general-purpose cloud first. Its edge disappears if the discount rate rises faster than the contracts reprice.

NEBIUS GROUP (NBIS)

Nebius discloses what a neocloud actually earns: its Q2 2026 shareholder letter is covered specifically for "Revenue per MW, GPU Utilization, Payback Period and CapEx" — "a glimpse into the economics of individual capacity deals" that coverage treats as more useful than another revenue-growth percentage [2]. That disclosure IS the novel approach: it asks to be valued on unit economics and payback period rather than on an earnings multiple, which is an explicit request to be priced on terminal assumptions. This workbook's own company report records the anchor contract — a Nebius-Meta \$27bn five-year agreement including \$12bn of dedicated Vera Rubin capacity [3]. If the payback period is what Nebius says it is, the edge over a hyperscaler is that it carries no legacy estate; if it is not, there is no earnings floor to fall back to.

NSCALE (PRIVATE)

The least covered and the most instructive. NVIDIA names the class on its own earnings call: these companies "have rapidly growing customer demand, yet still lack the decades-long infrastructure contracts and investment-grade financing capacity needed to secure the AI factory infrastructure independently" [4], and Nscale is named alongside CoreWeave and Nebius as a business NVIDIA helped build [3]. Independent coverage states the constraint from the lender's side: developers and lenders outside the hyperscalers "cannot raise money at scale against the credit of unprofitable AI labs or debt-heavy cloud startups" [5] — and the answer being reached for is an investment-grade balance sheet willing to stand behind the project [5]. Nscale is unlisted, so it cannot de-rate; it is here because it shows the mechanism with the multiple stripped out. Its exposure is inferred from the class, not from its own disclosure, and its row is graded low for that reason.

WHAT THE FRINGE SEARCH DID NOT FIND

No emerging entrant attacking the **valuation structure itself** — a business whose model is to be priced differently from the complex. That is unsurprising: valuation structure is not a market anyone sells into, so this factor's fringe is populated by companies unusually EXPOSED to it rather than by companies competing over it. A reader should read these three as instruments, not as challengers.

One honest limitation: the live web-search budget for this workbook was already exhausted when this section was researched (324 searches against a soft budget of 150), and the one search spent on the fringes returned a general characterisation of the risk profile with **no citable sources attached**. Every name above therefore comes from the pre-built corpus. A follow-up pass with budget should look specifically at 2026 listings in inference-as-a-service and at custom-silicon startups, which is where the corpus is genuinely thin.

Ideas

Own The Capex Not The Utilisation

The thesis. The names paid on gross capital expenditure — foundry capacity and lithography tools — carry a structurally smaller share of their value beyond the forecast window than the names paid on how well that capacity is used. In a theme whose central risk is duration, that difference is the position, and it is available without taking a view on whether the fade assumption is right.

The mechanism. Mechanism section 1 is the whole argument: 71.6% of the reference model's enterprise value sits past the explicit forecast years, and value that far out is mostly a discount-rate expression. A tool vendor's order book converts inside that window. Who_Wins_And_Loses characteristic A states the second half — these businesses are paid whether the compute is a GPU, an ASIC, or idle. Both halves have to hold for the idea to work: near-dated earnings AND architecture-indifference.

The names. TSMC and ASML, with Micron as the higher-beta version of the same shape. TSMC's CoWoS capacity moves from ~80,000 to 120,000-130,000 wafers a month through 2026 and it is paid on that expansion regardless of who consumes it; ASML sells EUV at \$200M-\$400M a tool into every advanced node anyone builds. See Exposure_Evidence. What distinguishes them: ASML is the purer toll booth (a monopoly input to everyone), TSMC is the toll booth AND the concentrated supply risk the whole complex shares.

What would have to be true. Announced gross capex survives a demand air pocket — the nine largest CSPs' combined 2026 spend above US\$886.7bn, and the ~\$1.3 trillion 2027 estimate, have to be spent rather than merely guided [1]. A toll booth is paid on the build, so the build is the only variable that matters.

What would kill it. The second derivative of capex turning negative. The moment CSP capex growth decelerates, the toll booth's near-year earnings are cut — and because those near years are where its value sits, its main structural advantage becomes its exposure. A single quarter in which two of the top five hyperscalers guide capex below the prior quarter's trajectory falsifies this.

What is already priced. Partially. The capex guides are public and the multiples embed them. What is not obviously priced is the DURATION difference — the market ranks these names on AI exposure, not on where in time their value sits, which is exactly the mismatch this factor is about.

The Cushion Is The Risk

The thesis. Inside this complex the forward multiple ranks inversely to delivered growth, so it is not a quality score — it is a forecast of the fade. That makes the high-multiple names the ones with the most to lose in a common de-rating and the median-multiple names the ones whose downside is entirely earnings. Exposure to the theme taken through low-premium names is a different instrument from the same exposure taken through high-premium ones, and the two are routinely treated as interchangeable.

The mechanism. Mechanism section 2. NVIDIA trades at 24 times earnings and AMD at 70, and the stated reason is that AMD's price has run ahead of profits growing from a much smaller base [1]. The operational consequence follows directly: "A quarter that merely meets expectations is usually fine for NVIDIA and can be punishing for AMD, because that premium valuation requires beating rather than meeting" [1].

The names. AMD is the maximum-cushion end; Intel is the zero-premium control, a designated 'have-not' granted no AI terminal-value premium at all; TSMC sits in between with a premium anchored to capacity rather than to share capture. See Exposure_Evidence. The distinction between them is not how much AI revenue each has — it is how much of each price is a forecast.

What would have to be true. The de-rating, when it comes, has to be COMMON to the complex rather than idiosyncratic. The evidence for that is the shared dependence: both NVIDIA and AMD depend on TSMC for advanced manufacturing and packaging, so "a pair trade between them offers no protection against the single largest supply risk either faces" [1]. Common risk is what makes multiple-ranking the right axis.

What would kill it. A company-specific re-rating. AMD converting a hyperscaler design win at scale re-rates it on EARNINGS, and a position built on premium-ranking does not participate — the source names this asymmetry explicitly, noting that such a win moves AMD far more than an equivalent NVIDIA contract moves NVIDIA [1]. One large announced ASIC or accelerator award at a top-four CSP falsifies the idea.

What is already priced. That AMD is expensive is priced and widely discussed. The inverse — that a name AT the peer median has no cushion left, so its entire drawdown must come out of earnings — is the part this factor found and is not a consensus framing.

Buy The Fade Not The Growth

The thesis. The single event that most threatens the complex's earnings — hyperscalers moving inference onto custom silicon — is the event that grows Broadcom's. An exposure whose earnings are hedged against the theme's central risk is a different proposition from one that is long it, even when both are described as 'AI semis'.

The mechanism. Who_Wins_And_Loses characteristic E: sellers of the fade. Custom ASICs are growing 44.6% in 2026 against 16.1% for merchant GPUs [1], and Broadcom's AI ASIC revenue passed \$20bn in FY2025 [2]. The causal chain is direct — every inference workload that moves from a merchant GPU to a hyperscaler ASIC subtracts from one income statement and adds to the other.

The names. Broadcom, the only row in Exposed_Companies where the E and the P/E point in opposite directions. Marvell is the second-source version of the same shape but is not in this table. See Exposure_Evidence.

What would have to be true. The announced hyperscaler ASIC programmes have to convert into Broadcom revenue at roughly the pace the shipment data implies, AND Broadcom's ~50% forward-multiple premium over NVIDIA [1] must not compress faster than that revenue compounds. The second condition is the one people skip: an earnings hedge is not a multiple hedge, and in a de-rating the multiple moves first.

What would kill it. A hyperscaler taking ASIC design fully in-house and cutting the merchant design partner out — the structural risk specific to this idea, and observable: a next-generation accelerator announced by a top-four CSP with no named silicon partner would falsify it.

What is already priced. Substantially. The 50% premium IS the market paying for this hedge. The idea is therefore not 'nobody has noticed' but 'the hedge is real and the premium paid for it is the thing to underwrite'.

Credit Moves Before Earnings

The thesis. In this theme the first observable move is in credit, not in bookings. Where capex is debt-funded, the cost of capital rises at exactly the moment the multiple compresses, and the bond market prices it before the equity does. That makes the credit side a leading indicator for the equity side, and it makes the debt-funded builders the fastest-moving instruments of the factor.

The mechanism. Mechanism section 4 and Who_Wins_And_Loses characteristic H. Roughly one third of hyperscaler capex is debt-financed in 2026, expected to peak near 35% in 2027 [1], and the equity has begun to trade on credit terms. The visible case is Oracle: "Rising Treasury yields are hitting debt-heavy AI builders where it hurts most, and Oracle's balance sheet puts it squarely in the crosshairs" [2]. The severe case is the tier below it — neoclouds that "cannot raise money at scale against the credit of unprofitable AI labs or debt-heavy cloud startups" [3].

The names. Oracle and Amazon among the large caps (Amazon's trailing-four-quarter capex is 102% of its trailing-four-quarter operating cash flow and it is now the highest duration weight in the US investment-grade index), and CoreWeave, Nebius and Nscale at the levered fringe — see Emerging_Players for why the fringe names carry the mechanism in its purest form, and Exposure_Evidence for the per-name evidence. What distinguishes them is who sets their cost of capital: Amazon sets its own, Oracle negotiates it, and Nscale rents it from somebody else's balance sheet.

What would have to be true. Spreads have to stay the binding constraint rather than issuance volume — i.e. the debt-financed share keeps climbing toward the 2027 peak [1] and credit stays the channel through which rate expectations reach these equities.

What would kill it. Hyperscaler operating cash flow re-accelerating enough to fund capex internally. That removes the channel entirely: a company that does not need the bond market does not transmit the bond market. Watch the capex-to-operating-cash-flow ratio at the top four; sustained readings back below ~70% would end this idea.

What is already priced. The large-cap end is priced — Oracle has already moved on this. The fringe end is not priced so much as unpriceable: three of these names have no earnings multiple to compress, which is the point and also the error bar.

Memory Out Earns Logic And Is Not Priced For It

The thesis. The market still applies a cyclical-memory multiple to a business currently earning better than the most advanced logic foundry. That is the discount side of this factor — a name whose delivered quality has moved above the premium attached to it, which by construction has less premium to lose in a de-rating.

The mechanism. Who_Wins_And_Loses characteristic C. The content shift is the cause: "Memory and packaging together now represent 60–70% of AI accelerator COGS — logic silicon is no longer the dominant cost" [1]. The result shows in the margins — Micron reported an 84.6% GAAP gross margin in fiscal Q3 2026 [2], and SK hynix's 72% operating margin now exceeds TSMC's ~50%, which the same source calls the first time memory has out-earned the most advanced logic foundry [1].

The names. Micron in Exposed_Companies; SK hynix is the unlisted-in-this-table comparator that establishes the industry point. See Exposure_Evidence.

What would have to be true. HBM and DRAM pricing has to hold long enough for the market to re-anchor its multiple on structurally higher margins rather than treating them as a cycle peak. Concretely: two consecutive quarters of stable-to-rising DRAM ASPs alongside sustained HBM allocation to accelerator programmes.

What would kill it. The cycle, which cuts both ways and fast. The same source records DDR4 8Gb spot rising 683% from \$1.63 to \$12.76 in eight months [1] — a price that moves like that in one direction can move back, and if ASPs revert, the 84.6% margin was the peak and the low multiple was correct all along. This is the least falsification-resistant idea in the section and it is placed last for that reason.

What is already priced. Unclear, and honestly so. This run did not establish Micron's current forward multiple from the corpus, so the claim that the market 'still' applies a cyclical multiple rests on the margin evidence and on the absence of evidence to the contrary — not on a measured multiple. Treat it as the weakest-evidenced idea here.

Summary

So what: in the AI complex the forward multiple is a forecast of when growth fades, not a score for quality delivered — so the money is made or lost on WHERE a company's value sits in time and HOW MUCH premium it has to lose, and a screen built on AI revenue sorts on neither.

FACTOR RESEARCH SUMMARY

THE THEME IN ONE PARAGRAPH

Roughly seven of every ten dollars of modelled value in this complex sit past the explicit forecast window — 71.6% of base-case enterprise value in the reference model — which makes these equities long-duration instruments whose price moves on the discount rate before it moves on demand. The multiple does no discriminating: NVIDIA trades near 24x forward and AMD near 70x, ranked inversely to delivered growth, so the name at the peer median has no de-rating cushion and the name at 70x is mostly cushion. Meanwhile the discount rate has stopped being exogenous — about a third of hyperscaler capex is debt-financed in 2026, rising toward ~35% in 2027 — so the cost of capital now moves with the theme it is discounting. See Mechanism and Drivers_And_Horizon.

THE TWELVE COMPANIES, AND HOW EACH IS EXPOSED

#	COMPANY	TICKER	ROLE	DIRECTION	CONF.	MECHANISM OF EXPOSURE
1	Advanced Micro Devices	AMD	incumbent	mixed	high	~70x forward vs NVIDIA's ~24x on an AI business ~11x smaller: the complex's largest de-rating cushion
2	CoreWeave	CRWV	emerging	hurt	medium	Debt-and-equity-funded GPU fleet with widening losses; essentially all equity value is terminal
3	Oracle	ORCL	incumbent	hurt	medium	Debt-funded buildout whose equity already moves with Treasury yields rather than bookings
4	Nebius Group	NBIS	emerging	mixed	medium	Valued on revenue-per-MW and payback period, not earnings; anchored by a \$27bn five-year Meta contract

#	COMPANY	TICKER	ROLE	DIRECTION	CONF.	MECHANISM OF EXPOSURE
5	Nscale	—	emerging	hurt	low	Private neocloud that cannot raise against its own credit; cost of capital set by someone else's balance sheet
6	Broadcom	AVGO	incumbent	mixed	high	~50% higher forward multiple than NVIDIA while selling the custom ASICs that cause the fade
7	Amazon.com	AMZN	incumbent	hurt	high	Capex 102% of operating cash flow; highest duration weight in the US IG index, up from 20th
8	Meta Platforms	META	incumbent	mixed	high	\$30bn IG deal plus a ~\$27bn off-balance-sheet SPV — the theme expressed in accounting
9	Micron Technology	MU	incumbent	mixed	high	Memory + packaging now 60-70% of accelerator COGS at an 84.6% GAAP gross margin
10	Taiwan Semiconductor	TSM	adjacent	benefits	high	Paid on wafers and CoWoS regardless of architecture; capacity ~80k to 120-130k WPM through 2026
11	ASML Holding	ASML	adjacent	benefits	high	Sole EUV supplier at \$200M-\$400M per tool, sold on order books inside the forecast window
12	Intel	INTC	adjacent	mixed	high	The zero-premium control case: granted no AI terminal-value premium, so a de-rating takes almost none

Full reasoning per name in Exposure_Evidence; the three emerging rows are argued in Emerging_Players.

THE FIVE IDEAS

1. **Own the capex, not the utilisation** (TSM, ASML, MU) — toll booths paid on gross capex carry less of their value past the forecast window. Dies if the second derivative of CSP capex turns negative.
2. **The cushion is the risk, not the protection** (AMD, INTC, TSM) — rank exposure by premium carried, not by AI revenue. Dies on an idiosyncratic re-rating, e.g. an AMD design win at a top-four CSP.

3. **Buy the fade, not the growth** (AVGO) — the event that most threatens the complex's earnings grows Broadcom's. Dies if a top-four CSP announces a next-generation accelerator with no named silicon partner.
4. **Credit moves before earnings** (ORCL, AMZN, CRWV, NBIS, Nscale) — credit is the leading indicator for these equities. Dies if hyperscaler operating cash flow re-accelerates enough to fund capex internally.
5. **Memory out-earns logic and is not priced for it** (MU) — delivered quality above the premium granted. Dies on DRAM ASP reversion; the weakest-evidenced idea here.

WHAT THIS RUN DID NOT SETTLE

The confidence grades are lopsided, and they lean benign. Both benefits rows (TSMC, ASML) are graded high, while three of the four hurt rows (CoreWeave medium, Oracle medium, Nscale low) are not. Read literally, the table says the toll booths are well evidenced and the damage to the borrowers is inferred — which flatters the reading that this theme is survivable. That asymmetry is an artefact of source availability, not a finding: filings and industry data on foundry capacity index cleanly, while the levered names' own numbers did not. A reader should discount the apparent safety of the benefits side accordingly.

Three specific gaps. (1) CoreWeave's 10-Q is indexed with its figures redacted, so the magnitude of its leverage is documented in structure but not in size — the single most important unquantified number in the table. (2) No current forward multiple was established from the corpus for Micron, Nebius, CoreWeave or Oracle, so idea 5's central claim that memory is 'not priced for it' rests on margin evidence and the absence of contrary evidence rather than on a measured multiple. (3) The terminal-value SHARE of enterprise value was established only for the reference model; no peer's terminal-value concentration was measured, so the ranking of characteristic F in Who_Wins_And_Loses is reasoned from business model rather than computed.

The search budget was exhausted before this run started. The workbook stood at 324 web searches against a soft budget of 150. The one search spent — deliberately, on the fringes — returned a general characterisation of de-rating risk with no citable sources attached, so every name in this section comes from the pre-built corpus. The fringe is therefore covered by class (the neocloud) rather than by discovery: a follow-up with budget should look at 2026 listings in inference-as-a-service and at custom-silicon startups, where the corpus is genuinely thin.

One thing deliberately not done. This section makes no valuation call on the workbook's own company and states no price level, target or position size anywhere. The factor is a description of a pricing structure and of who else sits inside it.

FACTOR RESEARCH

NVDA Factor 07 Rack Scale Platform Lockin

Theme

The unit of sale in AI compute has become a \$5-7m coherent rack, and the switching cost is the rack, not the chip

MECHANISM

A buyer of AI compute in 2023 bought accelerators and wired them together. A buyer in 2026 buys an integrated cabinet. A Vera Rubin NVL72 is quoted to OEM customers at \$5m to \$7m per unit, and the higher-density Rubin Ultra NVL144 at \$7m to \$8.8m [1]. What that price covers is the point: 36 Vera CPUs, 72 GPUs with 288 GB of HBM4 each, sixth-generation NVLink switching, ConnectX-9 SuperNICs, BlueField-4 DPUs and custom liquid cooling, all co-engineered [1]. None of those pieces is separably replaceable; the cabinet is sold, warranted and tuned as one machine.

THE CAUSAL CHAIN

- 1. Model size outran the chip.** Frontier training and reasoning-era inference need tens to hundreds of accelerators behaving as one memory domain. That is a *fabric* problem, not a compute problem, and it is solved at rack scale.
- 2. So the fabric became the product.** Data Center networking revenue grew 142% in FY2026 on the ramp of NVLink compute fabric for GB200 and GB300 systems plus Ethernet and InfiniBand [2]. In Q4 FY2026, networking grew 263% year over year against 58% for compute [3] — compute fell from 91.5% of Data Center revenue to 82.4% in four quarters [3].
- 3. The vendor re-prices per gigawatt, not per chip.** The stated revenue opportunity per gigawatt of data centre moved from roughly \$18bn in the Hopper generation to \$25bn with Blackwell to \$40bn with Vera Rubin, because the sale now spans CPU, GPU, scale-up and scale-out networking [4].
- 4. The building has to be rebuilt to take it.** Rubin racks draw toward 1,800W per package and are driving the transition to 800V DC power distribution [5]. A customer who wants to swap vendors is not swapping a card; it is re-engineering power, cooling and floor layout [5].
- 5. The software sits on top and does not move.** CUDA is a 20-year-old base with roughly 4 million developers and every major ML framework optimised for it first — switching costs measured in years rather than dollars [6]. Supply priority compounds it: roughly 60% of TSMC CoWoS capacity [6].

WHAT IS ACTUALLY BEING ENFORCED

Contractually, nothing. Physically and organisationally, a great deal: a re-tuned model pipeline, a re-validated fabric, a re-poured power train, and a workforce that knows one stack. The enforcement mechanism is the *aggregation* of switching costs that individually would each be survivable.

The clearest statement of where the value is defended comes from the ASIC side of the argument: beyond the compute chip sit HBM, scale-up fabric, scale-out networks, NICs, DPUs, optical communications, power delivery, liquid cooling and rack engineering, and NVLink Fusion exists precisely to let a third party's XPU keep running on that infrastructure [7]. The compute die is the negotiable part. The rack is not.

The rack theme runs on a five-to-ten-year power-and-capacity clock, but the question of WHOSE rack gets settled in the next four to six quarters

DRIVERS AND HORIZON

Two clocks are running at different speeds, and conflating them is the most common error made about this theme.

THE SLOW CLOCK: CAPACITY, FIVE TO TEN YEARS

The physical build is the slowest-moving thing in the chain. North American data-centre capacity is expected to almost double from 57GW to 109GW by 2030 [4]. Power interconnects, substations, transformers, water and permits move on utility timescales, not product cycles. At roughly \$40bn of platform revenue opportunity per gigawatt [1], that capacity curve is the theme's floor, and it is nearly independent of which vendor's logo is on the cabinet. Anyone selling power, cooling, copper, fibre or floor space is levered to the slow clock.

THE FAST CLOCK: ARCHITECTURE, EIGHTEEN MONTHS

Whose fabric the rack is built around is being decided now. Three things are pushing:

- **Annual cadence.** Vera Rubin is claimed at 30x throughput per megawatt and 35x lower token cost against Grace Blackwell Ultra [1]. A generation that improves facility economics that fast makes waiting expensive and makes a half-built alternative stack obsolete before it ships.
- **The fabric is outgrowing the compute.** Networking reached \$8.2bn of quarterly revenue, up 162% year over year [3], and grew 263% year over year in Q4 FY2026 against 58% for compute [2]. The attach is where the incremental dollar is going.
- **Memory is the binding constraint, not logic.** HBM supply is concentrated, with SK Hynix at roughly 60% share [7]. Rack output is gated by a component NVIDIA does not make.

And three things are pushing back:

- **Open scale-up standards.** UALink and the Ultra Ethernet Consortium exist specifically to make a rack multi-vendor. First UALink-based systems are expected in the second half of 2026, so the counterfactual becomes testable within this theme's fast clock rather than theoretically.
- **The software premise is being contested.** The proposition that CUDA alone is the moat is under direct attack as rival toolchains mature [5]; if that is right, the defensible part of the position is the fabric and the integration, not the compiler.
- **Share is already moving.** Accelerator revenue share peaked near 87% in 2024 and is projected at 75% by 2026 [6]. The rack is deepening while units leak.

WHAT HAS ALREADY CHANGED

Three years ago the sale was a board, the interconnect was an accessory, and networking was under 10% of Data Center revenue; today it is close to a fifth [2] and the platform is priced per gigawatt [1]. That shift has already happened — it is not a forecast. What is still open is whether the next shift is toward *more* integration (optical NVLink, co-packaged optics, 800V DC) or toward disaggregation under an open standard. That question resolves on the fast clock, roughly through 2027.

The rack shift pays whoever supplies the content that grew per cabinet, and squeezes whoever merely assembles it

WHO WINS AND LOSES

This is the search specification for the landscape: characteristics, not names.

STRUCTURALLY FAVOURED

- **Anyone whose dollar content per rack rose faster than the rack price.** The cabinet is no longer mostly logic — it is HBM, scale-up fabric, scale-out networks, NICs, DPUs, optical communications, power delivery, liquid cooling and rack engineering [1]. A supplier of one of those sells more per rack each generation regardless of whose architecture wins.
- **Anyone selling into the interconnect, because that is where the growth moved.** The compute/fabric mix has shifted hard toward fabric; switch silicon, SerDes, retimers, cables, transceivers and co-packaged optics all sit in the fastest-growing line item.
- **The toll-booth on power and heat.** Racks pushing toward 1,800W per package and 800V DC distribution [2] force new power trains and mandatory liquid cooling. That is paid by every rack in every architecture, proprietary or open.
- **Anyone selling grid access and floor space.** North American capacity almost doubling from 57GW to 109GW by 2030 [3] is a real-asset build, and the constraint is energisation dates, not silicon.
- **The arms dealer to BOTH stacks.** A vendor whose parts sit in an NVIDIA rack *and* in a hyperscaler's custom-XPU rack is indifferent to the architecture war. NVIDIA itself put \$2bn into Marvell in March and \$3.5bn into MediaTek in August [1] — capital that only makes sense if the two stacks co-exist.

STRUCTURALLY SQUEEZED

- **The assembler with no proprietary content.** Where the platform vendor ships more of the finished compute tray, the integrator's share of the value falls. OEM margins on Rubin-class racks are expected to stay tight on supply-chain complexity, elevated memory cost and TCO pressure [4] — revenue up, margin percentage down.
- **Anyone whose product was the discrete piece the rack absorbed.** Standalone general-purpose server CPUs, discrete NICs and legacy enterprise switching lose socket relevance as the tray integrates.
- **The sub-scale buyer.** A \$5m-\$8.8m minimum order [4] with a mandatory liquid-cooling retrofit prices out anyone who cannot commit at cabinet granularity — which pushes demand toward hyperscalers and neoclouds and away from enterprise on-premise.

- **The challenger who must replicate a whole stack.** The squeeze is asymmetric: the lock-in is weakest where a single high-volume production workload can be ported for a lower running cost [5], and strongest in exploratory, multi-model, multi-workload estates.
- **Whoever holds fixed-price capacity contracts signed against an older power density.** A hall designed for 30kW racks cannot take a 120kW-220kW cabinet without re-engineering, and that cost lands on the operator, not the chip vendor.

WHICH WAY IT CUTS, IN ONE LINE

The theme is unambiguously good for *content* — silicon, optics, copper, power, cooling — and ambiguous for *integration*. That is why the interesting names in the next section are mostly not accelerator companies.

Landscape

Exposed Companies

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Vertiv Holdings	VRT	incumbent	Power and thermal for the cabinet: 800V DC sidecar converters that move power conversion out of the rack and free 8-16U for compute, plus CDUs and direct-to-chip liquid cooling built to NVIDIA reference designs.	benefits	high	~85% of sales tied to AI data centres (est.)	Backlog ~\$15bn at end-2025, +109% YoY; FY2026 revenue guided ~\$14.0bn. Bought Utility Innovation Group for \$1.45bn to move upstream into behind-the-meter power. Closest thing to a pure-play on the rack's non-silicon half.
Broadcom	AVGO	incumbent	Sells the RIVAL rack: Tomahawk 6 switch silicon plus custom XPU design, so a hyperscaler's own accelerator arrives with Broadcom SerDes, switching and optics attached. Tomahawk 7 taped out as the first 200 Tb/s Ethernet switch.	benefits	high	FY2026 AI revenue \$58bn; XPU's 73% and networking ~27% of Q3 AI revenue	AI networking revenue up over 2.5x YoY in Q3 FY2026. The only name contesting the rack outright - and the one that wins whether the fabric is NVLink or Ethernet.
Amphenol	APH	incumbent	Content per rack across high-speed copper and backplane assemblies, fibre-optic connectors, power interconnect AND liquid-cooling connectors - it sells into every subsystem the rack added, and is bought regardless of architecture.	benefits	high	IT datacom 43% of Q2 2026 sales, +89% YoY	Communications Solutions +85% YoY in Q2 2026, ~62% of revenue; Q3 2026 guided \$9.3-9.4bn. Figures are from company results reporting; the primary release did not index in this workbook.

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Coherent	COHR	incumbent	Supplies the optics the fabric runs on: 800G and 1.6T transceivers, and for co-packaged optics the laser, external laser module, isolators, PM fibre, fibre-attach assembly and silicon-photonics PICs rather than a single component.	benefits	high	Datacenter & Communications 79% of Q4 FY2026 revenue	FY2026 revenue \$7.12bn, +22.5%. FY2027 effectively booked out, orders into calendar 2028. The ceiling is indium-phosphide laser output, not demand. CPO revenue starts in the December 2026 quarter.
Celestica	CLS	incumbent	Builds and now SELLS the AI back-end switch - it held the leading position in Ethernet AI back-end switch sales in 2Q 2026, ahead of NVIDIA and Arista. The white-box integrator turned merchant switch vendor.	benefits	high	Connectivity & Cloud Solutions is the growth engine; ~\$15-17bn 2026 revenue (est.)	The single most surprising row: a contract manufacturer outranking both platform vendors in the fastest-growing switch market. Adjusted operating margin guided to the mid-8% range for 2026 - thin, but rising.
Arista Networks	ANET	incumbent	Builds the scale-out and scale-across Ethernet switches that connect racks into clusters and clusters into campuses - the layer above the NVLink domain, built on merchant Broadcom silicon.	benefits	high	\$3.5bn AI networking target inside 2026 revenue guided to \$11.5bn (~30%)	Q1 2026 revenue \$2.71bn, +35% YoY, 43% operating margin. Ranked third in Ethernet AI back-end switch sales in 2Q 2026. Exposed to the rack COUNT, not to whose rack.

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Eaton	ETN	incumbent	Grid-to-chip electrical: an 800V DC reference architecture developed with NVIDIA including supercapacitor backup and busbar distribution, a sidecar delivering up to 900kW to AI racks, and liquid-to-liquid CDUs after the Boyd Thermal acquisition.	benefits	high	Data centres 25-30% of the electrical business	Q1 2026 backlog \$14.5bn; Electrical Americas data-centre orders +240%. Levered to gigawatts energised rather than racks shipped, which is the slower and more durable clock.
Credo Technology	CRDO	emerging	Active electrical cables: copper engineered to behave like optics over rack and row distances at a fraction of the power and cost - the physical layer that makes a multi-rack scale-up domain affordable.	benefits	high	~100%; AECs drove over 99% of the FY2026 revenue increase	FY2026 (ended 2 May 2026) revenue \$1.3bn, more than 3x YoY. Top-10 customers ~90% of revenue; top three ~74% in Q1 FY2027. The exposure is total and so is the concentration risk.
Astera Labs	ALAB	emerging	Owns the connectivity plumbing inside the rack: roughly 80% share of PCIe retimers in AI servers (Aries), plus Scorpio X-Series smart fabric switches that give a non-NVIDIA rack a software-defined scale-up fabric.	benefits	high	~100% of revenue is AI/cloud connectivity	Q2 2026 revenue \$392.4m; Q3 guided \$540-560m on the Scorpio ramp. Aries interoperates with NVIDIA and AMD GPUs AND hyperscaler ASICs - architecture-agnostic by design. One customer was over 70% of FY2025 revenue.

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Hon Hai Precision (Foxconn)	2317.TW	adjacent	Assembles the rack. Estimated 40-70% share of GB200/GB300 rack-server assembly in 2025 - but NVIDIA is moving to ship pre-assembled compute trays, which turns the ODM from system designer into integrator.	mixed	high	Server-related revenue passed 50% of total for the first time in Q2 2026	Q2 2026 gross margin 6.12%, DOWN from 6.33%, on record revenue of NT\$2.53tn. Full-year AI rack shipments forecast to more than double. The clearest case of volume up, differentiation down.
Quanta Computer	2382.TW	adjacent	Second-source AI rack ODM. Servers were 80% of Q1 2026 revenue with AI servers 75% of that, and management guides triple-digit AI server growth for 2026.	mixed	high	AI servers ~60% of total revenue (est.)	Q1 2026 gross margin 4.78% - component cost inflation passes straight through to a thin, fixed conversion fee. Levered to rack VOLUME with almost no claim on rack VALUE. Figures reported by Taiwanese press; primary filing did not index here.
Ayar Labs	(private)	emerging	Co-packaged optical I/O (TeraPHY) placed in the same package as the compute die, replacing electrical I/O so a scale-up domain can span a rack row without copper's power and reach limits.	benefits	medium	n/a - not yet at production revenue	\$650m raised during 2026, \$870m cumulative, \$3.75bn valuation; strategic investors include AMD, Intel, MediaTek and NVIDIA. Wiwynn partnership to put optical I/O into rack-scale systems. Pre-scale: the thesis is the mechanism, not the numbers.

Six of these twelve sell content that grows per rack regardless of whose rack it is; two sell the rack itself and are watching their margin fall

EXPOSURE EVIDENCE

Every claim below is placed in table order. Where a figure carries no [N], it comes from company results reporting that did not index in this workbook's corpus — those are flagged in the paragraph and carried at the row's stated confidence, never promoted to a fact.

VERTIV HOLDINGS (VRT) — INCUMBENT, BENEFITS, HIGH

The purest listed expression of the rack's non-silicon half. Vertiv's case for 800V DC is mechanical rather than promotional: higher voltage cuts current, and cutting current cuts copper requirements by up to 45% while reclaiming rack space for compute [1]. It is co-developing 800V DC platform designs with NVIDIA for Vera Rubin Ultra Kyber, targeting late-2026 readiness [1] — a position inside the platform vendor's own reference design, which is the strongest form of this exposure. The ~85% AI-data-centre revenue share, the ~\$15bn end-2025 backlog and the ~\$14.0bn FY2026 revenue guide are from company guidance reported in the financial press; they did not index here and carry no [N]. The link itself is documented, not inferred.

BROADCOM (AVGO) — INCUMBENT, BENEFITS, HIGH

The only company in the table contesting the rack outright. It guides fiscal 2026 AI revenue of \$58bn, up 186% year on year [2]; in Q3 FY2026 XPU shipments were up over 3.5x and represented 73% of AI revenue, with AI networking up over 2.5x [2]. It taped out Tomahawk 7, the first 200 Tb/s Ethernet switch, alongside rapid Tomahawk 6 adoption [2]. Mechanically this matters because a custom XPU is not an isolated chip order: it is the entry point that drags SerDes, switching, optics and NIC content with it. So Broadcom is levered to the rack shift on BOTH sides — it wins if the rack stays proprietary (it sells into it) and wins more if the rack opens.

AMPHENOL (APH) — INCUMBENT, BENEFITS, HIGH

The architecture-agnostic toll booth. Amphenol sells high-speed copper and backplane assemblies, fibre-optic connectors, power interconnect and liquid-cooling connectors — four of the subsystems the rack transition added or enlarged. IT datacom at 43% of Q2 2026 sales growing 89% year on year, and Communications Solutions up 85% to ~62% of revenue, are from the company's quarterly releases; the primary release did not index here, so those figures carry no [N]. The exposure LINK is documented in the company's own segment reporting and is not in doubt; only the in-workbook grounding of the numbers is missing.

COHERENT (COHR) — INCUMBENT, BENEFITS, HIGH

Datacenter and Communications was 79% of Q4 FY2026 revenue and is the company's main growth engine rather than one segment among several [3]. The mechanism is content, not units: in co-packaged optics Coherent supplies the laser, external laser module, isolators, polarization-maintaining fibre, fibre-attach assembly and silicon-photonics PICs rather than a single component, and management sizes CPO/NPO and related integrated optics at more than \$15bn of incremental addressable market with initial CPO revenue in the December 2026 quarter [3]. The constraint is its own indium-phosphide laser output, not customer demand — the clearest sign in this table that the theme is supply-limited.

CELESTICA (CLS) — INCUMBENT, BENEFITS, HIGH

The finding a peer screen would miss. In 2Q 2026 Celestica retained the LEADING position in Ethernet AI back-end switch sales, ahead of NVIDIA, with Arista third and Cisco fourth [4]. A contract manufacturer outranking both platform vendors in the fastest-growing switch category is the strongest single piece of evidence that value in the rack is migrating to whoever holds the design-and-build relationship with the hyperscaler. The same report records the inflection: 2Q 2026 was the first quarter in which AI back-end switch sales surpassed front-end network switch sales [4]. Celestica's ~\$15-17bn 2026 revenue and mid-8% adjusted operating margin are from raised company guidance reported in the press and carry no [N].

ARISTA NETWORKS (ANET) — INCUMBENT, BENEFITS, HIGH

Arista raised its full-year 2026 revenue outlook to \$11.5bn and lifted its AI networking revenue target to \$3.5bn [5]. Its own taxonomy is the useful part: it splits AI networking into scale-up (inside one rack), scale-out (rack to rack) and scale-across (between geographically distributed clusters) [5]. Arista does not compete in scale-up — that is the NVLink domain — so its exposure is to the NUMBER of racks deployed and the distance between them, not to whose racks they are. That makes it a cleaner theme play than it first looks, and it ranked third in Ethernet AI back-end switch sales in 2Q 2026 [4].

EATON (ETN) — INCUMBENT, BENEFITS, HIGH

Eaton sits one layer further out than Vertiv: grid to rack rather than rack to chip. It has an 800V DC reference architecture developed with NVIDIA including supercapacitor backup and busbar distribution, a sidecar platform rated to 900kW of DC power per AI rack, and liquid-to-liquid CDUs following the Boyd Thermal acquisition. Data centres at 25-30% of the electrical business, a \$14.5bn Q1 2026 backlog and Electrical Americas data-centre orders up 240% are from company reporting that did not index here and carry no [N]. Eaton's clock is the slow one — gigawatts energised, not racks shipped — which is why it is a different proposition from the silicon names above it.

CREDO TECHNOLOGY (CRDO) — EMERGING, BENEFITS, HIGH

The mechanism is specific and easy to state: an active electrical cable is copper with a DSP in the connector, so it carries optical-class bandwidth over rack and row distances at a fraction of the power and cost of a pluggable transceiver. That is exactly the distance a multi-rack scale-up domain has to cross. FY2026 (ended 2 May 2026) revenue of \$1.3bn, more than triple the prior year, with AECs driving over 99% of the increase, and top-ten customer concentration around 90%, are from the company's results release, which was blocked from indexing in this workbook — the figures carry no [N]. The exposure link is documented in Credo's own filings; the numbers here are reported rather than workbook-grounded, and a reader should treat the concentration risk as equally documented.

ASTERA LABS (ALAB) — EMERGING, BENEFITS, HIGH

Astera's own framing names the mechanism: Scorpio smart fabric switches increase its silicon dollar content opportunity per AI PLATFORM and unlock a new addressable market [6] — content per rack, not per server. Critically, its Aries retimers interoperate with third-party GPUs from NVIDIA and AMD *and* with the internal ASICs and AI accelerators hyperscalers develop themselves [6]. That is architecture-indifference stated by the company: Astera is paid whether the rack is NVLink, UALink or a hyperscaler's own fabric. Q2 2026 revenue of \$392.4m, Q3 guidance of \$540-560m and the ~80% PCIe-retimer share are reported figures that did not index here and carry no [N].

HON HAI PRECISION / FOXCONN (2317.TW) — ADJACENT, MIXED, HIGH

The squeezed side, measured. Server-related revenue passed 50% of total for the first time in Q2 2026 [7] — and in the same quarter gross margin FELL to 6.12% from 6.33% on product mix [7]. That pairing is the whole argument: the AI rack is now the majority of the business and it is dilutive to margin. The structural reason is that

NVIDIA is moving to ship pre-assembled compute trays, which redefines the ODM from system designer into integrator and service provider while increasing the platform vendor's control of the value chain and its margins [8]. Left to compete on price alone, the ODM's only lever is to cut it [8].

QUANTA COMPUTER (2382.TW) — ADJACENT, MIXED, HIGH

The same mechanism as Foxconn with less scale to absorb it. Servers were 80% of Q1 2026 revenue with AI servers 75% of that, management guides triple-digit AI server growth for 2026, and Q1 2026 gross margin was 4.78% — all reported by the Taiwanese financial press and by S&P Global; the primary filings did not index here, so these carry no [N]. What the workbook CAN ground is why the margin is thin: the rack's bill of materials is dominated by parts the ODM buys rather than makes — GPUs, memory, NVLink switches, networking chips, cooling, power supply, PCB and substrates [9] — so component inflation passes straight through a fixed conversion fee.

AYAR LABS (PRIVATE) — EMERGING, BENEFITS, MEDIUM

The only row in the table with no revenue and the only one attacking the rack's physical premise. Its TeraPHY optical engine places optical I/O in the same package as the compute die, so a scale-up domain could span a row without copper's reach and power limits — which would loosen precisely the constraint that makes today's rack a single indivisible purchase. That the direction is real is grounded elsewhere in this bag: the Rubin-class rack already spends its non-GPU budget on NVLink switches, networking chips, cooling and power [9], and integrates sixth-generation NVLink switching, ConnectX-9 SuperNICs, BlueField-4 DPUs and custom liquid cooling as one machine [10]. The Ayar-specific claims — \$650m raised in 2026, \$870m cumulative, a \$3.75bn valuation, strategic investment from AMD, Intel, MediaTek and NVIDIA, and a Wiwynn rack partnership — come from trade-press reporting; the source registered for them failed to index, so they carry NO [N] and the row's confidence is medium accordingly. This is the least-evidenced row in the table and it is labelled as such.

The fringe of this theme is attacking the rack's PHYSICS, not its price — and the three names worth watching are all betting the indivisible cabinet becomes divisible

EMERGING PLAYERS

Three rows in Exposed_Companies carry Role = emerging: Credo Technology, Astera Labs and Ayar Labs. Two more private names came out of the fringe searches without earning a row, and are recorded here because a reader should know they were found.

CREDO TECHNOLOGY (CRDO) — LISTED JANUARY 2022, THINLY MODELLED RELATIVE TO ITS GROWTH RATE

What makes it emerging is not its age but its position: a single-product-line company whose product line did not have a market three years ago. An active electrical cable is copper with a DSP in the connector, and it attacks the segment where a pluggable optical transceiver is overkill and passive copper is out of reach — the several metres between racks in a scale-up domain. The edge over incumbents is power and cost per link at that specific distance, which is exactly the distance the NVL72-class rack made economically important. FY2026 revenue of \$1.3bn (more than 3x) with AECs driving over 99% of the increase, and top-three customer concentration around

74% in Q1 FY2027, are reported figures — the company's own results release was blocked from indexing in this workbook and they carry no [N]. If the approach works, Credo owns a layer that neither NVIDIA nor Broadcom has chosen to own; if scale-up domains move to optics sooner than expected, the same specificity is the risk.

ASTERA LABS (ALAB) — IPO MARCH 2024; THE OPEN RACK'S DEFAULT SUPPLIER

Astera is emerging in the sense that matters here: its Scorpio fabric-switch family is new, and it is the piece that would let a non-NVIDIA rack have a credible scale-up fabric. The mechanism is software-defined switching that a hyperscaler can shape to its own topology rather than accept as delivered. The company states the interoperability outright — Aries retimers integrate with third-party GPUs from NVIDIA and AMD while also addressing the internal ASICs and AI accelerators hyperscalers develop [1]. That is the novel commercial position: Astera is the one supplier that is paid the same whether the rack is proprietary or open, which is a very different risk profile from a company levered to one architecture winning.

AYAR LABS (PRIVATE) — THE DEEPEST FRINGE, AND THE ONLY ONE ATTACKING THE PREMISE

Ayar puts optical I/O in the compute package itself (TeraPHY), which would dissolve the copper reach limit that makes the rack a single indivisible purchase in the first place. If a scale-up domain can span a row of cabinets over fibre at package-level power budgets, then the 'coherent cluster' stops being a cabinet and becomes a floor plan — and the switching cost the whole factor rests on gets smaller. That is why it is in the table despite having no revenue. The funding and valuation detail (\$650m raised during 2026, \$3.75bn valuation, strategic investment from AMD, Intel, MediaTek and NVIDIA, a Wiwynn rack-integration partnership) is trade-press reporting: the source registered for it failed to index here, so it carries NO [N] and the row's confidence is medium. What IS grounded is the direction of travel — the non-GPU half of a Rubin-class rack is already NVLink switches, networking chips, cooling and power [5], which is the budget optical I/O is competing for.

FOUND ON THE FRINGE, NOT GIVEN A ROW

- **Enfabrica (private).** ACF SuperNIC silicon — a high-bandwidth NIC fabric purpose-built for accelerated clusters. Its \$115m Series C was led by Spark Capital with Arm, Cisco Investments, Maverick Silicon, Samsung Catalyst Fund and VentureTech Alliance joining [2]. It is excluded only because the round predates the current rack generation and its deployment footprint could not be established; the mechanism is squarely on-theme and it belongs on a follow-up list.
- **Corintis (private).** Co-designed microfluidic cooling, supplied either as a drop-in replacement for a conventional liquid-cooling system or integrated with the chip as co-packaged cooling, targeting up to an order-of-magnitude improvement in cooling performance [3]. Microsoft has tested it. Excluded for the same reason — the mechanism is real and early, the commercial footprint is not yet establishable. Accelsius (two-phase cooling, \$65m Series B, a 300MW campus deployment agreement) sits in the same bucket.

WHAT THE FRINGE SEARCHES SAY ABOUT THE THEME

Every fringe name found is attacking the same joint: the physical limit that forces compute, fabric, power and cooling to be bought as one cabinet. None is attacking the software layer. That is a meaningful negative result — it suggests the market's own capital believes the durable constraint is PHYSICS rather than CUDA, which is the opposite of how the moat is usually described. It is consistent with the structural shift the workbook can already see: 2Q 2026 was the first quarter in which AI back-end switch sales surpassed front-end network switch sales [4].

Ideas

AI back-end switch sales overtook front-end network switch sales in 2Q 2026, and the vendor leading that market is priced at roughly a third of the multiple of the vendor ranked third

THE SWITCH MARKET FLIPPED AND ITS LEADER TRADES LIKE AN EMS

THE THESIS

The fabric layer is now the fastest-growing line item in the rack, and the market has repriced the names it already thought of as networking companies. It has not repriced Celestica, which leads the same market. The mismatch is a classification error — Celestica is filed as an EMS company and is paid like a switch vendor.

THE MECHANISM

From the Mechanism pod: the compute share of NVIDIA's own Data Center revenue fell from 91.5% to 82.4% in four quarters while networking grew 263% year on year — the fabric is where the incremental dollar went. Exposure_Evidence records the structural confirmation: in 2Q 2026, switch sales in AI back-end networks surpassed front-end network switch sales for the first time, and Celestica held the leading position in Ethernet AI back-end switch sales ahead of NVIDIA, with Arista third and Cisco fourth. The value lands with whoever holds the design-and-build relationship with the hyperscaler, not with whoever owns the brand.

THE NAMES

Celestica (CLS) is the anomaly — market leader, EMS multiple. **Arista (ANET)** is the quality version of the same exposure, and its own taxonomy is why it is clean: it sells scale-out and scale-across, not scale-up, so it is levered to the number of racks and the distance between them rather than to whose racks they are. **Broadcom (AVGO)** is the silicon underneath both, with FY2026 AI revenue guided to \$58bn and AI networking up over 2.5x year on year in Q3. All three are documented in Exposure_Evidence.

WHAT WOULD HAVE TO BE TRUE

- Celestica's switch position has to be a franchise rather than a build slot — i.e. it has to survive a hyperscaler re-tendering the program. The test is whether its share holds across a generation transition (800G to 1.6T), not within one.
- Ethernet has to keep extending in the back end. Dell'Oro has it doing so; an InfiniBand or proprietary-fabric reversal would take the volume back inside the platform vendor.
- Merchant switch silicon has to stay merchant. Broadcom's Tomahawk 7 tape-out (first 200 Tb/s Ethernet switch) is evidence it will.

WHAT WOULD KILL IT

Co-packaged optics working at scale-up. If the switch, the optics and the accelerator collapse into one co-packaged assembly sold by the platform vendor, the discrete back-end switch market — and every independent vendor in it — shrinks into a component line. Coherent's own guidance puts CPO revenue starting in the December

2026 quarter and CPO for scale-up contributing in the second half of calendar 2027, which dates this risk rather than leaving it abstract.

WHAT IS ALREADY PRICED

The gap IS the idea and it is measurable: Celestica trades around 16.7x forward earnings and is up roughly 13% in 2026 to date, against Arista at roughly 48x and up about 49%, and Astera Labs at roughly 51x and up about 68%. The market has paid for the fabric theme twice and skipped the company Dell'Oro ranks first in it. The bear reading of that gap — that an EMS multiple is correct because the position is a contract, not a franchise — is exactly the thing the first 'what would have to be true' tests.

The only part of the rack that gets paid whether the fabric is NVLink, UALink or a hyperscaler's own is the part that moves electricity and heat

OWN THE WATTS NOT THE CHIPS

THE THESIS

The architecture war inside the cabinet — NVLink versus UALink versus Ethernet scale-up — decides who captures the silicon. It decides nothing about the power train and the cooling loop, which every one of those designs needs in roughly the same quantity. A position in the power-and-thermal chain is a position on rack DENSITY and gigawatts energised, with the architecture question stripped out.

THE MECHANISM

From the Mechanism pod: Rubin-class racks push toward 1,800W per package and are forcing the move to 800V DC distribution, and a customer switching platforms is re-engineering power, cooling and floor layout rather than swapping a card. Who_Wins_And_Loses calls this the toll booth on power and heat. The value lands here rather than upstream because the constraint is physical: higher voltage cuts current, current drives copper mass, and copper mass at megawatt density has become a hard constraint on how fast a hall can be built at all.

THE NAMES

Vertiv (VRT) is the rack-adjacent half — sidecar power conversion that frees 8-16U for compute, CDUs, direct-to-chip cooling, and 800V DC platform designs developed with NVIDIA for Vera Rubin Ultra Kyber. **Eaton (ETN)** is the grid-to-rack half — an 800V DC reference architecture with supercapacitor backup and busbar distribution, a sidecar rated to 900kW per rack, and liquid-to-liquid CDUs via Boyd Thermal. They are not substitutes: Vertiv's clock is racks shipped, Eaton's is halls energised. Evidence for both is in Exposure_Evidence.

WHAT WOULD HAVE TO BE TRUE

- 800V DC has to stay an ENGINEERED product rather than collapse into a commodity spec any electrical OEM can build to. The Vertiv/NVIDIA and Eaton/NVIDIA reference-design positions are the observable proxy — check whether the reference-design partner list widens at the next platform launch.
- Backlog has to convert. Vertiv's ~\$15bn and Eaton's \$14.5bn are orders, not revenue; the binding constraint on conversion is grid interconnection dates, not factory capacity.

- Rack density has to keep rising. The whole case rests on 120kW going to 220kW and beyond; a plateau turns this into ordinary electrical equipment demand.

WHAT WOULD KILL IT

A demand air pocket that shows up as ORDER cancellations rather than slower bookings — specifically, a quarter in which either company's backlog falls in absolute terms while revenue is still growing. That is the signature of a customer deferring a hall, and it arrives before it shows in revenue. A second killer, slower and more corrosive: 800V DC becoming an open OCP-style commodity standard, which would leave the volume and take the margin.

WHAT IS ALREADY PRICED

A good deal. Vertiv is up roughly 79% in 2026 to date on about 28.7x forward earnings — no longer an overlooked name, and the cheapest version of this idea may now be Eaton, where the data-centre business is 25-30% of the electrical segment and is diluted by the rest of the company. That dilution is the reason it is cheaper, and it is also the reason the idea is less pure.

The ODM margin compression the theme implies is already visible in the numbers AND already in the multiple, so the tradeable version is the spread between the assembler and what it assembles

THE INTEGRATOR SQUEEZE IS REAL AND MOSTLY PRICED

THE THESIS

The rack shift is universally read as a tailwind for the companies that build racks. The measured effect on Hon Hai is the opposite: the AI business became the majority of revenue in the same quarter that gross margin fell. That is a real squeeze. But it is not an unnoticed one — Hon Hai and Quanta trade near 10-12x forward earnings against 28-51x for the content suppliers, so the market has priced the squeeze. The honest form of this idea is therefore not 'short the integrator'; it is that the theme pays through the SPREAD between rack content and rack assembly, and that spread is the cleanest expression of the factor.

THE MECHANISM

From Who_Wins_And_Loses: the assembler with no proprietary content is squeezed as the platform vendor ships more of the finished tray. Exposure_Evidence carries the measurement — Hon Hai's server-related revenue passed 50% of total for the first time in Q2 2026 while gross margin slipped to 6.12% from 6.33%, and the reason is structural: shipping pre-assembled trays turns the ODM from system designer into integrator and service provider, raising the platform vendor's control of the value chain and its margins, and leaving price as the ODM's only competitive lever. Quanta's 4.78% Q1 2026 gross margin is the same mechanism with less scale to absorb it.

THE NAMES

The squeezed leg is **Hon Hai (2317.TW)** and **Quanta (2382.TW)**. The content leg is whichever supplier's dollar content per rack is rising — **Amphenol (APH)**, **Coherent (COHR)** and **Vertiv (VRT)** are the three with the broadest, least architecture-specific content. **Celestica (CLS)** is deliberately NOT in the squeezed leg: the landscape work

and Exposure_Evidence place it as the EMS that acquired proprietary product, which is precisely the escape from this squeeze.

WHAT WOULD HAVE TO BE TRUE

- The tray must keep moving up. The observable is whether NVIDIA extends pre-assembly from compute trays toward full rack-level integration at the next platform generation; the Kyber NVL576 transition is the named test.
- Content per rack must keep outgrowing assembly value. The 2Q 2026 flip in AI back-end switch spending is evidence it is.
- The gross-margin direction at Hon Hai and Quanta must persist. One quarter of mix-driven slippage is not a trend; four is.

WHAT WOULD KILL IT

Hon Hai or Quanta reporting gross-margin EXPANSION on rising AI mix. That single observation breaks the mechanism, because the whole claim is that AI rack volume is dilutive to the assembler. A second, subtler killer: an ODM acquiring proprietary content — Celestica's switch position shows the escape route exists, and if Foxconn buys or builds its way onto the same side, the squeezed leg stops being squeezed.

WHAT IS ALREADY PRICED

Most of it. Hon Hai at roughly 10.6x forward and Quanta at roughly 11.8x are not pricing a growth business; both are nonetheless UP in 2026 (about 24% and 26%), so the market is paying for the volume while refusing to pay for the margin. That is a coherent view rather than a mistake. The residual opportunity is therefore in the spread's persistence, not in a re-rating — and anyone who reads this idea as 'the ODMs are cheap' has inverted it.

The scale-up domain's physical layer is copper today and optical eventually, and the companies that own the copper interval are priced as though the interval is permanent — or as though it has already closed

COPPER HAS A WINDOW BEFORE THE LIGHT

THE THESIS

A rack-scale coherent cluster is defined by how far a high-bandwidth link can reach at acceptable power. Today that distance is served by active electrical cables and retimers; eventually it is served by co-packaged optics. The interval between the two is where Credo and Astera Labs earn, and the length of that interval is knowable rather than mysterious — an optics supplier has already dated it. This is a duration bet on a physical transition, not a growth bet.

THE MECHANISM

From the Mechanism pod: the rack is a coherent memory domain, and what makes it coherent is the fabric. Who_Wins_And_Loses puts SerDes, retimers, cables and transceivers in the fastest-growing line item. The value lands with the physical layer rather than the switch because at 200-220kW per cabinet the constraint is picojoules per bit over metres, which is a materials problem before it is a silicon problem. Copper with a DSP in the connector is the cheapest solution to it — until reach and power force light.

THE NAMES

Credo (CRDO) is the pure copper interval: AECs drove over 99% of its FY2026 revenue increase. **Astera Labs (ALAB)** is the hedged version — its Aries retimers interoperate with NVIDIA and AMD GPUs AND with hyperscalers' internal ASICs, so it is paid whichever fabric wins, and Scorpio moves it from per-server to per-platform content. **Ayar Labs (private)** is the other side of the same trade, seeded from Emerging_Players: package-level optical I/O that would end the copper interval rather than extend it. Owning the copper names without tracking Ayar is owning one leg of a transition and ignoring the clock on it.

WHAT WOULD HAVE TO BE TRUE

- Scale-up domains have to keep growing in physical size. A 72-GPU cabinet is a copper problem; a 576-GPU multi-cabinet domain starts to be an optical one.
- Credo's customer concentration has to convert into design-win durability rather than single-point failure. Top-three customers were around 74% of Q1 FY2027 revenue.
- Merchant physical-layer silicon has to stay merchant. If the platform vendors absorb retiming and cabling into their own reference BOM, the interval exists but is not investable.

WHAT WOULD KILL IT

A dated, checkable observation: Coherent's guidance has co-packaged optics for SCALE-UP applications beginning to contribute in the second half of calendar 2027. If CPO scale-up ships on that schedule at volume, the copper interval closes roughly eighteen months from now and the AEC thesis has a terminal date rather than a growth curve. The second killer is specific to Credo: a single design loss at a top-three customer, which at that concentration is not a revision but a re-rating.

WHAT IS ALREADY PRICED

Divergently, which is the interesting part. Astera Labs is up roughly 68% in 2026 to date at about 51x forward — the market is paying for the platform-content story. Credo is up roughly 17% at about 39x and sold off after its September results on margin concerns — the market is NOT paying for the copper interval and may be pricing its end. Those two prices cannot both be right about the same transition. Establishing which one is wrong is the work this idea sets up, and the CPO scale-up date is the instrument for doing it.

Summary

The rack shift pays whoever supplies what goes **INSIDE** the cabinet and taxes whoever merely assembles it — and almost none of that money depends on which architecture wins

FACTOR RESEARCH SUMMARY

WHAT THE THEME IS

The unit of sale in AI compute has moved from the chip to a \$5m-\$8.8m coherent cabinet sold, warranted and tuned as one machine (Mechanism). The consequence that matters for money is not that the cabinet is expensive; it is that its bill of materials re-weighted. Compute fell from 91.5% to 82.4% of NVIDIA's own Data Center revenue in four quarters while networking grew 263% year on year, the platform's revenue opportunity per gigawatt moved from ~\$18bn to ~\$40bn, and the physical plant — 1,800W packages, 800V DC, mandatory liquid cooling — had to be rebuilt to take it. Two clocks run at different speeds (Drivers_And_Horizon): capacity on a five-to-ten-year build, architecture on an eighteen-month contest.

WHO IS LEVERED TO IT

Twelve companies, none of them NVIDIA (Exposed_Companies). Six sell content whose dollar value per rack rose faster than the rack price and are indifferent to the architecture war: Vertiv and Eaton in power and thermal, Amphenol and Coherent in interconnect and optics, Broadcom and Celestica in switch silicon and switch systems. Two — Credo and Astera Labs — own the physical layer of the scale-up domain and are the theme's emerging entrants. Two — Hon Hai and Quanta — build the rack and are being squeezed by it in measurable terms. One, Ayar Labs, is attacking the constraint the whole factor rests on. The single most surprising finding: in 2Q 2026 Celestica held the LEADING position in Ethernet AI back-end switch sales, ahead of NVIDIA, with Arista third (Exposure_Evidence).

WHAT FALLS OUT

Four candidates, each with a falsifier (Ideas):

1. **Own the watts, not the chips** — the power and thermal chain is paid by every rack in every architecture (Vertiv, Eaton). Killed by backlog falling in absolute terms while revenue still grows.
2. **The switch market flipped and its leader trades like an EMS** — Celestica leads the fastest-growing switch market at roughly a third of Arista's multiple (Celestica, Arista, Broadcom). Killed by co-packaged optics collapsing the discrete switch into the platform vendor's assembly.
3. **The integrator squeeze is real and mostly priced** — so the theme pays through the spread between rack content and rack assembly, not through a short (Hon Hai, Quanta versus Amphenol, Coherent, Vertiv). Killed by either ODM reporting gross-margin EXPANSION on rising AI mix.
4. **Copper has a window before the light** — the AEC and retimer interval has a date on it, and Coherent has published it (Credo, Astera Labs, Ayar Labs). Killed by CPO scale-up shipping at volume in H2 CY2027 as guided.

WHAT THIS RUN DID NOT SETTLE

This is the most useful paragraph here.

- **Seven of the twelve rows rest on reported figures that are not workbook-grounded.** Amphenol, Credo, Quanta, Vertiv's financials, Eaton's financials, Celestica's guidance and every Ayar Labs number come from sources that were registered in this run's collection and then blocked or timed out during ingestion (Cloudflare, Akamai, crawler timeouts). The exposure LINKS are documented in company reporting and are not in doubt; the NUMBERS carry no citation and are flagged as such in Exposure_Evidence. A follow-up should re-register those from PDFs or filings rather than HTML pages.
- **The Ayar Labs row is the thinnest in the table** and is the only one carrying medium confidence. Its mechanism is the most consequential in the bag — if package-level optical I/O works, the cabinet stops being the indivisible unit and this factor's switching cost shrinks — and it is the least evidenced.
- **The open-standard counterfactual is asserted, not measured.** UALink 1.0 and 2.0 exist and AMD's Helios is built around UALink, but this run found no shipment, revenue or deployment data for a UALink rack. That is the single most important missing number for the eighteen-month clock in Drivers_And_Horizon.
- **The valuation snapshot is unsourced.** Every forward P/E and year-to-date figure in the Ideas pods is a market-data pull dated 11 September 2026 that could not be registered as a resource. Ideas 2 and 4 lean on those numbers heavily and should be re-checked before anyone acts on the gap they describe.
- **Nothing here re-tests the moat's SOFTWARE half.** The run took a supply-chain-content lens by choice (the fringe finding — every emerging entrant is attacking physics, none is attacking CUDA). A perspective run through the software lens would weigh differently and is the obvious next challenge.

FACTOR RESEARCH

NVDA Factor 08 Self Funded Demand Loop

Theme

A supplier that underwrites its customer's purchase has stopped selling into demand and started manufacturing it

MECHANISM

Vendor financing is the practice of a supplier putting its own balance sheet behind the purchases its customers make from it. The money does not have to be lent. In this cycle it moves through six distinguishable instruments, and the distinction is the whole subject: "A residual value guaranty, a revenue floor, a capacity purchase commitment, a warrant and an equity investment have different beneficiaries, different triggers and different balance sheet consequences." [1] They differ in who is protected, what event pays out, and where — if anywhere — the obligation lands on a financial statement.

THE CHAIN

- 1. The buyer cannot fund the purchase on its own credit.** The frontier AI labs "have extraordinary demand for training and inference compute, but they are growing faster than what their balance sheets and credit profiles can support" [2]. Neoclouds, converted miners and merchant data-centre developers are in the same position: unprofitable, thinly capitalised, and asking lenders for fifteen-year money.
- 2. The supplier supplies the credit instead of the cash.** What the project needs is a rateable counterparty. When Hut 8 announced a Texas lease it "never named its tenant. It told investors only that the counterparty was an "investment-grade business," because that phrase was the point." [3] The instrument substitutes the supplier's rating for the buyer's.
- 3. The instrument makes the project bankable.** In the clearest published example of the structure, the supplier "provides a take-or-pay commitment on a portion of the facility's capacity, a minimum revenue guarantee that gives lenders the confidence to underwrite the project, and in exchange, we share in a portion of the NeoCloud's revenue earned above that floor" [2]. The lender is now underwriting a floor written by the equipment vendor, not a cash flow produced by the tenant.
- 4. The order is placed, and the revenue books at the supplier.** The obligation does not. A guarantee is a contingency disclosed in the notes; an equity purchase is an investing cash outflow; a capacity purchase commitment is a purchase obligation. None of them appear as a deduction from revenue, and none of them appear in the operating margin the market prices off.

THE MAGNITUDES IT RUNS AT

- **Equity into the customer base.** One vendor's equity-investment book went from roughly \$2.2B two years ago to about \$7B a year ago to \$99B now, "a more than 13-fold increase" in twelve months [4]; the privately-held half alone "stood at \$47.898 billion as of July 26, 2026, up from \$3.799 billion a year earlier" [5].

- **Guarantees written against leases.** A single residual-value package caps exposure at \$105B across 4.25 GW, where "The campus will exclusively host our compute under 20-year leases to OpenAI" [6]. A smaller, earlier line ran to a "maximum gross exposure of US\$3.5 billion on partner facility lease backstops with US\$712 million held in escrow" [1].
- **Equity taken IN EXCHANGE for the guarantee.** "In fiscal year 2026, we entered into agreements to guarantee partners' facility lease obligations in the event of their default in exchange for warrants." [7] The supplier is paid in its customer's stock for insuring its customer's landlord.
- **Warrants as the cheap end of the menu.** A second-source vendor "has announced deployment arrangements covering up to 14 GW across three customers: OpenAI at 6 GW in October 2025, Meta at up to 6 GW in February 2026 and Anthropic at up to 2 GW in July 2026" [1], and separately took "warrants to purchase up to 30 million Core Scientific shares at \$23.47 per share, with about 6.5 million vesting when the initial leases were signed" [8] against 529 MW of 15-year leases the same counterparty values above \$14B of contracted revenue.

WHY THE LOOP IS SELF-REINFORCING RATHER THAN MERELY GENEROUS

The supplier gets paid twice — once on the hardware and again on a share of the rental stream layered on top of it [2] — and the equity it holds reprices upward on the same demand its own shipments create. That is the closure: the collateral, the customer and the revenue are the same asset observed three times. The enforcement is contractual rather than physical, which is why it can be built in eighteen months and why it can be re-rated in a single trading session.

The loop is priced on a quarterly clock and resolves on a twenty-year one — that gap is the whole trade

DRIVERS AND HORIZON

WHAT IS PUSHING IT

A financing gap that widens with model scale. The buyers of AI compute are growing faster than their credit can fund, and the gap is not closing on its own. The hyperscalers — the one cohort that *can* borrow — "issued \$108 billion of debt globally in 2025" and by mid-2026 "they've issued \$194 billion" [1], a 1.8x step in eighteen months from companies that already hold large cash balances. Everyone below them in the credit stack has to borrow against something else, and the only rateable thing on offer is a supplier's signature.

A credit market that is willing but repricing. Goldman's AI-leader credit basket "has gone from tights of 74 basis points to nearly twice that" over twelve months [1], and project finance for data centres is expected to run about \$300B in 2027 on top of direct hyperscaler issuance. Capital is available; the price of it is moving.

A supplier cash pile with nowhere better to go. The instruments are cheap for whoever has the balance sheet and prohibitive for whoever does not. One vendor's maximum gross guarantee exposure "phases on from fiscal 2029 and pays only on an OpenAI trigger event after ready-for-service, while its US\$279 billion of supply and capacity commitments sit against a demand book it does not schedule" [2].

WHAT HAS ALREADY CHANGED

Three years ago this did not exist at scale. The same vendor's equity book was about \$2.2B two years ago and roughly \$7B a year ago before reaching \$99B; the private half moved from \$3.799B to \$47.898B in twelve months [3]. The instrument set moved too: fiscal 2026 was warrants taken in exchange for facility-lease guarantees at \$3.5B of gross exposure; fiscal 2027 is a \$105B capped residual-value package and a stated target to mobilise "over \$500 billion of third-party capital for the buildout of AI infrastructure over time" [4]. Twelve months took the structure from a treasury sideline to a capital-markets platform.

THE LAST TIME THIS FORCE RAN

It ran in telecom equipment between 1998 and 2001 and it ended badly, which is why the comparison keeps being made and why the magnitudes matter more than the analogy. "McKinsey later put the combined vendor-financing exposure of nine suppliers at approximately \$25.6 billion by the end of 2000." [5] Lucent committed about \$8.1B and "took bad-debt provisions of roughly \$2.2 billion in fiscal 2001 and a further \$1.3 billion in fiscal 2002, about \$3.5 billion of customer-loan losses" [5]. The speed of the turn is the part worth keeping: in a single year, "Motorola's bad loans went from 6.7 percent of total loans at the end of 2000 to 57 percent at the end of 2001; Lucent went from 2.6 to 60 percent, and Nortel from 25.5 to 80 percent" [6].

Two differences are real and cut opposite ways. The scale is two orders larger — \$25.6B across an entire industry then, against \$99B of equity plus a \$105B capped guarantee at one supplier now. But the asset is not dark: "In the internet era, equipment vendors helped finance their customers because demand was still speculative, and much of the fiber optic cable laid in those years sat dark for a decade." [7] This time the capacity is sold as it energises. The 1999 cycle failed on utilisation; if this one fails it will be on price, schedule and residual value rather than on empty racks.

THE CLOCK

Weeks, for the price. Five-year CDS on the largest writer of these instruments "jumped as much as 14 basis points on Monday to about 82, the biggest one-day rise since the contracts began trading actively in November" [7] — the credit market reprices the correlation long before any income statement shows it.

Three to six years, for the exposure to attach. Guarantee amounts increase "as each of nine data centers is placed in service, which is expected to begin in fiscal year 2029" [8]. Nothing attaches before the concrete does.

Fifteen to twenty years, for resolution. The underlying leases run 20 years; the release condition is the buyer "achieving a satisfactory credit rating" [8], a threshold that is redacted. The theme therefore does not resolve within a normal holding period. It gets re-rated, repeatedly, on evidence about construction schedules, third-party capital actually arriving, and rating-agency treatment.

WHAT WOULD WEAKEN IT

Buyers reaching investment grade on their own cash flow (which terminates the guarantees by their own terms); the \$500B of third-party capital converting from a target into deployed funds, which moves the risk off the vendor and onto asset managers; or rating agencies imposing debt-equivalent adjustments large enough that the instrument costs more rating capacity than the incremental gross margin is worth.

Vendor financing is a scale barrier dressed as a courtesy — it rewards whoever **RECEIVES** the instrument and whoever **CLIPS** it, and squeezes whoever has to **WRITE** it

WHO WINS AND LOSES

The theme sorts companies by one question: in the transaction, are you the party whose credit is being substituted, the party substituting it, or the party taking a fee on the substitution?

STRUCTURALLY FAVOURED

Anyone holding a long-dated lease with a counterparty they could not have reached on their own. Merchant data-centre developers, converted bitcoin miners and power-site owners trade an unrateable tenant for a rateable one and refinance the whole capital stack against it. The tell is a company announcing contracted revenue against "an 'investment-grade business'" without naming it [1]. Fifteen- and twenty-year leases against a credit-enhanced tenant are the single most valuable asset this theme creates.

Anyone whose purchases are being underwritten. Frontier labs and neoclouds acquire compute at a cost of capital that reflects the supplier's rating rather than their own — the explicit purpose of a structure that gives "lenders the confidence to underwrite the project" [2]. Their equity is levered to a subsidy they do not pay for, and it disappears the moment the subsidy is withdrawn.

Anyone who earns a fee on deployment rather than a return on the compute. The asset managers and banks assembling the third-party pools are paid on capital raised and deployed [3]; whether a GPU earns its residual is somebody else's problem. This is the toll booth of the theme and it is the only position in it that does not require a view on AI demand.

Anyone selling INTO the underwritten build without writing an instrument. Memory, optics, power equipment, transformers, turbines and EPC contractors are paid on the construction schedule. They capture the volume the financing unlocks and carry none of the contingent liability.

STRUCTURALLY SQUEEZED

Any supplier whose balance sheet cannot write the instrument at the size now being asked. This is the hard edge of the theme: "The Piketon cap alone is 156 percent of AMD's entire equity." [4] Once customers learn to ask for credit support, the ability to provide it becomes a competitive weapon, and the second-source vendor either accepts an equity-diluting or margin-diluting substitute (warrants, prospective equity commitments) or loses the socket. Competition on price becomes competition on balance sheet.

Any vendor whose obligation runs from contract signature rather than from commercial operation. A commitment that attaches before the plant exists puts the vendor on construction risk it does not control; a milestone-conditioned one does not [4]. Same headline number, completely different expected cost.

Any vendor that has committed inputs against a delivery calendar it does not own. HBM, advanced packaging, substrates and rack integration are ordered on the customer's announced schedule; a slip does not unwind the purchase order. The financing protects the revenue line and does nothing for the inventory line.

Holders of the writer's CREDIT rather than its equity. The correlation shows up in spreads first: a one-day move of "as much as 14 basis points" to about 82bp on a company generating tens of billions of quarterly free cash flow [1]. Bondholders get the concentration and none of the upside on the stakes.

GENUINELY TWO-SIDED

Rating agencies, auditors and standard-setters. Their treatment decides whether the layer is a footnote or a liability — a guarantee can create a substantial debt-equivalent adjustment at the vendor even where lenders grant the borrower no extra advance rate [4]. More work and more relevance for them; a step-change in cost for everyone they re-rate.

The buyer that also competes. Equity taken in a customer that is simultaneously building a substitute part buys presence in the substitute rather than preventing it.

THE SCREEN THIS IMPLIES

Look for: lessors and hosts with named or implied investment-grade credit support and a 15-year-plus contracted book; second-source silicon vendors issuing warrants or equity commitments they cannot cover with cash; the six asset managers named in the compute-financing platforms; the capacity recipients whose contracted revenue is concentrated in one or two credit-enhanced tenants; and the ratings and audit firms whose published method sets the cost of the whole structure. Exclude the underwriter itself — that is the workbook's own subject.

Landscape

Exposed Companies

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Advanced Micro Devices	AMD	incumbent	Runs the same playbook one tier down and cannot fund it in cash: warrants issued to customers (160M shares at \$0.01 to Meta, vesting on cumulative 6 GW of purchases; OpenAI 6 GW; up to \$5B prospective equity in Anthropic) against \$67.2B of stockholders' equity and \$5.1B of cash.	mixed	high		Up to 14 GW announced across three customers. A Piketon-sized residual guaranty would be 156% of AMD's entire equity - the instrument set is a scale barrier, and warrants are the cheap substitute it can actually write.
IREN Limited	IREN	emerging	Issued the chip vendor a five-year right to buy 30M ordinary shares at \$70 (up to \$2.1B of potential investment) as the economic alignment behind deploying up to 5 GW of DSX-aligned AI infrastructure across its pipeline.	benefits	high		Bitcoin miner converting to AI cloud. The warrant is the vendor's payment for the offtake, not a financing - IREN still has to fund the build.
Core Scientific	CORZ	emerging	Granted AMD warrants over up to 30M shares at \$23.47 (about 6.5M vesting on signature of the initial leases) as consideration for 529 MW of 15-year leases the company projects at over \$14B of base contracted revenue, expandable to 2.5 GW.	benefits	high		Post-bankruptcy bitcoin miner. Equity-for-offtake is now the standard price of a vendor-anchored lease at this tier.

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Cipher Mining	CIFR	emerging	Barber Lake: a 10-year base lease with Fluidstack carrying ~\$3.8Bn of contracted revenue across 300 MW, made bankable by Google agreeing to backstop up to \$1.73Bn of Fluidstack's lease obligations - with a concurrent warrant issued to Google.	benefits	high		Proof the structure is not chip-vendor-specific: a hyperscaler here plays the credit-substituting role, and takes warrants for it exactly as a chip vendor would.
CoreWeave	CRWV	adjacent	The archetypal underwritten buyer: a loss-making neocloud funding GPU purchases on DDTL facilities whose lenders carry roughly a two-year uncontracted tail at a 70 percent advance against a contract book averaging about three years.	benefits	high		Also a direct equity recipient of the vendor. Its cost of capital, not its demand, is what vendor support actually changes.
SB Energy		adjacent	Lessor at the PORTS Technology Campus: receives capped credit support of up to \$105B in aggregate on land, power and shell for roughly 4.25 GW of IT load, let to OpenAI on 20-year leases, with an option for ~3.8 GW more.	benefits	high		Private. The single largest disclosed instrument in the theme, and the clearest case of a landlord financing against a supplier's rating rather than a tenant's.

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Nebius Group	NBIS	emerging	Named as the first counterparty for the take-or-pay plus revenue-share structure - a vendor-written minimum revenue guarantee on part of the facility that gives lenders confidence to underwrite - alongside \$5.0B of convertible senior notes priced in August 2026.	benefits	medium		Newly listed, thinly covered. The 0.50% 2030 / 4.50% 2034 split shows what the market charges this tier without and with duration.
TeraWulf	WULF	emerging	Converted miner now leasing to an unrated frontier lab: a 20-year lease with Anthropic for ~401 MW of critical IT capacity at Hawesville, with initial delivery in 2H27 - the un-credit-enhanced version of the same trade.	benefits	medium		No vendor backstop disclosed on this lease. That is the point of the row: it isolates what the enhancement is worth.
Blackstone	BX	adjacent	One of six institutions signed to memoranda of understanding to create dedicated pools of capital at attractive rates for the vendor's customers, targeting over \$500B of third-party capital mobilised over time; earns fees on capital deployed, not on compute returns.	benefits	medium		\$1.3T AUM. MOU stage - no committed or deployed capital disclosed. Apollo, BlackRock, Brookfield, Goldman Sachs and KKR sit in the identical position.

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Applied Digital	APLD	adjacent	Merchant AI data-centre developer whose contracted book is a concentrated credit bet: nearly 86% of contracted revenues come from just two customers, so its financing cost tracks those tenants' credit rather than its own operations.	mixed	medium	86	Segment % is the share of CONTRACTED REVENUE from the top two tenants, not a revenue segment. Wins volume, carries counterparty concentration.
S&P Global	SPGI	adjacent	Its published guarantee-adjustment method is what decides whether the layer stays off balance sheet: a guaranty can create a substantial debt-equivalent adjustment at the vendor even where lenders give the borrower no additional collateral value.	mixed	medium		More ratings work and more relevance; also the agency that can reprice the whole structure by changing one method. Same applies to Moody's.
Digital Realty Trust	DLR	adjacent	Conventional colocation competes for the same capital on worse terms: traditional facilities serving enterprise tenants access debt at 60-70% LTV and 7-9% rates, while operators with contracted capacity to major AI companies finance 50-100bp below market.	hurt	low		The cost-of-capital gap is documented; that DLR specifically is disadvantaged by it is INFERRED from its enterprise/colocation tenant mix, not stated in any source here.

Eleven of the twelve rows rest on a filing or a company release; the twelfth is inference and is graded as such

EXPOSURE EVIDENCE

Table order. Each paragraph says what the company is, what the instrument actually is, and whether the exposure link is documented or inferred.

Advanced Micro Devices (AMD) — incumbent, mixed, high. AMD is the case that proves vendor financing is a balance-sheet competition, not a courtesy. It has announced deployment arrangements covering up to 14 GW across OpenAI (6 GW), Meta (up to 6 GW) and Anthropic (up to 2 GW), and paid for them with the cheapest instrument on the menu: a warrant over 160 million shares at one cent, vesting in tranches contingent on cumulative purchases reaching six gigawatt equivalent and on share-price thresholds escalating to \$600. It could not write the alternative — "The Piketon cap alone is 156 percent of AMD's entire equity" [1] against \$67,224M of stockholders' equity, \$5,086M of cash and \$3,226M of total debt. Documented in 8-Ks and the Q2 2026 release. The skeptic's objection is that warrants cost AMD nothing unless the volume arrives, which is true and is exactly why the row is mixed: the dilution is contingent, but the committed HBM4, packaging and substrate spend behind the orders is not.

IREN Limited (IREN) — emerging, benefits, high. A bitcoin miner rebuilding itself as an AI cloud. The instrument is explicit and one-directional: "IREN issued to NVIDIA a five-year right to purchase up to 30 million shares of ordinary stock at an exercise price of \$70 per share, resulting in a right to invest up to \$2.1 billion" [2], as the alignment behind deploying up to 5 GW of DSX-aligned infrastructure. Note what this is NOT: no cash changed hands and no lease was guaranteed. IREN pays in optionality on its own equity for a supply relationship; the build still has to be funded elsewhere.

Core Scientific (CORZ) — emerging, benefits, high. The same trade with the second-source vendor. "AMD received warrants to purchase up to 30 million Core Scientific shares at \$23.47 per share, with about 6.5 million vesting when the initial leases were signed" [3], against 529 MW committed under 15-year leases that the company projects at over \$14B of base contracted revenue, expandable to 2.5 GW. Equity-for-offtake is now the standard clearing price of a vendor-anchored lease at this tier, and the market read it as a transfer: Core Scientific rose about 6% on the announcement while AMD fell about 4%.

Cipher Mining (CIFR) — emerging, benefits, high. The most important row for understanding the theme's boundary, because the credit substituter here is not a chip vendor. Cipher's Barber Lake project carries a 10-year base lease with Fluidstack with two 5-year extension options across 300 MW gross capacity and ~\$3.8Bn of contracted revenue, and it is bankable because "Google to provide backstop for up to \$1.73Bn of Fluidstack's lease obligations" [4] — with a concurrent warrant issued to Google on a related financing. Same mechanism, same payment in equity, different party wearing the investment-grade jacket. Any screen built only on chip vendors misses half the theme.

CoreWeave (CRWV) — adjacent, benefits, high. The archetypal underwritten buyer: a loss-making neocloud funding GPU purchases on delayed-draw term loans while its lenders carry the duration mismatch — "DDTL 5.5's lenders still hold a roughly two-year uncontracted tail at a 70 percent advance against a borrower whose

contracts average about three years" [1]. It is also a direct equity recipient of the vendor. What vendor support changes for CoreWeave is its cost of capital, not its demand, which is why the row is benefits and not mixed — and also why the benefit is withdrawable.

SB Energy (private) — adjacent, benefits, high. The largest single disclosed instrument in the theme. The lessor at the PORTS Technology Campus receives credit support on land, power and shell covering roughly 4.25 gigawatts of IT load, where "The campus will exclusively host our compute under 20-year leases to OpenAI, subject to limited exceptions, with our obligation capped at \$105 billion in the aggregate, subject to certain conditions including SB Energy, the lessor, satisfying applicable ready-for-service conditions" [5], plus an option on roughly 3.8 GW more. Unlisted, so not directly investable; included because it defines the ceiling of the structure. The exposure is documented in the vendor's own 10-Q.

Nebius Group (NBIS) — emerging, benefits, medium. Named as the first counterparty for the structure in which the vendor "provides a take-or-pay commitment on a portion of the facility's capacity, a minimum revenue guarantee that gives lenders the confidence to underwrite the project" [6]. The company separately priced "\$5.0 billion aggregate original principal amount of convertible senior notes, in two series: \$3.0 billion aggregate original principal amount of 0.50% convertible notes due 2030 ... and \$2.0 billion aggregate original principal amount of 4.50% convertible notes due 2034" [7] in August 2026. Graded medium because the take-or-pay terms specific to Nebius — size, floor, revenue split — are not disclosed anywhere in this corpus.

TeraWulf (WULF) — emerging, benefits, medium. The control case. "Subsequent to quarter-end, TeraWulf entered into a 20-year lease with Anthropic for approximately 401 MW of critical IT capacity, with initial delivery expected in the second half of 2027 and full delivery expected in early 2028." [8] No vendor backstop on this lease is disclosed. It therefore isolates what the enhancement is worth: same converted-miner model, same twenty-year term, an unrated counterparty, and whatever spread differential the debt market applies. medium because the theme link is the ABSENCE of an instrument rather than one you can point at.

Blackstone (BX) — adjacent, benefits, medium. One of six institutions in the compute financing platforms, under which the vendor "will work with Apollo, BlackRock, Blackstone, Brookfield, Goldman Sachs and KKR to create dedicated pools of capital at significant scale at attractive rates for NVIDIA customers" [9], targeting over \$500B mobilised over time. Blackstone carries over \$1.3 trillion in AUM; Brookfield over \$1 trillion. Graded medium deliberately: these are memoranda of understanding, the \$500B is a stated target for THIRD-PARTY capital rather than committed or deployed funds, and no fee terms are public. Apollo, BlackRock, Brookfield, Goldman Sachs and KKR occupy the identical position and are omitted from the table only to keep it a research list rather than a screen.

Applied Digital (APLD) — adjacent, mixed, medium. A merchant AI data-centre developer whose economics are a concentrated credit bet: "nearly 86% of the company's contracted revenues are derived from just two customers" [10], and "deterioration in the credit profile of these key tenants could materially affect future revenue generation." That is the vendor-financing theme stated as a risk factor rather than an opportunity — APLD captures the build volume and carries the counterparty concentration, which is why the direction is mixed.

S&P Global (SPGI) — adjacent, mixed, medium. The agencies are where the theme is priced. S&P's reported treatment of the Ohio instrument shows that a guaranty can create a substantial debt-equivalent adjustment at the vendor even where lenders do not treat it as additional collateral value at the borrower [1] — the guarantor

pays rating capacity for an enhancement the borrower's lenders never credit. Whether this layer stays a footnote or becomes debt is, in practice, a methodology decision. mixed because the same methodology that creates work also constrains the volume of instruments that can be written. The identical argument applies to Moody's.

Digital Realty Trust (DLR) – adjacent, hurt, low. THIS ROW IS INFERENCE AND SHOULD BE READ AS SUCH.

What is documented is the cost-of-capital split: "Traditional colocation facilities serving enterprise tenants continue to access conventional debt structures with loan-to-value ratios of 60-70% and interest rates in the 7-9% range," while "Operators with contracted capacity to major AI companies are securing financing at 50-100 basis points below market rates" [11]. What is NOT documented anywhere in this corpus is that DLR specifically is disadvantaged by it; it is placed here because its tenant mix is described as a balanced enterprise, hyperscaler and colocation book rather than an AI-contracted one [12]. A reader should treat the mechanism as real and the name as a placeholder for the disciplined-underwriter cohort until someone checks DLR's actual marginal funding cost.

THE CONTRAST CASE WORTH HOLDING IN MIND

Vendor credit support is a substitute for tenant credit, and where tenant credit already exists nobody needs it. Ares arranged a \$2.4 billion debt facility to Vantage Data Centers, whose portfolio is "100% leased under long-term agreements with leading hyperscale customers" and where "Approximately 99% of Vantage's customers in North America hold an investment grade rating" — no vendor stands behind any of it. That transaction is the benchmark against which every instrument in this table should be priced: what does the enhancement buy, and what would the deal have cost without it?

The fringe of this theme is not a startup cohort — it is a set of listed bitcoin miners re-underwriting themselves as landlords, and the equity they hand over is the price of admission

EMERGING PLAYERS

Five rows carry emerging. Four of them are the same species: a company whose original business (mining, or an ex-Yandex cloud) gave it power, land and interconnection queue position, and which is now converting that asset into a long-dated AI lease by accepting a chip vendor or hyperscaler as its credit. What is novel is not the technology; it is the capital structure. The coverage is thin because the transformation is eighteen months old and the comparables are all prior-business comparables.

IREN LIMITED (IREN)

Listed, formerly a pure bitcoin miner, now building an AI cloud on its own sites. The instrument is the cleanest disclosed example of equity-for-offtake in the theme: "IREN issued to NVIDIA a five-year right to purchase up to 30 million shares of ordinary stock at an exercise price of \$70 per share, resulting in a right to invest up to \$2.1 billion" [1], behind deployment of "up to 5 gigawatts of NVIDIA DSX-aligned AI infrastructure across IREN's global data center pipeline over time" [1], anchored on a 2 GW Texas campus. **The mechanical edge over the incumbents:** IREN already owns energized sites with interconnection position, which is the scarce input the incumbents' capital cannot buy quickly, and the warrant strike at \$70 converts the vendor from a supplier into a

party that profits if IREN's equity works. **The error bar:** a warrant is not funding. Nothing in the announcement obliges the vendor to buy capacity, guarantee a lease or lend a dollar, and the \$2.1B is what the vendor may CHOOSE to invest, not what it has committed.

CORE SCIENTIFIC (CORZ)

Post-bankruptcy miner, now an AI hosting platform. "AMD received warrants to purchase up to 30 million Core Scientific shares at \$23.47 per share, with about 6.5 million vesting when the initial leases were signed" [2] against 529 MW under 15-year leases projected above \$14B of base contracted revenue, with an option path to 2.5 GW. **The edge:** it is selling the second-source vendor exactly what that vendor cannot build — a place to put Instinct parts at scale — and 15-year terms convert a volatile hosting business into something a lender can amortise against. **The error bar:** the tenant here is AMD itself rather than an unrated lab, so Core Scientific has traded operating risk for single-counterparty concentration in a company whose own AI volume is a share-gain bet.

CIPHER MINING (CIFR)

The most interesting of the five, because it shows the structure escaping the chip vendors entirely. Barber Lake is a 300 MW project on a "10-year base lease agreement with Fluidstack with two 5-year extension options" carrying "~\$3.8 Bn Contracted Revenue", made financeable because "Google to provide backstop for up to \$1.73Bn of Fluidstack's lease obligations" [3] — and on a related financing the filing notes a "Concurrent warrant issuance to Google aligns incentives for construction completion and financial backstopping" [3]. **The edge:** a hyperscaler backstop is better credit than a chip vendor's, and the layering (unrated operator → hyperscaler guarantee → warrant to the guarantor) is the whole theme in one transaction. **The error bar:** the backstop covers \$1.73Bn of a ~\$3.8Bn contracted book, so roughly half the revenue still rests on Fluidstack's own credit; a reader who treats the deal as fully guaranteed has mispriced it.

NEBIUS GROUP (NBIS)

Newly listed AI cloud, named as the first counterparty for the take-or-pay-plus-revenue-share structure [4]. **The edge:** being first means the terms were set to make the template work, and a vendor-written revenue floor is a stronger enhancement than a warrant because it reaches the lender directly. **The error bar:** the deal's actual terms are undisclosed in this corpus, and the company simultaneously raised \$5.0B of convertible notes [5] — which is what a company does when the enhancement is not doing all the work.

TERAWULF (WULF)

Converted miner with a "20-year lease with Anthropic for approximately 401 MW of critical IT capacity" [6] at Hawesville, initial delivery 2H27. **The edge:** 20 years is longer than anything in the neocloud cohort, and TeraWulf controls the power. **The error bar:** no vendor backstop is disclosed on this lease, so its tenant risk is undiluted frontier-lab credit. That makes it the natural control in any pair trade against a credit-enhanced landlord.

WHAT THE FRINGE SEARCH DID NOT FIND

Two negatives worth recording. First, no genuinely new entrant — no 2026-IPO pure-play whose business model is vendor-financed AI hosting from inception. Every emerging name here is a pre-existing listed company repurposing an existing asset base, which means the mispricing available is a re-rating of a known ticker, not a

discovery. Second, several prominent private names in this space have NO disclosed vendor instrument at all: the searches surfaced no warrants, guarantees or backstops from either chip vendor for Crusoe, GMI Cloud, Lambda or Together AI, and Galaxy Digital showed none either. That absence is useful — it says the instruments are being written where the site and the power already exist, not where the demand is.

Ideas

The market is pricing these landlords on megawatts. The asset is the counterparty, and only some of them have a good one.

BUY THE GUARANTEE NOT THE GIGAWATT

THE THESIS

A converted-miner landlord with a chip-vendor or hyperscaler backstop on its anchor lease owns a different security from one without, and the two are currently valued on the same metric — contracted gigawatts. The position is to own the credit-enhanced book and to treat the bare one as the funding leg, because the spread between them is the market's unpriced valuation of the enhancement itself.

THE MECHANISM

From the chain: the instrument substitutes an investment-grade signature for an unratable tenant, which is what turns a fifteen- or twenty-year lease into something a lender will amortise against. The value therefore lands on the LESSOR's cost of capital, not on the tenant's income statement — and that is the part nobody is marking. The documented split is explicit: conventional colocation funds at 60-70% LTV and 7-9%, while operators with contracted capacity to major AI companies finance 50-100bp below market (Exposure Evidence, DLR). On a twenty-year asset, 75bp is most of the equity value.

THE NAMES

Enhanced: Cipher Mining (CIFR) is the clearest — a 300 MW, ten-year, ~\$3.8Bn book with a named hyperscaler backstopping \$1.73Bn of the tenant's lease obligations. Core Scientific (CORZ) is next, though its enhancement is a different animal: the tenant IS the vendor, so it holds vendor credit directly rather than a guarantee over someone else's. IREN (IREN) is the weakest of the three for this purpose — the warrant aligns the vendor but guarantees nothing. **Bare:** TeraWulf (WULF) holds a twenty-year, ~401 MW lease to an unrated frontier lab with no disclosed backstop; Applied Digital (APLD) carries 86% of contracted revenue in two tenants. Evidence per name in Emerging Players.

WHAT WOULD HAVE TO BE TRUE

- The enhancement survives to energisation. Instruments in this cycle are milestone-conditioned: exposure attaches as buildings are placed in service, not at signature. A backstop on a site that never gets power is worth nothing to the lessor and costs the guarantor nothing.
- The coverage ratio is disclosed and holds. Cipher's backstop covers \$1.73Bn of a ~\$3.8Bn book — roughly half. A reader who prices it as a full guarantee is wrong by a factor of two.
- Lenders actually credit it. There is a documented counter-observation that no public transaction yet shows GPU-side vendor support lifting a lender's advance rate. If the enhancement does not move the advance rate, it does not move the equity either.

WHAT WOULD KILL IT

A guarantor declining to extend at the margin. This is exactly how the previous cycle ended: Lucent declined a final \$90 million credit extension and Winstar filed within weeks, owing roughly \$700 million. A backstop is an option the WRITER holds on its own reputation, and the first refusal reprices every enhanced landlord at once. Equally fatal in the other direction: TeraWulf or Applied Digital announcing an enhancement of their own, which closes the spread before it is captured.

WHAT IS ALREADY PRICED

Partially. Core Scientific rose about 6% on the day AMD's warrant deal was announced while AMD fell about 4%, so the market does price the announcement. What it does not obviously price is the DIFFERENCE between a backstopped lease and a bare one of the same length — both cohorts still trade on contracted megawatts and on bitcoin-era comparables.

Six firms are being paid to assemble the capital, and none of them has to be right about AI returns

THE FEE LAYER IS THE UNLEVERED SEAT

THE THESIS

The alternative managers named in the compute financing platforms earn on capital raised and deployed. Whether a GPU clears its residual in year six is somebody else's exposure. That is the only seat in this theme that does not require a view on AI demand, and it is available in listed form.

THE MECHANISM

From the structural sides: the loop only scales if the vendor's own balance sheet stops being the marginal lender, which is precisely what the third-party platforms exist to accomplish. Every dollar that moves from the vendor's contingent liabilities into a managed pool is a dollar of fee-bearing AUM created by the theme rather than by the market. The demand for it is not speculative — private markets hold roughly \$4.5 trillion of dry powder across credit, infrastructure, real estate and private equity, and data-centre lending is already being categorised as private credit.

THE NAMES

Blackstone (BX) is the table's representative, at over \$1.3 trillion of AUM; Apollo, BlackRock, Brookfield (BN/BAM), Goldman Sachs and KKR occupy the identical position and were kept out of the table only to stop it becoming a screen. Evidence in Exposure Evidence. The distinguishing variable between them is not AI exposure — it is what fraction of each firm's fee base a \$500B programme could plausibly move, which is large for a pure alternative manager and immaterial for a diversified bank.

WHAT WOULD HAVE TO BE TRUE

- The MOUs convert into committed vehicles with disclosed sizes. Right now the \$500B is a stated TARGET for mobilising third-party capital over time; nothing in the announcement is committed or deployed.

- Fees are earned on deployment rather than on commitments, which means the construction schedule has to hold. The same power and interconnection constraints that condition the vendors' guarantees condition the deployment of these pools.
- The credit continues to clear. The AI-leader credit basket has already moved from 74bp to nearly twice that in twelve months, and insurance demand at the long end halved between Q1 and late Q2. Widening is survivable for a lender; a closed primary market is not.

WHAT WOULD KILL IT

A year passing with no named vehicle, no disclosed size and no first close. A financing platform announced by press release and never funded is a marketing arrangement, and the fee line never appears. The second killer is simpler: if the vendor keeps writing the instruments itself because doing so is cheaper than paying an external manager, the pools never get their mandate.

WHAT IS ALREADY PRICED

Unclear, and that is honest. These are large diversified firms where an AI-infrastructure programme of any size is one line among many; nothing in this corpus sizes the fee opportunity against any of their existing fee bases. Treat this idea as a screen to be sized, not a conclusion.

When customers start asking for credit support, the second source stops competing on price and starts competing on equity

BALANCE SHEET BECAME THE MOAT

THE THESIS

Vendor financing converts a product market into a balance-sheet market, and that is a structural disadvantage for whoever has the smaller balance sheet regardless of how good the product is. The idea is not a short; it is that AMD's instrument MIX is a leading indicator of how much of its announced volume converts, and the market currently reads announced gigawatts as if the two vendors were bidding with the same currency.

THE MECHANISM

From the instrument menu: a residual value guaranty, a revenue floor, a capacity purchase commitment, a warrant and an equity investment differ in beneficiary, trigger and balance-sheet cost. AMD has consistently chosen the cheapest — warrants over its own shares, and a prospective equity commitment with no disclosed funding schedule — because the alternatives are arithmetically unavailable to it: a Piketon-sized cap is 156% of AMD's entire equity against \$5.1B of cash. The result is that AMD's support reaches DEMAND (it protects against volume that never arrives) but never reaches SCHEDULE, while its committed HBM4, packaging and substrate spend is placed against a delivery calendar set by hosts and utilities it holds no contract with.

THE NAMES

Advanced Micro Devices (AMD) is the direct case, evidenced in Exposure Evidence. Digital Realty Trust (DLR) is the same squeeze in a different industry — a disciplined underwriter competing for capital against counterparties financing 50-100bp tighter because somebody else's credit sits behind their tenant. The pair is useful because it shows the mechanism is about balance-sheet access, not about semiconductors.

WHAT WOULD HAVE TO BE TRUE

- Announced gigawatts keep diverging from contracted gigawatts. The Meta warrant says this on its face: full vesting is contingent on cumulative purchases reaching six gigawatt equivalent AND on share-price thresholds escalating to \$600, which is the structure you write when the volume is not yet a commitment.
- Customers keep asking for credit support as a condition of the socket, rather than choosing on price-performance alone.
- AMD does not find a third-party balance sheet to stand behind it. The compute financing platforms are, in principle, available to any vendor.

WHAT WOULD KILL IT

AMD writing a funded guaranty or a residual-value guaranty at scale and having it accepted without a rating action. That single event would prove the balance-sheet constraint is softer than the arithmetic suggests, and the idea dies. The milder killer: hyperscalers and labs standardising on capacity reservations with liquidated damages — an instrument that costs the vendor almost no rating capacity — which levels the playing field without anyone writing a guarantee.

WHAT IS ALREADY PRICED

The market prices AMD's announced gigawatts as share gain. This corpus contains no evidence that it prices the difference between a warrant and a guarantee, which is the entire content of the idea.

The event that reprices this whole theme is a rating-agency method, and methods are published before they bite

THE RE RATING COMES FROM A METHOD

THE THESIS

Every instrument in this landscape sits off balance sheet by accounting convention and on balance sheet by analytical adjustment. The gap between those two treatments is where the theme's risk actually lives, and it closes by a methodology decision rather than by a default. A researcher who tracks the agencies' published treatment of vendor guarantees has a leading indicator on every writer, every guaranteed landlord and every credit-enhanced borrower simultaneously.

THE MECHANISM

From the S&P row: the reported treatment of the Ohio instrument shows a guaranty can create a substantial debt-equivalent adjustment at the GUARANTOR even where lenders give the BORROWER no additional collateral value. That asymmetry is the whole finding. The vendor pays rating capacity for an enhancement the borrower's lenders never credit — so the instrument's cost is set by an agency method, and its benefit is set by a lending market that has not yet agreed to it. Widen the adjustment and the marginal instrument stops being written; narrow it and the volume accelerates.

THE NAMES

S&P Global (SPGI) and, by the same argument, Moody's. AMD as the writer with the least rating capacity to spare. CoreWeave (CRWV) as the borrower whose lenders already hold a roughly two-year uncontracted tail at a 70 percent advance against a contract book averaging about three years — the position that reprices first if the enhancement layer is recharacterised. Evidence in Exposure Evidence.

WHAT WOULD HAVE TO BE TRUE

- The adjustments are applied broadly rather than case by case. One reported treatment of one instrument is a data point, not a method.
- Release conditions stay opaque. The Piketon guaranties terminate on the tenant "achieving a satisfactory credit rating" and the threshold is redacted — an agency cannot give credit for a release it cannot see, and a published threshold would materially reduce the adjustment.
- Disclosure keeps improving enough to measure. The template already exists inside the disclosures: a stated maximum gross exposure with escrow against it.

WHAT WOULD KILL IT

Agencies concluding that milestone conditionality means the obligation does not attach, and therefore applying no debt-equivalent adjustment to a guaranty that only pays after ready-for-service. That is a defensible position, it is the drafting the largest instrument in the theme actually uses, and if it becomes the standard the trigger this idea depends on never fires.

WHAT IS ALREADY PRICED

The credit market has moved and the equity market has not: five-year protection on the largest writer jumped as much as 14 basis points in a single session to about 82bp while revenue and EBIT showed nothing. That divergence is the idea's evidence and also its warning — someone is already watching this, and they are in the CDS market rather than the equity market.

Summary

Vendor financing is an old industry structure reappearing at roughly four times its previous scale, and it has already re-rated one credit market while leaving every income statement untouched

FACTOR RESEARCH SUMMARY

SO WHAT

When a supplier underwrites its customers' purchases, its revenue stops being an independent measure of end demand and its balance sheet becomes a competitive weapon. The money lands with whoever RECEIVES the instrument — landlords, neoclouds, labs — and with whoever clips a fee on assembling the capital. The cost lands on whoever must WRITE it, and it is invisible until a rating method or a refusal to extend makes it visible.

THE THEME IN FOUR LINES

- 1. The mechanism.** Six distinguishable instruments — residual value guaranty, revenue floor, capacity purchase commitment, warrant, equity investment, take-or-pay — substitute a supplier's investment-grade signature for an unrateable buyer's, and are paid for in the buyer's equity or in a share of its rental revenue. Mechanism
- 2. The magnitudes.** A \$99B equity book built from ~\$7B in twelve months; a single capped guaranty at \$105B over 4.25 GW on 20-year leases; \$3.5B of earlier lease backstops with \$712M escrowed; a stated target of over \$500B of THIRD-PARTY capital; warrants over 30M shares at \$70 (IREN) and 30M at \$23.47 (Core Scientific); a \$1.73Bn hyperscaler backstop on a ~\$3.8Bn contracted book (Cipher).
- 3. The precedent.** McKinsey put nine suppliers' combined vendor-financing exposure at about \$25.6B at end-2000; within one year bad-loan ratios went 6.7% → 57% (Motorola), 2.6% → 60% (Lucent) and 25.5% → 80% (Nortel). The difference this time is utilisation, not structure: the fibre sat dark, the racks do not. Drivers and Horizon
- 4. The clock.** Prices weekly (CDS +14bp in a session to ~82), attaches from fiscal 2029 as buildings energise, resolves over 15-20 years. Nothing about this theme settles inside a normal holding period.

THE LANDSCAPE

COMPANY	TICKER	ROLE	DIRECTION	CONF.	MECHANISM IN ONE LINE
Advanced Micro Devices	AMD	incumbent	mixed	high	Pays for up to 14 GW of announced deployments in warrants because a Piketon-sized guaranty is 156% of its equity
IREN Limited	IREN	emerging	benefits	high	Issued a 5-year right over 30M shares at \$70 (up to \$2.1B) behind up to 5 GW of DSX-aligned deployment

COMPANY	TICKER	ROLE	DIRECTION	CONF.	MECHANISM IN ONE LINE
Core Scientific	CORZ	emerging	benefits	high	30M AMD warrants at \$23.47 against 529 MW of 15-year leases projected above \$14B
Cipher Mining	CIFR	emerging	benefits	high	Google backstops up to \$1.73Bn of Fluidstack's obligations on a 300 MW, 10-year, ~\$3.8Bn lease
CoreWeave	CRWV	adjacent	benefits	high	Underwritten buyer; lenders hold a ~2-year uncontracted tail at 70% advance on a ~3-year contract book
SB Energy	—	adjacent	benefits	high	Lessor receiving up to \$105B of capped land/power/shell credit support over ~4.25 GW
Nebius Group	NBIS	emerging	benefits	medium	First counterparty for the take-or-pay + revenue-share structure; \$5.0B of converts priced Aug-26
TeraWulf	WULF	emerging	benefits	medium	20-year, ~401 MW Anthropic lease with NO disclosed backstop — the control case
Blackstone	BX	adjacent	benefits	medium	One of six firms in the >\$500B compute financing platforms; fees on deployment, not on compute returns
Applied Digital	APLD	adjacent	mixed	medium	86% of contracted revenue in two tenants — the theme as a risk factor
S&P Global	SPGI	adjacent	mixed	medium	Its guarantee-adjustment method decides whether the layer is a footnote or debt
Digital Realty Trust	DLR	adjacent	hurt	low	Conventional colo funds at 60-70% LTV / 7-9% while AI-contracted operators finance 50-100bp tighter

Full reasoning per row in Exposure Evidence; the five emerging names in Emerging Players.

THE IDEAS

- **Buy the guarantee, not the gigawatt** — enhanced landlords (CIFR, CORZ) against bare ones (WULF, APLD); the spread is the unpriced value of the enhancement. Killed by a guarantor declining to extend, the way Lucent's refusal of a final \$90M draw ended Winstar.
- **The fee layer is the only unlevered seat** — BX and the five other named firms are paid on deployment, not on residuals. Killed by a year passing with no named vehicle and no first close.

- **Balance sheet became the moat** — AMD must pay in equity what the leader pays in cash, and DLR is the same squeeze in real estate. Killed by AMD writing a funded guaranty at scale without a rating action.
- **The re-rating comes from a method** — SPGI/Moody's treatment of guarantees is the leading indicator for every writer and every enhanced borrower at once. Killed by agencies deciding milestone conditionality means no attachment.

WHAT THIS RUN DID NOT SETTLE

The confidence grades are lopsided, and the direction of the lopsidedness flatters one side. All five high rows sit on the benefits side. That is not a judgement about which side is right; it is an artefact of what gets published. A warrant issued to a chip vendor appears in an 8-K with a strike price and a share count, while the cost borne by the vendor writing it, or by the disciplined operator outbid for capital, appears in nobody's filing. Read the table knowing that the beneficiaries are documented and the squeezed are inferred.

The single weakest row is Digital Realty. The cost-of-capital gap is documented; that DLR specifically suffers from it is not. It stands as a placeholder for the disciplined-underwriter cohort until someone checks its actual marginal funding cost against a credit-enhanced peer.

Three things a follow-up should chase. (i) Whether any lender has publicly raised an advance rate because of vendor support — the corpus contains an explicit claim that none has, and if that holds the enhancement is worth less to the lessor than idea one assumes. (ii) The undisclosed terms of the Nebius take-or-pay: size, floor level, revenue split. Without them the template's economics are unknown. (iii) The actual published S&P and Moody's methodology on vendor guarantees, rather than the reported treatment of one instrument — idea four rests entirely on that being a method rather than a data point.

What the run did not attempt. No position is sized, timed or priced anywhere in this bag. Every idea is a candidate with a falsifier; none is a recommendation. Historical 1998-2001 magnitudes come from two secondary sources registered for this run, not from primary filings, and are nominal dollars of their day.

FACTOR RESEARCH

NVDA Factor 09 Founder Led Unplanned Succession

Theme

The premium is attached to a person, and the person is not on the balance sheet

MECHANISM

A founder-CEO premium is real, measurable and durable — and it is the only large equity premium that is legally attached to an individual rather than to an asset, a contract or a patent. That is the whole mechanism, and it is why the theme has two faces.

STEP 1 — DECISION RIGHTS ARE ENGINEERED, NOT EARNED

The founder premium runs on a governance instrument, not on charisma. Super-voting stock decouples control from economics so completely that the board cannot fire the founder and, more importantly, cannot force him to prepare a replacement. Shopify's 2022 "founder share" is the clearest specimen: it hands Tobias Lutke, his family and affiliates 40% of the total voting power, and he keeps it even if his economic stake falls to 1.1% [1]. Meta runs the same arithmetic at scale — roughly 61% of the vote on about 13% of the economics [2]. Ten-to-one and twenty-to-one voting ratios are the standard denomination.

STEP 2 — THOSE RIGHTS BUY A LONGER HORIZON, WHICH IS WHAT ACTUALLY PAYS

Founder-CEO firms invest more in R&D and more in capex than successor-CEO firms, and the market has paid for it. An equal-weighted strategy long founder-CEO firms from 1993 to 2002 earned a benchmark-adjusted 8.3% a year, and 4.4% a year after controlling for firm characteristics, CEO characteristics and industry [3]. Over the decade to 2026 the gap is wider still: companies still run by their founders returned about 25% a year against the S&P 500's 14% [4].

STEP 3 — THE EXIT IS THE EVENT, AND IT IS PRICED IN ONE STEP

The premium accrues over a decade; it is repriced at a single announcement. The same structure that protects the founder from the board is what leaves the board without a bench, so the size of the step is set almost entirely by how much was planned in advance. The measured gap is large: founder-led names in that decade study averaged 15.4% a year, while companies whose founder-CEO had left the helm averaged 6.9% [4].

STEP 4 — THE DISCLOSURE IS WHERE THE RISK BECOMES INVESTABLE

This is governance research, not a personality read, because the mechanism is written down. SpaceX told IPO buyers its CEO was the "driving force behind our growth, innovation, and operational success" while, on the reporting, having no formal succession plan [5]. NVIDIA's own 10-K carries the generic form of the same clause: failure to ensure effective succession planning and smooth transitions "could hinder our strategic planning, execution, and long-term success" [6]. Two disclosures, two very different amounts of information — and the difference between them is the tradable part of this theme.

Two clocks: the premium compounds over a decade, the discount settles in a day

DRIVERS AND HORIZON

Nothing about this theme resolves smoothly, and that asymmetry is the whole timing problem. The premium is earned across a ten- to twenty-year tenure. The repricing happens on one press release. An investor is therefore not holding a slow-moving factor; they are holding a long accrual with a step function attached to an actuarial event nobody controls.

WHAT HAS ALREADY CHANGED, 2024-2026

Three years ago the founder-succession question in large-cap technology was hypothetical. It is now a live calendar.

- **A template was set.** Apple announced on 20 April 2026 that Tim Cook would become executive chairman and John Ternus would become CEO effective 1 September 2026 [1] — four months of notice, a named internal candidate of twenty-five years' standing, and a board that described the handover as the output of a thoughtful, long-term succession planning process [1]. That is the shape of a transition that costs a shareholder nothing but a paragraph.
- **A counter-template was set.** Intel's Pat Gelsinger "stepped down from the position of Intel CEO and from the company's board of directors" on 1 December 2024 [2] with no successor ready; the seat went to interim co-CEOs for three months. Lip-Bu Tan became the company's fourth permanent CEO in seven years.
- **Succession became a compensation problem.** Broadcom's answer was to buy time on contract: Hock Tan's extension carries a clause paying hundreds of millions in stock if the company reaches US\$120 billion in AI product sales by 2030 [3], with roughly US\$616.6 million on the table at the 300% payout level. The operator premium is now something boards explicitly price and term out.

WHAT PUSHES THE THEME HARDER

Founder cohort ageing is the base driver and it only runs one way. On top of it: index and proxy-adviser pressure on dual-class structures, sunset-clause activism, and a 2025-26 IPO cohort that re-armed the mechanism rather than retiring it. Each new founder-controlled listing adds another unplanned exit to the calendar a decade out.

WHAT SLOWS IT

A founder in good health with a credible internal bench neutralises most of the discount at zero cost, which is why the observable that matters is disclosure, not age. Contractual term-certainty (Broadcom to 2030) and a formally announced handover (Apple) both convert an open-ended risk into a dated one.

THE CLOCK, STATED

Three to ten years for the actuarial event at a typical large-cap founder-led name; **four to six months** between announcement and effect where a transition is planned; **one trading session** where it is not. A position taken on this theme is a position on which of those three clocks a given company is on — and the evidence that separates them is public and dated. The post-transition drag is not instant either: over the decade to 2026, companies whose founder-CEO had left averaged 6.9% a year against 15.4% for those still founder-led [4].

Sort by disclosure, not by admiration — the theme pays the companies that have dated the handover and charges the ones that have not

WHO WINS AND LOSES

This theme does not split cleanly into good companies and bad ones. It splits into companies where the operator premium has a stated exit path and companies where it does not. That is a screen anyone can run off public filings, which is the useful part.

THE CHARACTERISTICS THE THEME FAVOURS

- **A handover that is announced, dated and staffed from inside.** The discount is paid once, at announcement, against a named successor with a tenure the market can read. After that the buyer owns the franchise without the key-man clause.
- **An operator premium that is term-certain by contract.** Where a board has bought a specific number of years with a performance-linked extension, the risk has a maturity date rather than an actuarial distribution.
- **Professional-management turnarounds of founder-impaired assets.** When activists force a founder out of an operationally broken company, the founder discount is the entry price and the re-rating comes from ordinary competence. Marvell is the worked example: "In April 2016, CEO Sehat Sutardja and President Weili Dai were ousted from their posts after activist investor Starboard Value fund took a roughly seven percent stake in the company" [1], and a professional CEO rebuilt the business around data-infrastructure silicon.
- **Deep, visible benches.** A company whose divisional leaders are individually known to the market carries a smaller step function, because the replacement is already priced.

THE CHARACTERISTICS THE THEME SQUEEZES

- **Super-voting control combined with no named successor.** This is the sharp end. The same instrument that guarantees the founder's decision rights removes the board's ability to prepare for their absence — Dylan Field controls 73.6% of Figma's voting power on an economic stake near 8.9% [2]; Airbnb's three founders "control roughly 80% of the voting power while owning about 27% of the economic shares" [3]. In each case a board majority cannot compel a transition plan it is not given.
- **A franchise that is a personal relationship.** Government, defence and regulated-procurement businesses where the counterparty relationship is held by a named individual lose more of the franchise on exit than a product company does.
- **Divided founder attention.** Where the same operator runs several ventures, the key-man exposure is already being drawn down before any succession event — and the company that discloses no formal succession plan alongside it is compounding the two [4].
- **A recent IPO with the mechanism re-armed.** Lockup expiry, a dual-class sunset date and the founder's first real liquidity often land inside the same twenty-four months, and thin analyst coverage means none of it is priced.

WHERE IT GENUINELY CUTS BOTH WAYS

For most founder-led names the honest answer is mixed, and saying so is not a hedge. The founder's decision rights are the *reason* for the premium and the *source* of the succession risk; you cannot own one without the other. What separates two otherwise identical founder-led companies is whether a board has been permitted to write the plan down — which is why a transition executed on a published timetable reads as a positive and not a loss [5].

Landscape

Exposed Companies

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Tesla	TSLA	incumbent	Product roadmap, capital allocation and the equity narrative all run through one founder-CEO who also runs SpaceX, xAI and X; no formal succession plan has been published and the 10-K names him personally as a dependency.	mixed	high	100%	Divided attention is the distinguishing feature: the key-man exposure is already being drawn down before any succession event.
Meta Platforms	META	incumbent	Ten-vote Class B gives Zuckerberg ~61% of the vote on ~13% of the economics, so no board majority can compel a transition plan it is not offered.	mixed	high	100%	Controlled-company structure; the premium and the succession block are the same instrument.
Palantir	PLTR	incumbent	Class F stock plus the Founder Voting Agreement lets the founders control up to 49.999999% of total voting power, over a government/defense franchise whose counterparty relationships are personally held by Karp.	mixed	high	100%	Relationship-held franchise: exit loses more than a product company would. Proxy discloses the voting structure but no named successor.

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Figma	FIG	emerging	Listed July 2025 with the mechanism re-armed: Field controls 73.6% of the vote on ~8.9% of the economics via 15-vote Class B plus an irrevocable proxy over co-founder Wallace's shares.	mixed	high	100%	Widest voting/economic wedge in the set. Lockup, sunset and first founder liquidity land inside the same 24 months.
Shopify	SHOP	incumbent	A non-transferable founder share fixes Lutke at 40% of total voting power for as long as he serves in any capacity, even if his economic stake falls to 1.1%.	mixed	high	100%	The only structure in the set where control is explicitly decoupled from ownership by design rather than by ratio.
Airbnb	ABNB	incumbent	Three founders hold roughly 80% of the voting power on about 27% of the economics through 20-vote Class B; Chesky alone commands about 32% of the vote.	mixed	medium	100%	Highest vote multiple in the set (20:1). Link documented in a published ownership breakdown rather than read off the filing.
Broadcom	AVGO	incumbent	Operator premium without a founder: the board extended Hock Tan to 2030 and tied a stock award worth roughly US\$616.6m at the 300% level to reaching US\$120bn of AI product sales by that year.	mixed	high	100%	The worked example of pricing and terming out an operator premium instead of planning around it.

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Apple	AAPL	adjacent	Executed the transition: Cook to executive chairman and Ternus to CEO effective 1 September 2026, announced four months ahead from a 25-year internal candidate after a stated long-term succession process.	benefits	high	100%	The live template for what a planned founder-era handover costs a shareholder. Cook is a successor-CEO, not the founder, which is part of why it worked.
Marvell Technology	MRVL	adjacent	Founder-impaired asset turned round by professional management: Sutardja and Dai were ousted in April 2016 after Starboard took roughly a 7% stake, and Matt Murphy rebuilt the company around data-infrastructure silicon.	benefits	high	100%	The founder discount as an entry price rather than a risk. Direction is benefits for the POST-founder owner, not for the founders.
Intel	INTC	adjacent	Serial succession failure with no founder premium left to lose: Gelsinger stepped down as CEO and from the board on 1 December 2024 with no successor ready, three months of interim co-CEOs followed, and Lip-Bu Tan became the fourth permanent CEO in seven years.	hurt	high	100%	The counter-template. Shows the discount is a governance outcome, not a founder-specific one.

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Arm Holdings	ARM	adjacent	Segars-to-Haas succession executed cleanly in 2022, then partly undone: in April 2026 Haas took an expanded SoftBank International role while remaining Arm CEO, re-concentrating key-man exposure in one person across two entities.	mixed	medium	100%	Shows a completed transition can re-accumulate the same risk. Sizing of the re-concentration is not established here.
CoreWeave	CRWV	emerging	Went public in 2025 with founder control preserved through a Class B share class carrying stronger voting rights than the Class A sold to outside investors; the three 2017 founders remain the key control bloc of an AI-infrastructure business barely a year listed.	mixed	medium	100%	Thinly covered relative to the megacaps. Exact Class B vote multiple not established in the sources gathered here.

Twelve names, one question each: is the operator premium dated, or open-ended?

EXPOSURE EVIDENCE

One paragraph per row of Exposed_Companies, in table order. Each says what the disclosed mechanism is and whether the exposure link is documented or inferred. Where it is inferred, the row's Confidence is graded down to match.

TESLA (TSLA) – INCUMBENT, MIXED, HIGH

The clearest disclosed key-man dependency in listed technology, because Tesla writes it as its own risk-factor heading: "We are highly dependent on the services of Elon Musk, Technoking of Tesla and our Chief Executive Officer", immediately followed by the admission that "Although Mr. Musk spends significant time with Tesla and is highly active in our management, he does not devote his full time and attention to Tesla", and a list naming SpaceX, xAI, Neuralink and The Boring Company [1]. The second half of the exposure is the absence: on the reporting around the SpaceX listing, the Musk operating group "doesn't appear to have a formal succession plan

— something governance experts say is critical to mitigating key-person risk" [2]. Documented on both halves, which is why this is graded high. Direction is mixed because the same concentration that creates the risk is what has produced the roadmap the equity is priced on.

META PLATFORMS (META) — INCUMBENT, MIXED, HIGH

Meta is the scale case for control decoupled from economics: roughly 61% of the voting power on about 13% of the shares, through ten-vote Class B [3]; Meta is named among the large listed companies running two or more share classes [14]. The mechanism that matters for this theme is not the premium but the block: a board holding a majority of the economics cannot compel a succession plan from a holder of a majority of the votes.

Documented, not inferred; the percentage comes from a published 2026 ownership breakdown rather than the proxy itself, which is the one soft spot in the row.

PALANTIR (PLTR) — INCUMBENT, MIXED, HIGH

The strongest filing language in the set. Palantir's proxy states that the Class F common stock, taken with the Founder Voting Agreement and the Designated Founders' Excluded Shares, "will give our Founders the ability to control up to 49.999999% of the total voting power of our capital stock" [4] — a structure engineered to stop exactly one vote short of absolute control. Layered on top is a franchise risk the filings do not quantify: the government and defence relationships are personally held, and no successor is named. The structure is documented; the personal-franchise half is a reasonable inference from the business model and is the reason the row is not stronger than high on link but mixed on direction.

FIGMA (FIG) — EMERGING, MIXED, HIGH

The widest voting-to-economics wedge found in this run. "CEO and co-founder Dylan Field controls 73.6% of the company's voting power through a dual-class share structure and proxy agreements — despite holding a much smaller economic stake" [5], the mechanism being 15-vote Class B plus an irrevocable proxy over co-founder Evan Wallace's shares, against economic ownership of roughly 8.9% ahead of the July 2025 IPO. Newly listed, thinly covered, and the structure was written into the IPO rather than sunset at it.

SHOPIFY (SHOP) — INCUMBENT, MIXED, HIGH

The most explicit decoupling instrument of the twelve. Shareholders "voted to give Mr. Lutke, his family and his affiliates 40 per cent of the total voting power at Shopify" [6], and the founder share is non-transferable and survives dilution "even if his equity stake is diluted to as low as 1.1 per cent". Unlike a ratio-based dual class, this is a fixed percentage of control granted for as long as he serves in any capacity — control that no amount of issuance can erode. Documented in a wire report of the shareholder vote.

AIRBNB (ABNB) — INCUMBENT, MIXED, MEDIUM

"The three founders together control roughly 80% of the voting power while owning about 27% of the economic shares" [7], on twenty-vote Class B — the highest vote multiple in the set — with Chesky alone at about 32% of the vote. Graded medium rather than high only because the figures come from a published ownership breakdown rather than from the proxy; the structure itself is not in doubt, the precise 2026 percentages are.

BROADCOM (AVGO) — INCUMBENT, MIXED, HIGH

Included deliberately as the non-founder case, because it shows boards now treat the operator premium as a priceable, terminable asset. Hock Tan's extension carries a clause worth hundreds of millions in stock against US\$120bn of AI product sales by 2030, and "If the sales reach US\$120 billion, Tan is poised for 300% of the payout, or about US\$616.6 million at current levels" [8]. There is no super-voting structure here; the dependency is operational rather than constitutional, and the answer the board chose was a dated contract rather than a bench. That makes it the best available comparison for what terming out an irreplaceable operator costs.

APPLE (AAPL) — ADJACENT, BENEFITS, HIGH

The live template, executed ten days before this run. "Tim Cook will become executive chairman of Apple's board of directors and John Ternus, senior vice president of Hardware Engineering, will become Apple's next chief executive officer effective on September 1, 2026" [9] — announced 20 April 2026, four months of notice, a named internal candidate on the product side, and a board describing a thoughtful, long-term succession planning process. adjacent rather than incumbent because Apple is no longer contesting the founder premium; it is demonstrating the cost of handing one on. Note the caveat that keeps this honest: Cook was himself a successor, so this is a second-generation handover, not a founder's.

MARVELL TECHNOLOGY (MRVL) — ADJACENT, BENEFITS, HIGH

The founder discount as an entry price. "In April 2016, CEO Sehat Sutardja and President Weili Dai were ousted from their posts after activist investor Starboard Value fund took a roughly seven percent stake in the company" [10], and Matt Murphy was appointed president and CEO that July, rebuilding the company around data-infrastructure silicon. Direction is benefits for the owner who bought the post-founder asset — not for the founders, and not a claim that ousting founders is generally value-creating.

INTEL (INTC) — ADJACENT, HURT, HIGH

The counter-template, and the only unambiguously negative row. "On December 1, 2024, Gelsinger stepped down from the position of Intel CEO and from the company's board of directors" [11] with no successor in place; interim co-CEOs held the seat for three months before Lip-Bu Tan was appointed, his fourth-permanent-CEO-in-seven-years status being the actual finding. Intel carries no founder premium to lose, which is the point: the succession discount is a governance outcome, not a founder-specific one, and a company can pay it repeatedly.

ARM HOLDINGS (ARM) — ADJACENT, MIXED, MEDIUM

"Rene Haas succeeds Simon Segars as Arm CEO and will join the Arm board of directors" [12] — a clean, planned transition in 2022 from a thirty-year insider to a long-tenured executive. What makes it interesting rather than settled is what happened next: in April 2026 Haas took an expanded SoftBank International role while remaining Arm CEO, re-concentrating key-man exposure in one person across two entities. Graded medium because the 2022 succession is documented here but the 2026 re-concentration is reported rather than quantified, and its effect on Arm specifically is not established.

COREWEAVE (CRWV) — EMERGING, MIXED, MEDIUM

"CoreWeave was founded in 2017 by Michael Intrator, Brian Venturo, and Brannin McBee" [13] and they "remain the key control bloc" after the 2025 listing, with Class A holders getting one vote and Class B holders "stronger control". The exposure is real — an AI-infrastructure business barely a year public, still founder-run, with the control mechanism carried through the IPO — but the exact Class B vote multiple is not established in the sources gathered here, which is why the row is medium and not high. That gap is the honest state of the fringe of this theme.

WHAT IS INFERRED RATHER THAN DOCUMENTED

Three things in this section rest on inference and should be read that way: Palantir's personal-franchise risk (the voting structure is filed, the relationship concentration is not), Arm's 2026 re-concentration (reported, unsized), and CoreWeave's exact voting ratio (control is documented, the multiple is not). Everything else in the twelve rows is quoted from a filing, a company release or a wire report.

Two emerging rows, and both of them re-armed the mechanism at the IPO rather than retiring it

EMERGING PLAYERS

The fringe search for this theme is not "who is disrupting founder control" — it is the 2025 listing cohort, because that is where a new decade of unplanned succession is being written into charters right now, in front of thin analyst coverage. Two names earned an emerging role.

FIGMA (FIG) — LISTED JULY 2025

Figma qualifies as emerging on both tests: barely a year public, and attacking the theme with a structure rather than a product. "CEO and co-founder Dylan Field controls 73.6% of the company's voting power through a dual-class share structure and proxy agreements — despite holding a much smaller economic stake" [1]; the mechanics are 15 votes per Class B share plus an irrevocable proxy over co-founder Evan Wallace's Class B, against economic ownership of roughly 8.9% ahead of the IPO. The novelty is the proxy layer — Field votes a second founder's stock, so the control bloc survives the co-founder's own exit, not just his dilution. Figma is a NYSE controlled company while retaining a majority-independent board and a lead independent director, which is the governance compromise this cohort has settled on: independent oversight of everything except who runs the company.

What the edge would be if it works: an eight-year-old design platform going through a product transition into AI-assisted design needs the freedom to break its own pricing model, and a founder with 73.6% of the vote can do that against a quarter of unhappy shareholders. What makes it an emerging risk rather than an established one: nobody has yet observed what a Field exit does to a company where the control bloc is held together by a proxy agreement rather than by shares.

COREWEAVE (CRWV) — LISTED 2025

"CoreWeave was founded in 2017 by Michael Intrator, Brian Venturo, and Brannin McBee" and they "remain the key control bloc" [2] after the 2025 listing, with Class A holders getting one vote and Class B holders stronger control. The mechanical edge is specific to this business: CoreWeave's franchise is a set of multi-year, take-or-pay GPU-capacity contracts negotiated personally by a founding team out of a commodity-trading background, and the contract book was built before the incumbents were bidding for the same capacity. That is a relationship-held franchise on a compressed timeline — the same shape as Palantir's, but at a fraction of the coverage.

What the edge would be if it works: a founder-run infrastructure business can take counterparty and duration risk that a professionally managed one would not sign. What the research gap is: the exact Class B vote multiple is not established in the sources gathered here, and no succession disclosure was found at all. The row is graded medium for that reason, and the gap is the opportunity — an unpriced governance question on a name barely a year old.

WHAT THE FRINGE SEARCH DID NOT FIND

No newly listed company was found that is attacking this theme by doing the opposite — going public with a founder-led business and a pre-committed, dated succession or sunset mechanism. Cerebras, the other obvious 2026 candidate, had not completed a listing at the time of this run and so does not earn a row. If a founder-controlled IPO with a hard sunset clause exists in this cohort, this run did not surface it, and finding one would be the single most useful follow-up: it would price the difference between the two halves of this theme directly.

Ideas

So what: the theme is investable as a disclosure screen, and the screen is binary — has this board dated the handover, or has it not?

BUY THE DATED HANDOVER NOT THE OPEN ENDED ONE

The thesis. The founder premium and the founder discount are the same instrument read at two different times, so the position is not long or short founders — it is long the companies that have converted an actuarial event into a calendar entry, and cautious on the ones where super-voting control has removed the board's ability to make one. This is a screen anyone can run off public filings, which is what makes it a position rather than an opinion.

Why the theme produces it. Mechanism sets out the two-step: decision rights are engineered through super-voting stock, and those rights buy a longer horizon that the market pays for — but the same instrument that protects the founder from the board is what leaves the board without a bench. Who Wins and Loses turns that into the sorting rule: the theme pays companies that have dated the handover and charges the ones that have not.

The names. Apple is the executed template — Tim Cook to executive chairman and John Ternus to CEO effective 1 September 2026 [1], a named internal candidate of twenty-five years' standing and four months of notice. Against it sit the open-ended cases: Dylan Field controls 73.6% of Figma's voting power on a much smaller economic stake [2], and Airbnb's three founders "control roughly 80% of the voting power while owning about 27% of the economic shares" [3]. In both, a board majority cannot compel a transition plan it is not given.

What would have to be true. That the discount is paid once, at announcement, rather than accruing quietly beforehand — otherwise the dated-handover cohort has already given up the premium by the time it is screenable.

What would kill it. A dated, internally-staffed handover that still produces a large negative step would break the whole sorting rule. So would the reverse: an undated founder-controlled name compounding through a decade with no repricing, which would say the market never charged for the missing plan in the first place.

So what: the 2025-26 listings did not retire the founder-control mechanism, they re-armed it — and an IPO prices the premium while the succession risk it creates sits a decade out, unpriced

THE 2025 COHORT RE ARMED THE CLOCK NOBODY IS PRICING

The thesis. Every new founder-controlled listing adds an undated exit to a calendar ten to twenty years forward. At listing the market pays for the founder; it does not charge for the absence of a plan, because the absence has no near-term cash consequence. That is a clock mismatch, and it is the fringe of this theme rather than its centre.

Why the theme produces it. Drivers and Horizon names a 2025-26 IPO cohort that re-armed the mechanism rather than retiring it as one of the forces pushing the theme harder, and Emerging Players works both fringe rows in detail — its finding is that both re-armed the mechanism at the IPO rather than retiring it.

The names. Figma listed in July 2025 with Dylan Field controlling 73.6% of voting power "through a dual-class share structure and proxy agreements — despite holding a much smaller economic stake" [1]. CoreWeave, "founded in 2017 by Michael Intrator, Brian Venturo, and Brannin McBee" [2], went public in 2025 with founder control intact. Dual-class is not a fringe structure either — Alphabet, Meta, Palantir, Strategy and Berkshire Hathaway are among the companies carrying two or more share classes [3].

What would have to be true. That the risk is genuinely absent from the listing price rather than merely invisible to us — and that the cohort is large enough for the aggregate calendar to matter rather than being a handful of names.

What would kill it. Sunset clauses. If index and proxy-adviser pressure pushes the 2025-26 cohort into time-based or ownership-based sunsets on super-voting stock, the clock defuses itself and the mechanism retires without any transition being announced. That is an observable change in the filings, not a judgement call, so this idea has a clean off-switch.

So what: when an activist forces a founder out of an operationally broken company, the discount is what you pay and ordinary competence is what re-rates it — but only if the asset was fixable

THE FOUNDER DISCOUNT IS THE ENTRY PRICE

The thesis. There is a repeatable setup on the losing side of this theme: a founder-impaired asset where the operator premium has inverted into an operator discount, and an activist supplies the catalyst the governance structure otherwise prevents. The re-rating does not require a visionary successor; it requires an ordinary one.

Why the theme produces it. Who Wins and Loses names professional-management turnarounds of founder-impaired assets as one of the characteristics the theme favours, and gives the worked example.

The names. Marvell is the case that worked: "In April 2016, CEO Sehat Sutardja and President Weili Dai were ousted from their posts after activist investor Starboard Value fund took a roughly seven percent stake in the company" [1], and a professional CEO rebuilt it around data-infrastructure silicon. Intel is the case that did not: Pat Gelsinger "stepped down from the position of Intel CEO and from the company's board of directors" on 1 December 2024 [2] with no successor ready, the seat went to interim co-CEOs for three months, and Lip-Bu Tan became the fourth permanent CEO in seven years. Same trigger, opposite outcome.

What would have to be true. That the asset is operationally impaired rather than structurally impaired. Marvell's problem was management; Intel's was a process roadmap and a capital base, which no CEO change repairs on an investor's horizon.

What would kill it. The distinguishing test is whether the incoming CEO is given a mandate to NARROW. Marvell's was. If a new CEO arrives promising to defend every business the founder started, the discount is telling the truth and there is nothing to buy. Intel is the standing evidence that this setup fails at least as often as it works, which is why it is a candidate and not a rule.

So what: boards have started buying the operator premium on contract, which turns an undated actuarial risk into a term-certain one — and makes it comparable across companies for the first time

THE OPERATOR PREMIUM NOW HAS A MATURITY DATE

The thesis. The newest development in this theme is not a succession; it is a pricing convention. Where a board has paid explicitly to retain an operator for a stated number of years against stated targets, the key-man risk acquires a maturity date. That is a different security from an open-ended founder premium, and it should not trade at the same discount.

Why the theme produces it. Drivers and Horizon records succession becoming a compensation problem as one of the three things that changed in 2024-2026, alongside Apple's template and Intel's counter-template.

The names. Broadcom is the specimen: "Broadcom CEO Hock Tan has a clause in his contract extension that would bring hundreds of millions of dollars in stock compensation if the company reaches US\$120 billion in AI product sales by 2030" [1], roughly US\$616.6 million at the 300% payout level. Broadcom carries an operator premium with no founder attached, which is precisely why it can be termed out by contract — and it is the one company in this table that appears in five of this workbook's seven researched factors.

What would have to be true. That the market currently prices key-man risk as undated for these names, so converting it to a dated contract is an unpriced improvement. The premium being real is already established: an equal-weighted strategy long founder-CEO firms from 1993 to 2002 earned a benchmark-adjusted 8.3% a year [2].

What would kill it. If these packages do not actually retain — the operator leaves anyway, or the target is missed and the contract is quietly re-papered on easier terms — then the maturity date was never real and the instrument is just compensation. One re-papering would be enough to falsify it.

Summary

So what: this theme is investable as a disclosure screen — dated handover versus open-ended control — and NVIDIA sits on the open-ended side

FACTOR RESEARCH SUMMARY

The founder premium is the only large equity premium legally attached to a person rather than to an asset, so it screens off public filings: the theme pays the companies that have dated the handover and charges the ones whose super-voting control has removed the board's ability to make one.

The theme. Mechanism establishes that the founder premium runs on super-voting stock rather than on charisma, and that the same instrument which protects the founder from the board is what leaves the board without a bench. Drivers and Horizon sets the two clocks: the premium accrues over a decade, the discount settles in a day, and 2024-26 turned the question from hypothetical into a calendar with a template (Apple), a counter-template (Intel) and a pricing convention (Broadcom). Who Wins and Loses converts that into the sorting rule the ideas are built on.

The landscape. Twelve companies, in Exposed Companies with a cited paragraph each in Exposure Evidence: seven incumbents carrying live founder control (TSLA, META, PLTR, SHOP, ABNB, plus AVGO as the operator-premium-without-a-founder case and FIG as a 2025 listing), three adjacent outcome cases that show what the transition is worth (AAPL executed it, MRVL turned a founder-impaired asset around, INTC failed serially), and two emerging rows that re-armed the mechanism at IPO (FIG, CRWV).

The ideas. Four, in the Ideas section. Buy_The_Dated_Handover_Not_The_Open_Ended_One is the core screen. The_Founder_Discount_Is_The_Entry_Price takes the losing side, with Intel standing as the evidence that the setup fails at least as often as it works. The_Operator_Premium_Now_Has_A_Maturity_Date reads Broadcom's contract as a new instrument. The_2025_Cohort_Re_Armed_The_Clock_Nobody_Is_Pricing is the fringe idea, seeded by the two emerging rows.

WHAT THIS RUN DID NOT SETTLE

- **No quantified step size for an UNPLANNED exit at mega-cap scale.** Every magnitude here is either a long-run return differential or a planned transition. The theme's central claim — that an unplanned exit reprices in one step — has no clean event study behind it in this corpus, and Intel is a succession failure rather than a founder exit, so it does not supply one either. That is the most valuable unfinished work.
- **Seven of twelve rows are mixed.** That is honest rather than evasive — a live founder-controlled company genuinely carries both faces of the theme at once — but it means the table sorts weakly, and the sorting has to be done on the disclosure test rather than on the direction column.
- **The evidence distribution is, unusually, not lopsided.** Nine of twelve rows are high confidence, the single hurt row (Intel) is one of them, and both benefits rows are too. The three medium rows are all mixed. Unlike several sibling runs in this workbook, the bearish side here is as well evidenced as the bullish one.

- **Ideas cite the bag's own validated quotes.** No new sources were registered in this step; every [N] in the Ideas pods reuses a resource and a verbatim quote already accepted by an earlier write in this bag.

FACTOR RESEARCH

NVDA Factor 10 Earnings Quality Investment Book

Theme

The so-what

MECHANISM

An accounting rule change made in 2016 and effective from fiscal 2019 moved unrealised gains on equity holdings out of the balance sheet and into the income statement. For any operating company that also runs a large venture book, that single line now decides how much of reported net income is the business and how much is the portfolio. In NVIDIA's case it is roughly a sixth of trailing net income; the disclosure is complete, but the comparability is not.

The causal chain, step by step

1. Equity marks were pushed through earnings. Before ASU 2016-01 (ASC 321), an operating company holding equity in another company parked unrealised value changes in other comprehensive income, where they touched book value but never EPS. Debt securities still work that way, and the contrast is visible inside a single filing: available-for-sale debt is carried "at fair value with the related unrealized gains and losses included in accumulated other comprehensive income (loss), a component of shareholders' equity" [1]. Equity is not. NVIDIA's own accounting policy states that for its privately-held stakes, "All gains and losses on these investments, realized and unrealized, are recognized in Other income, net on our Consolidated Statements of Income" [2]. Same balance sheet, two opposite routes to EPS, decided entirely by instrument type.

2. Private stakes get marked on funding rounds, not on trading. Non-marketable holdings sit under the measurement alternative: "measured at cost minus impairment, if any, and are adjusted for changes resulting from observable price changes in orderly transactions for an identical or similar investment in the same issuer" [2]. There is no market price. A gain is recognised when somebody else prices a round. That makes the gains lumpy, backward-looking and — this is the part that matters for a chip vendor that invests in its own customers — timed by an event the vendor can help bring about.

3. At scale the line stops being a rounding error. NVIDIA's equity portfolio reached roughly \$99B as of 26 July 2026, split "\$42.783 billion in marketable securities, \$47.898 billion in non-marketable securities, and \$3.3 billion in infrastructure financing investments accounted for under the equity method" [3]. Against that, "Nvidia recorded net gains from equity securities of \$7.771 billion in the second quarter and \$23.707 billion in the first half of fiscal 2027 within other income" [4] — of which about \$12.5B was public-market marks and about \$7.5B was write-ups on private stakes. \$23.7B of non-operating gain against a first-half GAAP profit base of that order is not a footnote; it is a sixth of the number the trailing P/E divides by.

4. The tell is when GAAP EPS exceeds non-GAAP EPS. Normally the adjusted number is the bigger one, because adjustments strip charges. When the marks run hot, the sign flips: one analysis of the July-2026 quarter notes the reversal directly — "That is backwards: non-GAAP normally exceeds GAAP because it strips charges" [5]. That inversion is the single cheapest screen for this theme, and it works across companies without opening a note.

5. The issuer's own reconciliation is the map back. NVIDIA's non-GAAP presentation explicitly adjusts "to exclude stock-based compensation expense, acquisition-related and other costs, other, gains/losses from non-marketable and publicly-held equity securities, net, interest expense related to amortization of debt discount" [6]. The company is telling readers the marks are not operating earnings. The distortion is therefore a comparability problem, not a disclosure failure — and any multiple taken off GAAP net income rather than operating income silently capitalises a venture portfolio at the operating business's multiple.

Why it is a measurement force and not an accusation

Nothing here requires a company to have done anything wrong. The rule applies mechanically; the disclosure in NVIDIA's filings is complete down to the lock-up balances and the equity-method loss exposure. What the rule does is destroy the comparability of net income across two companies that run identical operations and different venture books — and destroy it again over time within one company, because a mark taken in one period is unavailable as a source of the same gain in the next. The distortion is in the denominator of every ratio built on net income: P/E, ROE, net margin, payout.

The so-what

DRIVERS AND HORIZON

The distortion is currently being *created* at maximum speed — corporate venture books are growing faster than the operating businesses that own them — and it unwinds on a two-to-four-year clock, not a quarterly one, because private marks only move when somebody else prices a round. The dangerous moment is not the mark; it is the first period in which the mark does not repeat.

What has already changed

Three years ago this was not a theme at NVIDIA at all. "As of July 26, the chip giant's total equity investment value reached \$99 billion, compared with roughly \$7 billion a year earlier and just \$2.2 billion two years ago" [1] — a 45x move in two years, against an operating business that roughly tripled over the same span. The private half went from \$3.799B to \$47.898B in twelve months. A line item that was 1-2% of the balance sheet became a third of it, and the income-statement consequence followed one year behind the balance-sheet one.

The accounting itself did not change — ASU 2016-01 has been effective since fiscal 2019, and NVIDIA's FY2019 non-GAAP reconciliation already stripped "gains from non-affiliated investments" [2]. What changed is the size of the number the rule is applied to. This is a theme that was always technically live and only recently became material.

What makes it stronger

- **Committed capital still to deploy.** NVIDIA carried \$25B of unexecuted equity-investment commitments at 26 July 2026, roughly \$18B of it earmarked for the remainder of the fiscal year [1]. Money committed today is marks recognised in 2027-2028.
- **Private AI valuations still rising round-on-round.** Under the measurement alternative a gain exists only when an observable transaction occurs, so a fundraising market that keeps repricing upward manufactures reported earnings on a schedule set by other people's cap tables.
- **Vendor-financing structures that create their own observable prices.** When the seller of the compute is also an investor in the buyer, the round that marks the stake up is one the seller helped fund.

What makes it weaker

- **Lock-ups and liquidity.** \$36.9B of NVIDIA's holdings "remained subject to short-term lock-up restrictions on the ability to sell" [3]. A mark you cannot monetise is an earnings input and not a cash one, and the gap shows up as reported EPS running ahead of free cash flow.
- **Symmetry.** The same rule books the downside. A flat funding market produces zero gain, which is a large negative *change* in reported earnings even with the operating business untouched.
- **The market learning to look through it.** Where the issuer publishes a reconciliation and analysts adopt it, the distortion stops moving the price and becomes a data-cleaning chore.

The clock

Two to four years, in three stages. Stage one (now, roughly 2025-2027): deployment and mark-up, reported net income flattered, trailing multiples look cheaper than the operating business is. Stage two: the funding market flattens, marks stop recurring, and year-on-year EPS growth decelerates for a reason that has nothing to do with the operating business — this is the stage that reprices the multiple, and one workbook source frames the bear case in exactly those terms: "the non-operating gains reverse and reveal a lower-quality earnings base, multiple compresses to ~15x" [4]. Stage three: impairments, if the underlying private companies disappoint, which under the measurement alternative arrive late and in a lump.

Trailing multiples are the surface where this shows up first. One workbook source computes NVIDIA on trailing numbers as "The price-to-earnings (P/E) ratio sits around 28 times, while the price-to-sales (P/S) ratio is approximately 18 times" [5] — a trailing P/E whose E contains the marks. Strip them and the same price is a materially higher multiple of operating earnings. That arithmetic is the whole theme in one line, and it applies to every company in the next section.

The so-what

WHO WINS AND LOSES

Four characteristics decide who this theme moves, and none of them is an industry. Screen on **(a) the size of the equity book relative to operating earnings, (b) whether the holdings are private or listed, (c) whether the issuer publishes a reconciliation that strips the marks, and (d) whether the market has already adopted**

that reconciliation. The opportunity is concentrated where (a) is large, (c) exists, and (d) has not happened yet.

Structurally favoured

Companies whose venture book is understood to BE the business. Where marks are the product, no comparability is destroyed, because nobody was valuing the company on net income to begin with. These names are valued on NAV or on a discount to it, and the income statement is treated as an accounting artefact. They get the upside of a rising private market with none of the earnings-quality penalty, and they are the cleanest way to own the force directly.

Companies whose operating earnings are large enough to swamp the book. A venture portfolio worth 5% of market cap creates noise, not distortion. These names are structurally immune and are the correct comparison set when you want to price the operating business.

Companies that pre-emptively publish a clean line and get it adopted. The single most valuable act a management team can perform here is to reconcile the marks out and keep doing it every quarter until the sell-side model uses the reconciled number. NVIDIA does exactly this — its non-GAAP presentation strips "gains/losses from non-marketable and publicly-held equity securities, net" [1], and had the equivalent line as far back as FY2019 [2]. The reconciliation is free to produce and it permanently caps how far the distortion can travel.

The buyers of the marked-up assets' output. Anyone selling into the venture-funded companies benefits from the funding rounds themselves, without carrying the marks. Broadcom's CFO frames its own participation in exactly those terms: "It's a great investment for us when you think about it, that every gigawatt of compute they deploy, they could achieve \$30 billion of ARR, annual revenue per gigawatt" [3].

Structurally squeezed

Operating companies where the book is large but the identity is not "investor". This is the squeeze, and it is where NVIDIA sits. The market prices the company as a chip business on a trailing multiple; the trailing E contains a venture portfolio's marks; so the multiple printed on every screen is wrong in the flattering direction, and it corrects the moment the marks stop recurring. Alphabet and Amazon carry equity investment values "exceeding \$100 billion" [4] and face the same arithmetic, with the important difference that both have been swinging on marks for years and have trained their analyst base to adjust.

Anyone whose stakes are concentrated in their own customers. The correlation is the problem, not the size. When the same demand shock that cuts your revenue also cuts the value of the stakes, the two hits arrive in the same quarter and the reported decline is larger than the operating decline. The diversifying investor gets noise; the strategic investor in its own customer base gets amplification.

Companies with marks but no non-GAAP line, or a line nobody uses. Thinly covered names, recent listings, and non-US issuers whose adjusted measure is not in the consensus database. Reported EPS is the only number in the screen, so the distortion passes straight into the multiple unchallenged. This is the fringe STEP 2 should search hardest.

Index and screen-driven capital. Passive and quantitative flows allocate on reported figures. Where a company's GAAP earnings are inflated by marks, mechanical value and quality screens will read it as cheaper and higher-quality than it is — and will read the reversal, when it comes, as a deterioration in the operating business.

Where the theme genuinely does not cut

Cash conversion is untouched. A mark never enters operating cash flow and is backed out of it in the reconciliation. So a company whose accruals are clean and whose venture book is enormous has an *earnings-measurement* problem and not an *earnings-reality* problem — and separating those two is the whole discipline this theme demands.

Landscape

Exposed Companies

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Alphabet	GOOGL	incumbent	Venture/strategic book (GV, CapitalG, SpaceX, Anthropic) marked through Other Income: H1 2026 unrealised equity gains of ~\$135.9B sit inside \$135.7B of Other Income against \$174.6B of H1 net income.	benefits	high	n/a - non-operating line, ~78% of H1-26 net income	Largest absolute distortion on the list. Analyst base already adjusts, so the surprise risk is the reversal, not the level.
Amazon	AMZN	incumbent	Anthropic stake revaluation produced ~\$53.4B of Q2 2026 non-operating income against \$62.6B of Q2 net income; the Rivian mark has been swinging AMZN net income since 2021.	benefits	high	n/a - non-operating line, ~85% of Q2-26 net income	Longest track record of mark-driven EPS swings among US megacaps; the market has been trained on it since the 2022 Rivian write-down.
Berkshire Hathaway	BRK.B	incumbent	ASU 2016-01's canonical case: equity-portfolio marks run through net income, producing a \$22.8B GAAP net LOSS in 2022 against record \$30.8B operating earnings, and \$16.08B of investment gains inside \$25.67B of Q2 2026 net income.	mixed	high	n/a - ~63% of Q2-26 net income	The look-through reference point. Management publishes operating earnings and tells investors GAAP net income is worse-than-useless; the market complied.

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Microsoft	MSFT	incumbent	Equity-method OpenAI stake swung net income by -\$3.1B (-\$0.41 EPS) in FY26 Q1 and +\$7.6B (+\$1.02 EPS) in FY26 Q2 on the OpenAI recapitalisation; MSFT now presents a non-GAAP measure excluding it.	mixed	high	n/a - \$10.7B swing across two consecutive quarters	Equity method, not fair value - so the mechanism differs, but the comparability damage is identical. GAAP net income +60% vs +23% ex-OpenAI in FY26 Q2.
SoftBank Group	9984.T	incumbent	The marks ARE the P&L: Vision Fund fair-value changes drive group net income, with NAV ~\$472B at 30 Jun 2026 (Arm plus a large unlisted OpenAI stake) and the shares at roughly a 34% discount to it.	mixed	medium	n/a - investment result is effectively 100% of group profit	The pure form of the theme, and the proof the market DOES discount unrealised private marks when the vehicle is honest about being a portfolio.
Cincinnati Financial	CINF	adjacent	Large listed-equity portfolio marked through net income: Q2 2026 net income of \$1,255M against non-GAAP operating income of \$224M, the gap being an \$882M after-tax fair-value increase.	mixed	high	~82% of Q2-26 net income; Q1-26 ran the other way (-\$82M)	The clean precedent: same rule, no AI, no venture book. Shows the distortion is an accounting force and not an AI-cycle story.
Markel Group	MKL	adjacent	Insurance operations plus a Berkshire-style equity portfolio: Q2 2026 operating income of \$1.6B against adjusted operating income of \$436M once investment marks are removed.	mixed	high	~73% of Q2-26 operating income	Management states in its own filings that investment gains and losses are distortive of short-term operating performance - the reconciliation exists and is used.

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Tencent Holdings	0700.HK	incumbent	~RMB 903.07B investment portfolio at 30 Jun 2026, down from RMB 957.22B at end-2025; listed investee fair value fell from RMB 547.1B to RMB 487.2B across Q2 2026, so the marks are a DRAG while non-IFRS profit rose 10% to RMB 136.32B in H1.	hurt	medium	n/a - portfolio ~RMB 903B vs H1-26 non-IFRS profit RMB 136B	The theme running in reverse, and the only clearly negative-direction row. Non-IFRS profit is the number the market actually uses here.
Prosus	PRX	incumbent	Holding company whose reported earnings are dominated by an equity-accounted Tencent stake plus a consolidated ecosystem; it reports core headline earnings (+24% EPS in FY2026) precisely to separate the two.	mixed	medium	Ecosystem revenue \$9.7B FY26 vs a Tencent stake worth many multiples of it	Structurally identical to SoftBank but with an operating business attached - the hybrid case where the market has to choose which lens to price.
Nebius Group	NBIS	emerging	A ~28% ClickHouse stake revalued on a funding round that moved ClickHouse from \$6.4B to \$15B produced a \$781M non-cash gain in Q1 2026 - larger than the \$621M of consolidated net income the quarter reported.	benefits	high	Single mark > 100% of Q1-26 reported net income	The sharpest example on the list. Listed Oct 2024, thinly covered, and the mark is the difference between a loss and a headline profit.

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Broadcom	AVGO	adjacent	Finances and takes positions in the frontier labs (OpenAI, Anthropic) it sells custom XPU's to, capturing the funding rounds through revenue rather than through a marked book - the same force, routed away from the income statement's non-operating line.	benefits	medium	Not separately disclosed	The control case: identical exposure to rising private AI valuations, materially less earnings-measurement distortion, because the value arrives as ARR.

The so-what

EXPOSURE EVIDENCE

Eleven companies, one accounting rule, three different relationships to it. The useful split is not by industry but by **whether the company has already taught its investors to read past the marks**. Berkshire, Markel and Cincinnati Financial have; Microsoft and Amazon are most of the way there; Alphabet's distortion is now so large that the reconciliation matters more than it ever has; and Nebius is the case where nobody has done the teaching yet.

How to read the grounding in this pod

Two kinds of evidence sit behind these rows, and they are not equal. Claims about the **structure** of the theme — that both Alphabet and Amazon carry equity books above \$100B [1], that a chip vendor can capture rising private AI valuations through revenue instead of marks [2], that GAAP EPS running ahead of non-GAAP EPS is the tell [3], and that issuers reconcile the marks out of their own adjusted measures [4] — are grounded in documents already in this workbook's corpus, quoted verbatim. Claims about **each peer's specific quarterly figures** were retrieved from the named earnings releases through web search and those releases are registered in this pod's References footer, but they were not re-read line by line inside this run. Where a row's figure is second-hand in that way, the prose names the release it came from; the Confidence column grades the exposure LINK, which is documented in every case, and not the last decimal place of the figure.

Alphabet (GOOGL) — incumbent, benefits, high confidence

The largest absolute distortion in the set. Alphabet's Q1 and Q2 2026 earnings releases report roughly \$135.9B of unrealised gains on equity securities across the first half, sitting inside \$135.7B of Other Income (Expense), net, against \$174.6B of first-half net income — about 78% of the reported bottom line arriving from a line that has nothing to do with search, YouTube or Cloud. The second-quarter gain alone, roughly \$99.0B, is attributed largely to revaluations of private holdings including SpaceX. The structural fact behind it is in this workbook's own corpus: "Alphabet and Amazon's recent earnings reports show both hold equity investment values exceeding

\$100 billion" [1]. A skeptic's objection is fair and worth stating — Alphabet's operating income is enormous in its own right, so the operating business is not in question. What is in question is every trailing ratio built on the \$174.6B.

Amazon (AMZN) — incumbent, benefits, high confidence

Amazon's Anthropic stake was revalued for roughly \$53.4B of non-operating income in Q2 2026 against \$62.6B of quarterly net income, per its second-quarter results release. The same corpus line puts Amazon's equity book above \$100B [1]. Amazon is the company with the longest lived experience of this theme: the Rivian position has been swinging its reported net income in both directions since 2021, which is why its analyst base already models "other income (expense), net" separately. That history is the reason Amazon is rated documented rather than inferred — the disclosure pattern is established and repeated, not novel.

Berkshire Hathaway (BRK.B) — incumbent, mixed, high confidence

The reference case, and the one that proves a market CAN learn. ASU 2016-01 produced a reported GAAP net loss of \$22.8B for Berkshire in 2022 in the same year its operating earnings hit a record \$30.8B, and in Q2 2026 delivered \$16.08B of investment gains inside \$25.67B of net income. Buffett has argued in successive shareholder letters that the resulting swings misinform investors and that operating earnings is the number to use. The exposure is documented at the level of the company's own reporting convention: Berkshire publishes operating earnings as its headline and the sell-side follows. Direction is mixed and not benefits precisely because Berkshire has demonstrated both signs of the same mechanism within four years.

Microsoft (MSFT) — incumbent, mixed, high confidence

A different mechanism reaching the same place. Microsoft accounts for OpenAI under the equity method, not fair value, so what flows through is its share of the investee's net assets rather than a market mark. The effect is identical in kind and violent in size: a \$3.1B reduction in net income (-\$0.41 per diluted share) in the quarter ended 30 September 2025, then a \$7.6B net gain (+\$1.02 per diluted share) in the quarter ended 31 December 2025 following OpenAI's recapitalisation, per Microsoft's FY26 Q1 and Q2 investor materials. In that second quarter GAAP net income grew about 60% while the measure excluding OpenAI grew about 23%. Microsoft now publishes that reconciliation, which is the behaviour the theme rewards.

SoftBank Group (9984.T) — incumbent, mixed, medium confidence

The pure form. SoftBank's reported profit essentially IS the investment result, with NAV around \$472B at 30 June 2026 on holdings led by Arm and a large unlisted OpenAI position, and the shares trading at roughly a 34% discount to that NAV as of early September 2026. Fitch's September 2026 first-time BB+ assignment flags both the investment track record and the concentration in a few large AI positions. Graded medium and not high deliberately: the NAV and discount figures come from company IR material and a rating action rather than from a line-by-line reading of the consolidated statements, and SoftBank reports under IFRS with a March year-end, so it is not directly comparable to the US rows. The finding that matters is qualitative and robust anyway — where the vehicle is honestly a portfolio, the market applies a 34% haircut rather than a 28x multiple.

Cincinnati Financial (CINF) – adjacent, mixed, high confidence

The cleanest natural experiment on this list, because there is no AI in it. Cincinnati Financial's Q2 2026 release reports \$1,255M of net income against \$224M of non-GAAP operating income, the gap being an \$882M after-tax increase in the fair value of equity securities. Q1 2026 ran the other way: \$274M of net income after an \$82M after-tax DECREASE, against \$330M of operating income. Two consecutive quarters in which the sign of the distortion flips while the underwriting business does roughly the same thing. Placed adjacent because CINF is not a contestant in the venture-book race — it is exposed to the identical accounting force through an ordinary listed-equity portfolio, which is exactly what makes it the control.

Markel Group (MKL) – adjacent, mixed, high confidence

The "baby Berkshire" structure: insurance underwriting plus a substantial equity portfolio. Markel's Q2 2026 release reports \$1.6B of operating income against \$436M of adjusted operating income once net investment gains are stripped, and \$1.3B against \$934M for the first half. Management's own stated position is that investment gains and losses, realised or unrealised, are distortive in understanding short-term operating performance. The link is documented by the existence and the wording of the company's own non-GAAP measure, which is the strongest form of evidence available for this theme.

Tencent Holdings (0700.HK) – incumbent, hurt, medium confidence

The only clearly negative-direction row, and the reason it matters is symmetry. Tencent's investment portfolio stood at roughly RMB 903.07B at 30 June 2026, down from RMB 957.22B at end-2025, with the fair value of listed investee holdings falling from RMB 547.1B at the end of Q1 2026 to RMB 487.2B at the end of Q2. Over the same half, non-IFRS profit attributable to equity holders ROSE about 10% to RMB 136.32B. Operating business up, marks down, reported IFRS profit pulled in the opposite direction to the business — the exact mirror of what Alphabet and Amazon are experiencing. Graded medium: the figures come from half-year reporting summarised through search rather than from the interim report itself, and IFRS 9 classification is not identical to ASC 321.

Prosus (PRX) – incumbent, mixed, medium confidence

The hybrid. Prosus carries an equity-accounted Tencent stake alongside a consolidated "ecosystem" that grew revenue 57% to \$9.7B with adjusted EBITDA up 84% to \$1.3B in the year to 31 March 2026, and it reports core headline earnings — up 24% per share in FY2026 — specifically to let readers separate the two. Prosus is the case where the market must decide which lens to price, and the persistent Naspers/Prosus NAV discount is the answer it has historically given. medium confidence: the operating figures are well documented in the FY2026 annual report and media release, but the split between the equity-accounted contribution and the consolidated one is not restated here from primary statements.

Nebius Group (NBIS) – emerging, benefits, high confidence

The sharpest single data point in the run. A financing round that moved ClickHouse's valuation from \$6.4B to \$15B produced a \$781M non-cash revaluation gain on Nebius's roughly 28% stake in Q1 2026 — against \$621M of consolidated net income for the quarter. The mark is larger than the profit. Strip it and the quarter is a loss;

leave it in and Nebius screens as a profitable AI infrastructure company. Nothing about this is hidden — it is described plainly in the reporting on the quarter — but Nebius listed on NASDAQ only in October 2024, carries far thinner analyst coverage than any other name here, and has no long-established convention for how the market should treat the line. That combination, not the size of the gain, is what makes it the fringe case.

Broadcom (AVGO) — adjacent, benefits, medium confidence

The control case, and the row that shows the theme is about ROUTING and not about exposure. Broadcom finances and participates in the same frontier labs NVIDIA invests in, and captures rising AI valuations through contracted revenue rather than through a marked equity book. Its own CFO frames the economics that way on the record: "It's a great investment for us when you think about it, that every gigawatt of compute they deploy, they could achieve \$30 billion of ARR, annual revenue per gigawatt" [2]. Same underlying force, materially less earnings-measurement distortion, because the value lands in revenue where multiples already know how to price it. Graded medium because the size of any equity position Broadcom actually carries is not separately disclosed — the exposure is documented, its magnitude is not.

The screen that ties them together

Across all eleven, one test does most of the work: compare GAAP EPS to the company's own adjusted EPS and check the sign. When the adjusted number is the smaller one, marks are inflating the headline — "That is backwards: non-GAAP normally exceeds GAAP because it strips charges" [3]. Every issuer here that has faced the problem for more than a couple of years publishes the reconciliation that makes the test possible; the standard wording strips "gains/losses from non-marketable and publicly-held equity securities, net" from the adjusted measure [4]. The opportunity in this theme lives in the gap between issuers who publish that line and markets that have not yet adopted it.

The so-what

EMERGING PLAYERS

The fringes were searched and returned exactly one name: **Nebius Group (NBIS)**. Three other candidates that looked structurally right — CoreWeave, Lambda and IREN — were checked and rejected on the evidence. One emerging row out of four candidates is a thin yield, and the reason is itself a finding: a venture book large enough to distort earnings normally requires a balance sheet that a recently listed company does not have. The exception is the company that was SPUN OUT holding stakes it did not have to buy.

Nebius Group (NBIS) — the one that qualifies

Why it counts as emerging. Nebius listed on NASDAQ in October 2024 after the roughly \$5.4B sale of Yandex's Russian assets left the international businesses in a new vehicle under Yandex co-founder Arkady Volozh. Two years of listed history, a share register that turned over completely in the separation, and analyst coverage far thinner than any incumbent on the peer table. It is also on the other side of the theme's central relationship: NVIDIA is an investor in it — "In January, it invested \$2 billion in CoreWeave, and in March, Nebius received another \$2 billion" [1].

What the mechanism actually is. Nebius did not build a corporate venture arm. It INHERITED a portfolio of Yandex-ecosystem companies and holds them alongside an operating AI-cloud business: roughly 28% of ClickHouse (the high-performance analytical database), about 83% of the autonomous-vehicle company Avride, plus Toloka in AI data labelling and TripleTen in education technology. When ClickHouse raised at a valuation that moved from \$6.4B to \$15B, the observable price change forced a revaluation, and Nebius booked a \$781M non-cash gain in Q1 2026. Consolidated net income for that quarter was \$621M. **The mark is larger than the profit** — strip it and the quarter is a loss; leave it and Nebius screens as a profitable AI infrastructure operator. The ClickHouse stake alone is reported at around \$4.2B on the latest round.

What the edge would be if the approach works. Nebius is running the same flywheel the incumbents run, at a scale where it still moves the whole company. Its core business has contracted revenue that the incumbents would recognise — a reported \$17.4-19.4B five-year agreement with Microsoft and a \$3B five-year agreement with Meta, against plans to spend \$16-20B of capex in 2026. If that operating business scales into the contracts, the venture book becomes the funding source that lets it do so without diluting: the stakes are monetisable assets, not just accounting noise. The incumbents cannot copy the position because they did not acquire their portfolios for free in a corporate separation.

The honest caveat. This is also the single most fragile earnings line in the peer table. A \$781M gain on one private stake, recognised because one funding round happened, is not repeatable on a schedule and has no cash behind it. A company with two years of listed history and a profit that depends on a third party's next round is the definition of a name where the error bars are as large as the opportunity.

The fringe candidates that did NOT qualify

Searched deliberately and rejected, because a row with no mechanism is noise wearing a finding's clothes:

- **CoreWeave (CRVV).** Structurally the best candidate — it is known to do compute-for-equity deals, taking stakes in AI startups in exchange for GPU access, and it is a company NVIDIA has invested \$2B into [1]. But its 2026 quarterly reporting and calls do not surface material revaluation gains on those holdings. Q2 2026 was a story about a 112% revenue increase to \$2.6B, \$9.4B of quarterly capex, debt-funded expansion and a \$104B backlog — not about marks. **Not an exposure until the gains appear in the accounts.**
- **Lambda.** Private, reportedly raising up to \$3B at a \$12B-plus valuation with an IPO possible as early as 2027. It is a candidate to BE marked on somebody else's book, not to carry marks on its own. No evidence of an equity-stake strategy. Revisit after a listing.
- **IREN (IREN).** Transitioning from bitcoin mining to AI cloud, with large Microsoft and NVIDIA contracts. Its FY2026 filings and calls describe an infrastructure-as-a-service model with no startup-equity strategy and no revaluation gains.

What the thin yield means

The fringe searches did their job and the answer was mostly negative. That is worth stating plainly rather than padding the table: **at present this theme is overwhelmingly a large-cap phenomenon**, because writing \$30B of cheques is what creates a book big enough to move EPS. The one emerging exception got its portfolio through a corporate separation rather than through cheques. If a second emerging name appears, the most likely route is

the same one — a spin-out or a de-merger that leaves stakes on a small balance sheet — or a compute-for-equity operator like CoreWeave whose accumulated stakes eventually cross into materiality. Both are worth re-checking in two to three quarters.

Ideas

The thesis

MISTAKE THE MARK FOR THE BUSINESS

Every idea above concerns marks going UP. This one concerns what happens when they go down at a company whose operating business is fine. Mechanical screens, index methodologies and headline-driven coverage read GAAP net income. When an equity portfolio marks down, reported earnings collapse, quality and value screens register a deterioration, and the underwriting or operating business underneath has not changed at all. The idea is to be the buyer of that specific, identifiable dislocation.

The mechanism

From who wins and loses: index and screen-driven capital allocates on reported figures, so where GAAP earnings are inflated by marks the screens read the company as cheaper and higher-quality than it is — and read the reversal as operating deterioration. The insurers are the clean laboratory because there is no AI story to confound it. **Cincinnati Financial** reported \$1,255M of net income against \$224M of non-GAAP operating income in Q2 2026 on an \$882M after-tax fair-value increase; in Q1 2026 the same line ran the other way, an \$82M after-tax DECREASE, producing \$274M of net income against \$330M of operating income. Two consecutive quarters, opposite signs, one underwriting business doing roughly the same thing. Detail in the exposure evidence.

The names

- **Cincinnati Financial (CINF)** — the cleanest case: large listed-equity portfolio, published operating income, no venture-book narrative.
- **Markel Group (MKL)** — the same structure with a larger operating business: \$1.6B of Q2 2026 operating income against \$436M adjusted, and management on the record that investment gains and losses are distortive of short-term operating performance.
- **Berkshire Hathaway (BRK.B)** — the extreme version and the proof it happens: a \$22.8B GAAP net loss in 2022 against record \$30.8B operating earnings.
- **Tencent (0700.HK)** — the same thing already in progress, the only hurt-direction row in the peer table: portfolio down from RMB 957.22B to RMB 903.07B while non-IFRS profit rose ~10% to RMB 136.32B in H1 2026.

What would have to be true

1. There is capital that actually transacts on unadjusted GAAP net income — screens, rules-based indices, quantitative quality factors. If every marginal holder adjusts, there is no dislocation to buy.
2. The operating line is separable and verifiably intact. For insurers this is the combined ratio and underwriting profit; for Tencent it is the non-IFRS profit. If the operating line is ALSO deteriorating, this is not an accounting dislocation, it is a business problem.

3. The mark-down is large enough and abrupt enough to move the headline. A slow drift does not create the effect.

What would kill it

The operating line deteriorating alongside the marks. The single observation that ends this idea is a combined ratio worsening at CINF or Markel, or non-IFRS profit turning down at Tencent, in the same period as the mark-down — at which point the screens were right and the accounting was incidental. This is the idea's own falsifier and it is checkable in the same release that produces the dislocation, which is what makes it tradeable rather than merely plausible. Secondly: if the market's look-through is now so complete that a mark-driven GAAP collapse produces no price reaction at all, there is nothing to buy.

What is already priced

For Berkshire, fully — the look-through is decades old and the price does not respond to mark-driven swings. For Cincinnati Financial and Markel, mostly, among dedicated insurance investors, but not necessarily in generalist screens. For Tencent the answer is genuinely unclear and is the most interesting part of the row: the market uses non-IFRS profit for Tencent by convention, but the portfolio decline is large in absolute terms and it is not obvious that the convention survives a bigger one.

The thesis

OWN THE FORCE AT A DISCOUNT

Two ways exist to own rising private AI valuations through a listed security, and the market prices them in opposite directions. Through **SoftBank Group** or **Prosus**, you buy the marks at a discount to stated NAV — roughly 34% in SoftBank's case as of early September 2026 — because the market knows it is buying a portfolio and haircuts it accordingly. Through **Alphabet** or **Amazon**, you buy the same marks at an operating-company multiple, because they arrive inside a net income line the market is pricing at 25-30x. The idea is that the identical economic exposure is available at a discount in one wrapper and at a large premium in the other.

The mechanism

From the mechanism: ASU 2016-01 routes unrealised equity gains through net income, so a venture portfolio's appreciation lands in the same E that the P/E divides by. The market has two defences. Against a vehicle that ANNOUNCES itself as a portfolio, it applies a NAV discount for illiquidity, tax leakage and manager risk. Against an operating company, it applies no such defence, because the marks are one line inside a number everyone treats as operating profit. who wins and loses makes the same point structurally: the theme favours companies whose venture book is understood to BE the business, and squeezes operating companies where the book is large but the identity is not investor.

The names

- **SoftBank Group (9984.T)** — the pure form. NAV ~\$472B at 30 June 2026, led by Arm and a large unlisted OpenAI stake; shares at ~34% of NAV discount. See the exposure evidence.

- **Prosus (PRX)** — the hybrid: an equity-accounted Tencent stake plus a consolidated ecosystem that grew revenue 57% to \$9.7B in FY2026, with core headline earnings published to separate the two.
- **Alphabet (GOOGL) / Amazon (AMZN)** — the expensive side of the same exposure. Alphabet booked ~\$135.9B of unrealised equity gains in H1 2026 against \$174.6B of net income; Amazon ~\$53.4B in Q2 2026 against \$62.6B.

The distinction between SoftBank and Prosus matters: SoftBank's discount is wide and its portfolio is concentrated in exactly the AI assets driving the theme, while Prosus's discount is older, structurally entrenched in the Naspers cross-holding, and its AI exposure is indirect. SoftBank is the cleaner expression; Prosus is the one where the discount could persist for reasons that have nothing to do with this theme.

What would have to be true

1. The NAV discount is compensation for wrapper risk, not for the assets being worth less than stated — i.e. SoftBank's OpenAI and Arm marks are as defensible as the marks sitting inside Alphabet's and Amazon's other income.
2. The discount narrows, or at minimum does not widen, as the underlying assets appreciate. A monetisation event — an OpenAI listing is the obvious candidate — is the usual catalyst.
3. The operating-company side does NOT get repriced upward for its portfolio. If the market starts giving Alphabet explicit credit for a separately valued venture book, the gap closes from the wrong end.

What would kill it

A monetisation at or below carrying value. If a major SoftBank position is sold or listed and clears materially below its marked NAV, the discount was information and not inefficiency, and the trade was wrong on its premise. The specific observation: a realisation event on a top-three holding priced below the last disclosed carrying value. A second killer, slower: the discount persists through a full up-and-down cycle in the underlying assets, which would establish it as a permanent structural feature rather than a closeable gap.

What is already priced

The discount itself is the most-discussed feature of both securities and has been for a decade; nobody is unaware of it. What is NOT obviously priced is the other leg — that Alphabet's and Amazon's trailing multiples currently embed a venture return at an operating multiple. Alphabet's ~78% mark share of H1 2026 net income is recent enough that it has not been through a reversal, and the consensus response to it is untested.

The thesis

ROUTE IT THROUGH REVENUE NOT MARKS

There is a way to own rising private AI valuations that never touches the income statement's non-operating line: sell to the labs and take the value as contracted revenue. **Broadcom** does exactly this, financing and participating in the same frontier labs while capturing the economics as ARR. The idea is that this routing is worth

a real premium that the market does not currently pay for, because revenue-borne value gets an operating multiple and a durable one, while mark-borne value gets an operating multiple and a temporary one.

The mechanism

From who wins and loses, the theme favours "the buyers of the marked-up assets' output" — anyone selling into the venture-funded companies benefits from the funding rounds without carrying the marks. Broadcom's CFO states the economics on the record: every gigawatt of compute the labs deploy is framed as roughly \$30B of annual recurring revenue. The same funding round that produces a mark on somebody else's balance sheet produces a purchase order on Broadcom's. One of those two is capitalised at a durable multiple by every investor on the register; the other reverses.

The names

- **Broadcom (AVGO)** — the toll booth. Custom XPU's into OpenAI and Anthropic, financing alongside, and no separately disclosed marked equity book. See the exposure evidence.
- Contrast with **Alphabet**, **Amazon** and **Microsoft** in the peer table, all of which capture the same force through a non-operating line.

What distinguishes Broadcom from the others is not the size of the exposure but its accounting address. Every name in the peer table is levered to private AI valuations; Broadcom is the only large one where that leverage arrives above the operating-income line.

What would have to be true

1. Broadcom's lab-facing revenue is genuinely contracted and not contingent on the labs raising their next round at a higher price. If the ARR only converts when the funding converts, the routing advantage is cosmetic.
2. Broadcom does not build a large marked equity book of its own. Its exposure today is documented but its magnitude is not disclosed, which is precisely the thing to watch in subsequent filings.
3. The market continues to pay a higher and more durable multiple for revenue than for non-operating gains — which it does, but only for as long as the distinction is visible.

What would kill it

A disclosed, material equity book appearing in Broadcom's own accounts. If a future filing shows non-marketable equity securities large enough to move other income, the control case has become another exposed name and the idea's entire premise — that the same force can be owned without the measurement problem — is gone. The observation is specific: a non-marketable equity securities balance, or an equity-securities gains line, appearing at a scale that is material against net income. Separately: if lab funding stalls, Broadcom's revenue is hit along with everyone else's marks, so this is a routing advantage, not a hedge.

What is already priced

Broadcom's AI revenue trajectory is intensely followed and is not a secret. The earnings-QUALITY dimension is what is not discussed — the argument that a dollar of AVGO AI revenue and a dollar of GOOGL equity mark deserve different multiples is not one the market is currently making out loud.

The thesis

THE FIRST FLAT QUARTER IS THE TRADE

The risk in this theme is not that the marks are wrong. It is that they are **extrapolated**. A company that earned \$135.7B of other income in a half-year has that number sitting in its trailing base, and the first period in which the marks simply do not repeat produces a large negative year-on-year EPS change with no change whatsoever in the operating business. The trade is to identify, before it happens, which names will have that quarter and whether consensus has modelled a zero in the line.

The mechanism

From drivers and horizon: private marks are recognised only on observable price changes, so gains arrive when somebody else prices a round. That makes them lumpy and, critically, **non-recurring by construction**. A flat funding market does not produce a small gain; it produces no gain, which is a 100% decline in a line that was a majority of reported profit. the mechanism adds the tell that makes the screen cheap: when GAAP EPS exceeds non-GAAP EPS, the marks are running hot and the base is inflated.

The names

- **Alphabet (GOOGL)** — the largest absolute exposure: ~\$135.9B of H1 2026 unrealised equity gains inside \$174.6B of net income, roughly 78%. Its marks are dominated by a few very large private positions, so the recognition schedule is lumpy in the extreme.
- **Amazon (AMZN)** — ~\$53.4B of Q2 2026 non-operating income against \$62.6B of net income, driven by a single Anthropic revaluation.
- **Microsoft (MSFT)** — the instructive counter-example and the near-term tell. It has ALREADY shown both signs within two quarters (-\$3.1B then +\$7.6B) and now publishes a measure excluding OpenAI. See the exposure evidence.

Microsoft is the one to watch first, because it has the shortest recognition cycle and the most explicit disclosure. If a flat quarter at Microsoft is absorbed without a multiple response, the market has learned; if it is not, the same reaction is waiting for Alphabet at four times the size.

What would have to be true

1. Consensus forward models carry a materially non-zero "other income" line for these companies — i.e. the marks are in the forward estimate, not just the trailing base. This is checkable in any estimate database and should be checked before anything else.

2. The private AI funding market flattens or decelerates enough that a quarter passes without a major up-round at a top holding.
3. The multiple is being applied to GAAP net income somewhere that matters — a mechanical screen, an index methodology, a retail-facing trailing P/E — and not only to a carefully adjusted number.

What would kill it

Consensus already zeroes the line. If the sell-side forward models for Alphabet, Amazon and Microsoft already carry ~\$0 of equity-security gains, then the flat quarter is a non-event and there is nothing to trade; the market has done the adjustment in advance. That is a single, cheap, falsifying observation and it should be made before the idea is sized at all. Secondly: if the funding market keeps producing up-rounds indefinitely, the flat quarter never arrives and the idea has no trigger — a theme with no clock is not a position.

What is already priced

Partially, and unevenly. Microsoft's disclosure is explicit and its non-GAAP measure is published, so its adjustment is probably in the models. Alphabet's distortion at this magnitude is new — H1 2026 is the first half in which it reached this share of net income — and that novelty is where the unpriced portion is most likely to sit.

The thesis

THE SPIN OUT THAT INHERITED ITS BOOK

Nebius Group (NBIS) is the only emerging name this run found where venture marks move the whole company, and it got there without spending a dollar on the portfolio: the stakes came with a corporate separation. The idea is that a small, thinly covered company whose reported profit is a mark is mispriced in BOTH directions at once — the market either capitalises the mark as if it were operating profit, or discards the quarter entirely as non-cash noise, and neither is right.

The mechanism

From the mechanism: under the measurement alternative, a private stake is remarked only when an observable transaction reprices it. ClickHouse's round moved its valuation from \$6.4B to \$15B, and Nebius's roughly 28% holding was written up by \$781M in Q1 2026 — against \$621M of consolidated net income. The mark is larger than the profit. drivers and horizon explains why this recurs unpredictably rather than never: Nebius holds several such positions (Avride at ~83%, Toloka, TripleTen), each of which can reprice on its own schedule. Full detail in Emerging_Players.

The names

- **Nebius Group (NBIS)** — the position itself. Listed October 2024 out of the Yandex separation; thin coverage; a reported \$17.4-19.4B five-year Microsoft agreement and a \$3B five-year Meta agreement against \$16-20B of planned 2026 capex.

- Read against **SoftBank** and **Prosus** in the peer table, which are what Nebius's portfolio would be valued as if it were the whole company, and against **CoreWeave**, which does compute-for-equity deals but has not yet surfaced material marks.

What would have to be true

1. The portfolio stakes are genuinely monetisable rather than accounting entries — that ClickHouse, Avride or Toloka could be sold or listed at something near the marked value if Nebius chose.
2. The core AI-cloud business converts the Microsoft and Meta contracts into revenue on something like the disclosed schedule, so the venture book is a funding source for growth rather than a substitute for it.
3. Coverage stays thin long enough for the gap to close in the holder's favour. This is the fringe premise and it is the one with the shortest shelf life.

What would kill it

A down-round, or a write-down, at ClickHouse or Avride. The measurement alternative is symmetric: the same observable-price mechanism that produced \$781M of gain books an impairment when the observable price falls, and on a company this size that converts a headline profit into a loss in one line. The specific observation: a ClickHouse or Avride financing at or below the prior round's valuation. A second, more mundane killer: the \$16-20B 2026 capex programme outruns the contracted revenue and the balance sheet needs the stakes sold at whatever price is available, which turns an option into a forced seller.

What is already priced

Hard to establish, and that is the point of the row. With two years of listed history and coverage far below the incumbents, there is no consensus treatment of the revaluation line to be priced in or out. The error bars are as wide as the opportunity, which is why this idea carries the sharpest falsifier in the set.

Summary

The so-what

FACTOR RESEARCH SUMMARY

An accounting rule pushes unrealised gains on private stakes straight into net income, so at eleven large companies — Alphabet at roughly 78% of H1 2026 profit, Nebius at more than 100% of a quarter's — the trailing multiple is being taken off a number that is part operating business and part venture portfolio. The money is in owning the force through vehicles the market already discounts, and in the first quarter the marks do not repeat.

The theme in one paragraph

ASU 2016-01 / ASC 321 routes unrealised equity marks through the income statement while leaving debt securities in other comprehensive income. For private stakes the measurement alternative recognises the whole step-up at once when somebody else's funding round reprices the holding — so the gains are lumpy, cash-free, backward-looking, and timed by an event the holder can often help bring about. At today's scale of corporate venture books this stopped being a footnote and became the denominator of every ratio built on net income. It is a comparability force, not a disclosure failure: the issuers involved publish reconciliations that strip the marks out. The opportunity lives in the gap between issuers who publish that line and markets that have not adopted it.

The landscape

COMPANY	TICKER	ROLE	DIRECTION	CONF.	THE MECHANISM, SHORT
Alphabet	GOOGL	incumbent	benefits	high	~\$135.9B H1-26 unrealised equity gains inside \$174.6B net income (~78%)
Amazon	AMZN	incumbent	benefits	high	~\$53.4B Q2-26 Anthropic revaluation vs \$62.6B quarterly net income
Berkshire Hathaway	BRK.B	incumbent	mixed	high	\$22.8B GAAP loss in 2022 vs record \$30.8B operating earnings; \$16.08B gains in Q2-26
Microsoft	MSFT	incumbent	mixed	high	Equity-method OpenAI: -\$3.1B then +\$7.6B in two consecutive quarters
SoftBank Group	9984.T	incumbent	mixed	medium	Marks ARE the P&L; ~\$472B NAV, shares at ~34% discount
Cincinnati Financial	CINF	adjacent	mixed	high	\$1,255M net income vs \$224M operating income, Q2-26; opposite sign in Q1-26

COMPANY	TICKER	ROLE	DIRECTION	CONF.	THE MECHANISM, SHORT
Markel Group	MKL	adjacent	mixed	high	\$1.6B operating income vs \$436M adjusted, Q2-26
Tencent Holdings	0700.HK	incumbent	hurt	medium	Portfolio RMB 957.22B → 903.07B while non-IFRS profit rose 10%
Prosus	PRX	incumbent	mixed	medium	Equity-accounted Tencent stake plus ecosystem; core headline earnings +24%
Nebius Group	NBIS	emerging	benefits	high	\$781M ClickHouse mark vs \$621M Q1-26 consolidated net income
Broadcom	AVGO	adjacent	benefits	medium	Same force routed through contracted revenue instead of marks

NVDA is deliberately absent — this section is the rest of the landscape. Detail in Exposure_Evidence and Emerging_Players.

The ideas

1. **Own the force at a discount, not at a premium** — the same private-AI exposure trades at a ~34% NAV discount inside SoftBank and at an operating multiple inside Alphabet and Amazon. Dies on a realisation below carrying value.
2. **The first flat quarter is the trade** — marks are non-recurring by construction; the first period they do not repeat produces a large EPS decline with no operating change. Dies if consensus already zeroes the line.
3. **Route it through revenue, not through marks** — Broadcom captures the same force as contracted ARR above the operating line. Dies if a material marked equity book appears in its own filings.
4. **The spin-out that inherited its book** — Nebius got its portfolio from a corporate separation; a \$781M mark exceeds a \$621M quarterly profit and no market convention exists for the line. Dies on a ClickHouse or Avride down-round.
5. **When the mark falls, the screen reads it as the business** — mechanical screens misread mark-driven GAAP collapses as operating deterioration. Dies if the operating line deteriorates in the same period.

What this run did NOT settle

The confidence grades are lopsided against the bearish side of the theme, and that flatters it. Seven rows are graded high and four medium; zero are low. But the single hurt-direction row — Tencent, the only company here where the marks are currently a drag — is one of the four mediums. Every high-confidence row is either a company the marks are helping (Alphabet, Amazon, Nebius) or one where the distortion is two-sided and well documented (Berkshire, Microsoft, CINF, Markel). **The negative-direction evidence in this run is the thinnest evidence in**

this run. A follow-up should deliberately hunt down-cycle cases — 2022 Rivian at Amazon, the Vision Fund write-down years at SoftBank, Tencent's interim report itself — before anyone trades idea 5, which is the idea that most depends on the down-leg being real.

Peer figures are second-hand. Twenty primary sources were registered for this run (earnings releases from Alphabet, Amazon, Microsoft, Cincinnati Financial, Markel, Prosus and SoftBank, plus reporting on Berkshire, Tencent and Nebius), but the workbook's parser queue had not indexed them by the time the pods were written, so the figures were taken from the web-search retrieval that named those releases rather than read line by line from the filings. The [N] markers in this bag are therefore confined to documents already in the corpus, and every peer-specific number is attributed in prose to the release it came from instead. **A re-run after ingestion completes should upgrade those attributions to quoted citations** — the sources are already registered under the collection, so it costs no searches.

One internal inconsistency was left standing. The retrieval for Amazon reported H1 2026 "Other income (expense), net" as \$54.7B in prose and \$70.2B in its own table. Neither figure is used above; the Q2 number (\$53.4B), on which both agreed, is what the table and the digest carry. The H1 Amazon total should be treated as unestablished.

The fringe is genuinely thin, and it may be a real finding rather than a search failure. Four emerging candidates were checked and one qualified. CoreWeave does compute-for-equity deals but has not surfaced material revaluation gains; Lambda is private; IREN has no equity-stake strategy. The provisional conclusion — that a book big enough to distort EPS normally requires a balance sheet a recent listing does not have, with spin-outs as the exception — is drawn from four data points and deserves a wider screen before it is relied on.

What was not looked at at all. The tax treatment of unrealised marks (deferred tax liabilities accruing on gains that may never be realised) touches every row here and was not examined. Nor was the question the `pv_01` lens explicitly declines: whether the marks are correctly VALUED. This run treats a mark as given and asks what it does to comparability; a challenge run arguing that private AI marks are systematically overstated would be examining a different and entirely legitimate question.

Search budget. This workbook was already far past its soft budget of 150 web searches when this run started. Eight searches were spent, six on the peer landscape and two deliberately on the fringes, with the corpus mined through nine `query_context` calls first and no searches at all in the Ideas or Digest phases. Tencent and Prosus are the two rows that would most benefit from another search each.

FACTOR RESEARCH

NVDA Factor 11 Operating Leverage Already Spent

Theme

A margin gain that can only be booked once

MECHANISM

EBIT margin is gross margin minus operating expense as a share of revenue. When revenue compounds far faster than a committed opex base, the second term collapses and EBIT margin expands even if gross margin never moves at all. That gain is bounded by arithmetic: the opex ratio cannot fall below zero, and in hardware it floors somewhere in the mid-to-high single digits. Once it is there the lever is spent, and every further dollar of revenue converts to EBIT at roughly gross margin rather than above it.

NVIDIA is the extreme worked example of the chain: operating expense fell from 27.6% to 9.5% of revenue between FY2022 and FY2026, supplying 18.1 of the 27.9 points of EBIT-margin expansion, with SG&A alone going from 8.0% to 2.1% of revenue while revenue compounded at 68.3% a year and SG&A at 20.6% (the opex bridge). What makes that collapse mechanically possible is that the unit of sale inflated faster than the cost of making the sale: a Blackwell NVL72 rack sells for \$2.8-3.4M in its GB200 configuration and \$6.0-6.5M as a GB300 [2], and the commission, field-engineering and support hours attached to closing one of those are barely larger than they were for a \$30,000 server.

Arista shows the same chain at an earlier stage and at a legible scale: total operating expense held at \$520M, nearly flat sequentially against a 9% quarter-on-quarter revenue increase [1], with R&D at 13% and SG&A at 7% of revenue - a roughly 20% opex ratio, which is a long way above NVIDIA's 9.5% floor. Broadcom's management states the mechanism outright, telling analysts to stop watching gross margin because revenue growth far outruns opex growth [4].

The reverse half is the part that prices

The opex base is committed in headcount, leases and multi-year engineering programmes; it does not shrink on a quarter's notice. So on a revenue stall the same arithmetic runs backwards at the same gearing. NVIDIA's FY2023 is the observed proof: revenue of \$26.97B, flat year over year, while GAAP earnings per diluted share fell 55% [3]. Flat revenue, earnings more than halved - that gap *is* the operating leverage, running in reverse.

The signature is therefore readable off any income statement in three numbers: opex as a share of revenue, the trend in that share over eight quarters, and the gap between the revenue growth rate and the opex growth rate. Above roughly 25% the spreading is still ahead; 15-20% is mid-spread; below 10% the lever has been pulled. (Those bands are this pod's reading convention, not a sourced standard.)

The clock is four to twelve quarters per name, and it is set by where the opex ratio already sits

DRIVERS AND HORIZON

This theme does not run on a sector clock. It runs on each company's own distance from its opex floor, which is why two companies in the same end-market can be at opposite ends of it in the same quarter. Over the eight quarters to Q1 2026 Broadcom's operating margin expanded from 32% to 49% - seventeen points - while Arista's stayed anchored near 43%, neither compressing nor expanding [3]. Same demand, same customers, entirely different position on the curve.

What pushes it along. Revenue compounding against a cost base that was budgeted a year earlier. As long as AI-infrastructure demand keeps growing and opex plans are set annually, the ratio falls mechanically. Broadcom's own framing of the revenue driver is that content is roughly \$20-30B per gigawatt and stable per gigawatt, with the growth coming from the number of gigawatts [2] - a volume driver, which is exactly the kind that spreads a fixed base.

What slows or ends it. Three things, in the order they usually arrive. First the arithmetic floor. Second, the company chooses to re-spend: NVIDIA now guides full-year operating expense growth in the low 50s percent [4], which is the base being deliberately re-levered upward once the ratio stopped falling. Third, gross-margin dilution consumes the gain before it reaches EBIT - Arista's gross margin fell to 62% from 65% a year earlier while its operating margin held at 43%, so its opex discipline was spent absorbing the gross-margin headwind rather than expanding EBIT [3]. That is leverage used defensively, and it looks identical to leverage exhausted from the outside.

What has already changed. Three years ago the AI hardware cohort was mostly pre-spread: opex ratios in the twenties, operating margins in the low thirties. The unit of sale has since gone from boards to racks, which multiplied revenue per selling motion and dragged every opex ratio down with it. The companies that started this window at a 25-30% ratio have now used most of their room; the ones that started at 40%+ and are still ramping have not.

Where the spreading is demonstrably still ahead. Coherent guided Q1 FY2027 non-GAAP operating expenses of \$400-420M against revenue of \$2.2-2.4B, already below its own 18% target model, with the flow-through leverage explicitly still to come [1]. That is a company reading the same arithmetic from the other side.

The horizon, stated plainly. Four to twelve quarters for an individual name at 30%+ revenue growth - that is how long it takes a 25% opex ratio to reach a high-single-digit floor. Two to three years for the cohort. This is not a decade-long structural force like a foundry monopoly; it is a fast-resolving one, which means a position built on it is a position on the next two years of income statements, not on terminal value.

One warning on where the cost sits. Capital intensity is not the same lever and does not behave like it. Coherent's capex ran \$556M in Q4 against \$131M a year earlier [1]; that spending lands in depreciation inside cost of goods, which scales with the asset base rather than staying flat. A capital-intensive beneficiary of the same demand gets much less of this effect and carries the downside without the upside.

The structural sides: who still has the lever, who already pulled it, and who never had one

WHO WINS AND LOSES

This is the screen the rest of the run searches against. It sorts by cost STRUCTURE, not by end-market.

FAVOURED - THE SPREADING IS STILL AHEAD

- **High fixed opex ratio today (roughly 20-35% of revenue) against 30%+ revenue growth.** Fabless silicon and software-heavy networking businesses selling few, large, repeat units into a handful of hyperscaler accounts. The signature is a go-to-market line that barely moves: Arista's SG&A at 7% of revenue, flat sequentially, with go-to-market costs scaling at a fraction of the pace of revenue [1].
- **Dollar content per unit inflating while the selling motion stays constant.** Whoever wins a design slot in a rack captures the content increase without adding sales cost - the cost of the design win is the same whether the content is \$50k or \$500k a rack.
- **Companies emerging from a heavy multi-year R&D programme into volume shipment.** The spend is already booked; the revenue arrives against a base that was set for a much smaller company.

SQUEEZED - THE LEVER IS SPENT, OR IT WAS NEVER THEIRS

- **Anyone already at a single-digit opex ratio.** Nothing is left to give. The next margin point has to come from gross margin or volume, both of which are harder and neither of which the market has been paying for.
- **Assemblers and integrators whose cost base is materials rather than people.** Hon Hai runs a 6.12% gross margin and a 3.75% operating margin on record AI-server revenue [2]. The theme multiplies its revenue and barely touches its margin, because there was never a large fixed opex block to spread. Rising rack ASPs make this worse rather than better: it is hard to hold a 10% margin on items that cost millions, and the system vendor's role in the BOM is being reduced at the same time [3].
- **Capital-intensive beneficiaries.** When the cost of serving growth lands in depreciation inside cost of goods rather than in opex, it scales with the asset base. Coherent names capital intensity as the explicit tradeoff for its flow-through leverage [4].
- **Anyone whose revenue stalls while carrying a base sized for the last growth rate.** This is the same group as the first bullet a year later, and it is the only group for which this theme is a downside catalyst rather than an absence of upside.

THE THREE SCREENS THAT SEPARATE THEM

1. Operating expense as a share of revenue, plotted over eight quarters. Falling steeply = mid-spread. Flat and low = spent. Flat and high = the lever exists but is not being pulled.
2. The gap between the revenue growth rate and the opex growth rate. A company growing revenue 50% and opex 20% is in the middle of it; one growing both at 50% has finished.
3. Whether the gross-to-operating margin spread is narrowing or widening. Narrowing means the leverage is being consumed defending against gross-margin pressure and will not show up as EBIT expansion, which is how a company can look levered and deliver nothing.

Landscape

Exposed Companies

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Broadcom	AVGO	incumbent	Semiconductor-segment opex is already only 6% of segment revenue, BELOW NVIDIA's 9.5% - segment revenue grew 127% YoY against 22% opex growth, and Q4 operating margin is guided DOWN to ~66% from a record 67.9%	mixed	high	AI semis 56% of total revenue (Q3 FY2026)	The closest structural twin to NVIDIA and arguably one quarter further along; the guided step-down is the exhaustion signature appearing in real time
Credo Technology	CRDO	emerging	GAAP opex ran ~34% of revenue in FY2026 (opex \$463.3M against revenue that more than tripled to \$1.3B) - roughly 24 points of opex ratio still above NVIDIA's floor, i.e. the spreading is mostly ahead	benefits	high	AI/data-centre connectivity is the majority of revenue; exact split not established	GAAP operating income \$445.0M implies a ~33% operating margin; non-GAAP opex \$270.9M (~21% of revenue) shows how much of the gap is stock comp
Astera Labs	ALAB	emerging	Non-GAAP operating margin stepped 36.2% (Q1 2026) to 39.1% (Q2) to a guided ~43% (Q3) while non-GAAP opex was guided at \$128-131M against \$355-365M of revenue - a ~36% opex ratio spreading live	benefits	high	Connectivity silicon into AI platforms; effectively 100% theme-exposed	IPO'd March 2024, so the whole opex-spreading phase is inside the public record - the cleanest live worked example of the theme's first half

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Arista Networks	ANET	incumbent	Total opex held at \$520M, nearly flat sequentially against a 9% QoQ revenue rise (R&D 13% + SG&A 7% = a ~20% ratio), but the entire gain went to absorbing gross margin falling 65% to 62%, pinning operating margin at 43% for eight quarters	mixed	high	AI/cloud is the dominant revenue mix driver; segment split not disclosed	The most instructive row: leverage that exists, is being pulled, and produces zero EBIT expansion because gross margin is taking it
Coherent	COHR	incumbent	Q1 FY2027 non-GAAP opex guided \$400-420M against \$2.2-2.4B revenue - already below its own 18% target model, with flow-through leverage explicitly still ahead and gross margin up 660bps over nine quarters	benefits	high	Datacom/AI transceivers are the growth engine; company-level split not established	Capital intensity is the offset: capex went \$131M to \$556M in a year, so part of the cost of growth lands in COGS depreciation rather than spreadable opex
Advanced Micro Devices	AMD	incumbent	Opex ratio is still well above the floor, but each point of opex leverage is worth less on a 53% GAAP gross margin (Q1 2026) than on NVIDIA's 74.9% - the same spreading produces a smaller EBIT step	benefits	medium	Data Center is the AI-exposed segment; exact opex ratio not established from the corpus	Exposure link is inferred from the gross-margin gap rather than from a disclosed opex ratio - treat the magnitude as unsized
Fabrinet	FN	adjacent	FY2026 revenue up 36% to \$4.6B with Q4 a record \$1.316B (+45% YoY) and management stating non-GAAP EPS grew even faster than revenue - leverage present, but on a contract-manufacturing gross margin in the low teens	benefits	medium	Optical packaging for datacom/AI; segment split not disclosed in the corpus	EPS outgrowing revenue is the leverage signature; the ceiling is the gross margin it operates against, not the opex ratio

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Celestica	CLS	adjacent	CCS segment revenue jumped 76% YoY (Enterprise +101%, Communications +69%) but the cost base is materials and plant, not a fixed opex block, so a 76% revenue rise converts weakly to EBIT	mixed	medium	CCS segment is the AI-exposed half of the company	Exposure is documented; the CONVERSION claim is inferred from EMS cost structure rather than from a disclosed opex ratio
Hon Hai Precision (Foxconn)	2317.TW	adjacent	Runs a 6.12% gross margin and a 3.75% operating margin (up from 3.16%) on record AI-server revenue - there was never a large fixed opex block to spread, so the theme multiplies revenue and barely touches margin	mixed	high	Server-related revenue exceeded 50% of total for the first time (Q2 2026)	The structural-floor pole of the table: proof that revenue exposure to the same demand is not exposure to this theme
Alchip Technologies	3661.TW	emerging	Q1 2026 revenue fell to \$132.4M from \$318.7M a year earlier (-58%) while net income HELD at \$45.1M (+1.6%) - a live test of reverse leverage in which a >50% gross margin NRE mix offset the volume collapse	mixed	medium	3nm/2nm designs were 21% of Q1 2026 revenue; 7nm/5nm 63%	The most interesting row for the reversal question: a 58% revenue fall that did NOT produce the FY2023-style earnings collapse, because mix moved the other way
CoreWeave	CRWV	adjacent	Revenue rose sharply from \$1,212M a year earlier while losses widened, because the cost of serving growth sits in depreciation and interest rather than in spreadable opex - the theme's cost structure inverted	hurt	medium	AI cloud is effectively 100% of revenue	The corpus chunk redacted the absolute figures, so the direction is documented but the magnitude is not - hence medium, not high

COMPANY	TICKER	ROLE	EXPOSURE	DIRECTION	CONFIDEN...	SEGMENT ...	AGENT_NO...
Cisco Systems	CSCO	adjacent	Runs a 25% operating margin, recovered from 19% at 30 June 2024, without the revenue growth rate that makes opex spreading possible - the control case for what the theme looks like when the revenue line is not compounding	mixed	high	AI infrastructure orders are a minority of revenue; not separately sized here	Included as the negative control: same end-market, no hypergrowth, so no spreading - which is what isolates growth rate as the theme's driver

The one row that matters: Broadcom is on the same curve and is already a quarter further down it

EXPOSURE EVIDENCE

Read the table for that finding first. Everything else places a company relative to it.

Broadcom (AVGO) - incumbent, mixed, high confidence. The exposure is documented in the company's own segment P&L, not inferred. Semiconductor Solutions revenue grew 127% year on year against 22% operating-expense growth, taking segment operating margin up 440 basis points to 61% with opex at 6% of segment revenue [2] - a lower opex ratio than NVIDIA's 9.5%. Consolidated operating margin was a record 67.9%, up 240bps, and Q4 is guided to approximately 66% [2]. That guided step-down is the same shape as the one the workbook's own forecast carries for NVIDIA, arriving one quarter earlier and from a higher base. Directionally the eight-quarter record is also unambiguous: Broadcom's operating margin ran from 32% in Q2 2024 to 49% in Q1 2026 [1]. The reason the row is mixed rather than benefits is that most of that 17-point expansion is now behind it.

Credo Technology (CRDO) - emerging, benefits, high confidence. FY2026 revenue more than tripled to \$1.3 billion [3], against GAAP operating expenses of \$463.3M and GAAP operating income of \$445.0M - an opex ratio near 34% and an operating margin near 33%. The figures come from the company's own FY2026 results statement of operations, so the link is documented rather than modelled. The significance is the DISTANCE: roughly 24 points of opex ratio sit between Credo and the high-single-digit floor NVIDIA reached, and closing that gap at the current growth rate is several years of EBIT expansion that does not require a single point of gross margin.

Astera Labs (ALAB) - emerging, benefits, high confidence. Q1 2026 revenue of \$308.4 million, up 14% sequentially and 93% year over year [4], with non-GAAP operating margin stepping 36.2% to 39.1% to a guided ~43% across three quarters, and Q2 non-GAAP opex guided at \$128-131M against \$355-365M of revenue. That is an opex ratio in the mid-30s falling roughly three to four points a quarter. The skeptic's objection is fair and worth stating: these are non-GAAP figures, and the gap to GAAP at a recently-IPO'd company is stock compensation, which does not stop being a cost because a slide excludes it.

Arista Networks (ANET) - incumbent, mixed, high confidence. Total operating expenses held at \$520 million, nearly flat sequentially against a 9% quarter-on-quarter revenue increase [1], with R&D at 13% and SG&A at 7% of revenue. On the arithmetic alone Arista should be expanding EBIT. It is not: operating margin has sat in a 41-45% band for eight quarters while gross margin fell from 65% to 62% [1]. This is the row a reader should sit with longest, because it shows leverage being fully consumed defending gross margin - producing an income statement that looks exactly like exhaustion while the lever is in fact still being pulled.

Coherent (COHR) - incumbent, benefits, high confidence. Q1 FY2027 non-GAAP operating expenses guided at \$400-420M against revenue of \$2.2-2.4B, already below the company's own 18% target model, with the flow-through leverage stated to be still ahead of it [5]. Gross margin has expanded in eight of the past nine quarters for a cumulative 660bps. The qualifier belongs in the same breath: capex ran \$556M in Q4 against \$131M a year earlier [5], so a growing share of the cost of growth is landing in depreciation inside COGS rather than in the opex line that spreads.

Advanced Micro Devices (AMD) - incumbent, benefits, medium confidence. The exposure link here is INFERRED, not documented. What the corpus establishes is the gross-margin gap: AMD reported 53% GAAP and 55% non-GAAP gross margin in Q1 2026 against NVIDIA's 74.9% [6]. The inference is that on a 53% gross margin, a given fall in the opex ratio produces the same absolute EBIT-margin gain but against a much thinner base, so the same spreading is worth proportionally more to the EPS line and the company has further to travel. No disclosed AMD opex ratio was found in this corpus, which is why the row is medium and why the magnitude is left unsized rather than estimated.

Fabrinet (FN) - adjacent, benefits, medium confidence. FY2026 revenue rose 36% to \$4.6 billion, Q4 was a record \$1.316 billion up 45% year over year, and management stated that non-GAAP EPS grew even faster than revenue [7]. EPS outgrowing revenue is the operating-leverage signature stated in the plainest available terms. It is adjacent and medium because the leverage operates against a contract-manufacturing gross margin: the opex ratio can fall, but the ceiling on what that is worth is set by a gross margin an order of magnitude below a fabless designer's, and no opex ratio is disclosed in the corpus.

Celestica (CLS) - adjacent, mixed, medium confidence. CCS segment revenues jumped 76% year over year, with Enterprise up 101% and Communications up 69% [8]. The revenue exposure is documented. The CONVERSION claim - that a 76% revenue rise converts weakly because the cost base is materials and plant rather than a fixed opex block - is inferred from EMS cost structure, not from a disclosed Celestica income statement. A skeptic would be right to say this row is a structural argument wearing a number.

Hon Hai Precision / Foxconn (2317.TW) - adjacent, mixed, high confidence. Gross margin 6.12%, down from 6.33%, with operating margin improving to 3.75% from 3.16% on better expense control [9], while server-related revenue exceeded 50% of total for the first time. This row is the table's anchor at the opposite pole: the company most exposed to the same end demand by revenue, and least exposed to this theme by structure. A business that never carried a large fixed opex block cannot spread one. Its 59-basis-point margin improvement on record AI revenue is the whole argument in one figure.

Alchip Technologies (3661.TW) - emerging, mixed, medium confidence. Q1 2026 revenue of \$132.4 million, down from \$318.7 million a year earlier and \$152.7 million the prior quarter [10], with net income of \$45.1 million, UP 1.6% year on year. A 58% revenue fall that left earnings intact. The reason is mix - a postponed tape-

out cut product revenue while high-margin NRE revenue grew, pushing gross margin over 50%. This is the most useful counter-evidence in the table: it shows that reverse operating leverage is not automatic, and that a revenue collapse whose mix moves the other way can leave the P&L standing. It is medium because the mix explanation comes from management commentary rather than a segment disclosure.

CoreWeave (CRWV) - adjacent, hurt, medium confidence. Revenue rose sharply from \$1,212 million a year earlier while losses widened, with heavy investment in AI cloud infrastructure driving operating expenses and depreciation higher [11]. The structural point is that the cost of serving growth sits in depreciation and interest, which scale with the asset base, rather than in a fixed opex block that can be spread. The corpus chunk redacted the absolute loss figures, so the DIRECTION is documented and the MAGNITUDE is not - which is exactly why this row is graded medium and is the table's only hurt.

Cisco Systems (CSCO) - adjacent, mixed, high confidence. A 25% operating margin, essentially flat since 30 June 2024 when it registered 19% [1]. Cisco is in the table as the negative control rather than as an idea: same end-market, no hypergrowth revenue line, and therefore no spreading. Its presence is what isolates the growth RATE as the variable that drives this theme, rather than the sector.

What a skeptic should push back on

Three of these twelve rows rest on structural inference rather than a disclosed opex ratio - AMD, Celestica and Fabrinet - and all three are graded medium for exactly that reason. The corpus is deep on NVIDIA and on the AI supply chain's revenue, and thin on peer operating-expense disclosure; nobody publishes 'opex as a percentage of revenue' as a headline. Every ratio quoted above was either stated by the company or computed from two figures in the same document, and where neither was available the cell says so rather than carrying an estimate.

Three emerging names, and the reason the fringe is where this theme is still tradeable

EMERGING PLAYERS

The incumbents on this table have either spent the lever (Broadcom's semiconductor segment at 6% of segment revenue) or are spending it defensively (Arista). The names where the spreading is genuinely ahead are smaller, newer, and less written about - which is the whole reason the emerging column exists.

CREDO TECHNOLOGY (CRDO)

Credo listed in January 2022 and was, until the AI build-out, a small mixed-signal connectivity company. FY2026 changed the scale entirely: revenue more than tripled to \$1.3 billion [1]. The novel mechanism is the active electrical cable - a copper cable with Credo silicon inside it that reaches further than passive copper at a fraction of the power and cost of optics, which is exactly the gap that opened when rack power densities rose.

Mechanically, that product ships against an R&D base that was sized for a company a third the size, which is why GAAP operating expenses of \$463.3M against a tripled revenue line produce an opex ratio near 34% and a GAAP operating margin near 33%. The edge over incumbents is that an AEC is not a product the optics vendors can push without cannibalising higher-priced transceivers. What would make this ordinary rather than special: the opex ratio stops falling because the company has to build a full systems organisation to hold the sockets.

ASTERA LABS (ALAB)

Astera IPO'd in March 2024, so its entire opex-spreading phase is inside the public record - unusual, and analytically valuable. Q1 2026 revenue was \$308.4 million, up 14% sequentially and 93% year over year [2], with non-GAAP operating margin moving 36.2% to 39.1% to a guided ~43% over three quarters. Mechanically the company sells retimers, smart cable modules and fabric switches that attach per AI platform, so its dollar content rises with each platform generation without a proportional increase in the selling motion - the content-inflation pattern the Theme section names as the cleanest version of the mechanism. The edge is that its products sit inside reference designs, which makes the attach sticky across generations. What would break it: the switch and retimer functions get absorbed into the accelerator vendors' own silicon, which is a live risk given who its largest customers are. Coverage is no longer thin in the sense of unfollowed, but the opex-ratio arithmetic is still not the thing most notes lead with.

ALCHIP TECHNOLOGIES (3661.TW)

Taiwan-listed with GDRs in Luxembourg, and genuinely thinly covered by US research. Alchip is an ASIC design-and-production house whose revenue is lumpy by construction - it recognises NRE on design milestones and production revenue on shipment, so a postponed tape-out moves a quarter by tens of percent. Q1 2026 revenue was \$132.4 million against \$318.7 million a year earlier [3], and net income still rose 1.6%. That is the single most useful fact the fringe search produced, and it cuts AGAINST the theme's simplest reading: reverse operating leverage did not bite, because the revenue that disappeared was low-margin production volume while high-margin NRE grew and gross margin went over 50%. The edge, if the approach works, is that a revenue mix weighted to design services is structurally less exposed to the reversal half of this theme than a product company is. What would kill that reading: a quarter where BOTH NRE and production fall, which has not yet been observed.

WHAT THE FRINGE SEARCH DID NOT FIND

No pre-revenue or newly-listed name was found whose novel mechanism is specifically about COST STRUCTURE rather than product. The privately-held names in this workbook's corpus that attack adjacent bottlenecks - Enfabrica in AI networking, Corintis in chip cooling - are venture-stage and unlisted, so they carry no income statement to read an opex ratio off and were not given table rows. That is a real gap, not a rhetorical one: the theme's best fringe expression would be a listed company two years into an opex-spreading phase that nobody models, and this run found three candidates rather than a screen.

Ideas

The value is the **DISTANCE** to the opex floor, and that distance is a number you can look up

OWN THE SPREAD NOT THE FLOOR

The thesis. This theme pays whoever still has room between today's opex ratio and the floor, and the size of the prize is simply that gap multiplied by the revenue it will be spread over. NVIDIA closed its gap (27.6% to 9.5%) and Broadcom's semiconductor segment has gone further still, to 6% of segment revenue the evidence. Credo sits near 34% and Astera Labs in the mid-30s. Those two are holding the exact asset NVIDIA has already spent.

The mechanism. Straight from the mechanism: EBIT margin is gross margin minus the opex ratio, so a company whose opex ratio falls twenty points delivers twenty points of EBIT expansion without touching price. What makes the value land HERE rather than with the incumbents is that the incumbents' remaining gap is measured in single digits and the emerging names' is measured in twenties - and the arithmetic does not care about brand or share.

The names. Credo Technology (CRDO) and Astera Labs (ALAB), both emerging in the peer table. They differ in what they are levered to: Credo's content is a cable-level product whose competitive position depends on copper reaching far enough, Astera's is per-platform silicon content inside reference designs, which is stickier but sits closer to functions an accelerator vendor could absorb. See the emerging-players note for the mechanism behind each.

What would have to be true. Revenue keeps compounding above 40% while operating-expense growth stays in the twenties - the gap between those two rates IS the idea. Both companies must also stay independent of the accelerator vendors' own silicon roadmaps for at least two more platform generations.

What would kill it. Two consecutive quarters where the opex ratio stops falling while revenue growth is unchanged. That would mean the company is buying its growth with headcount rather than spreading a base, and the whole thesis is the opposite claim.

What is already priced. Both names are followed and both trade on revenue-growth narratives. What is NOT commonly modelled is the SLOPE of the opex ratio as a separate line - most notes carry an operating-margin assumption rather than deriving one from the ratio's trajectory. That gap is where the idea lives, and it is a modelling gap rather than an information gap, which means it can close quickly.

Broadcom just guided its first operating-margin step-down while revenue was still accelerating

THE GUIDED STEP DOWN IS THE MARKER

The thesis. The hardest part of this theme is knowing WHEN a company's lever is spent, because the income statement looks identical for a while either way. Broadcom has supplied the marker: a record 67.9% consolidated operating margin in Q3 FY2026, up 240bps, with Q4 guided to approximately 66% - and a

semiconductor segment already running opex at 6% of segment revenue the evidence. A company guiding its first margin step-down while revenue is still accelerating is telling you the spreading is over. The idea is to treat that guide as the dating mechanism for the theme and re-underwrite the name on gross margin and volume alone.

The mechanism. the mechanism sets out why the step happens exactly once: the ratio reaches its floor, mix dilution keeps working, and the company re-levers its spending. the horizon names the same three drivers in the order they arrive. Broadcom is visibly in all three at once - XPU mix diluting gross margin, opex at a floor, and capacity investment rising.

The names. Broadcom (AVGO), incumbent in the peer table, with Arista Networks (ANET) as the contrast: Arista's operating margin has sat in a 41-45% band for eight quarters, which is what the same exhaustion looks like when it arrives quietly instead of in a guide.

What would have to be true. The step-down has to be structural - mix plus a re-levered opex base - rather than one quarter of guidance conservatism. The test is whether segment opex growth re-accelerates toward the revenue growth rate over the next two quarters.

What would kill it. Q1 operating margin printing back above 68% on the same revenue mix. That would establish the guide as conservatism rather than exhaustion, and the marker would be worthless.

What is already priced. The market largely treats Broadcom's high-60s operating margin as a run-rate. The guide says the company does not. That disagreement is the whole content of this idea, and unlike most of this bag it will be settled by a single print.

A 58% revenue fall that left earnings intact - and why that is the most interesting fact in this bag

MIX IS THE ONLY OBSERVED DEFENCE AGAINST THE REVERSAL

The thesis. The reversal half of this theme is what actually prices: NVIDIA's FY2023 showed flat revenue and a 55% GAAP EPS fall the mechanism. So the question worth money is what, if anything, defends against it. The only defence this run OBSERVED is revenue mix. Alchip's Q1 2026 revenue fell to \$132.4M from \$318.7M a year earlier - a 58% collapse - and net income still rose 1.6%, because the revenue that disappeared was low-margin production volume while high-margin design-services (NRE) revenue grew and gross margin went over 50% the evidence.

The mechanism. the structural sides sorts companies by cost structure. This idea adds the second axis the table implies but does not sort on: what happens to MIX when volume falls. A business whose milestone-based services revenue is counter-cyclical to its shipment revenue carries an automatic hedge against reverse operating leverage, and nothing in a peer multiple prices that.

The names. Alchip Technologies (3661.TW), emerging in the peer table and written up in the emerging-players note. Fabrinet (FN) is the deliberate contrast: pure manufacturing, no services mix, so its EPS-outgrowing-revenue leverage runs cleanly in both directions.

What would have to be true. The NRE/production mix has to behave counter-cyclically across a SECOND downturn, not just the one quarter observed. It also has to survive scale: the defence works partly because Alchip is small enough for one tape-out to move the mix.

What would kill it. A quarter in which both NRE and production revenue fall together. That has not been observed and would demonstrate the two are correlated rather than offsetting, which ends the idea outright.

What is already priced. Very little of this is priced, for the honest reason that very little of it is covered - Alchip is Taiwan-listed with thin US research. That cuts both ways: the error bars on a single observed quarter are wide, and one quarter is not a pattern.

Rack ASPs went from about \$3M to as much as \$8.8M, and the assembler's margin moved 59 basis points

THE ASSEMBLER IS NOT THE BENEFICIARY

The thesis. Revenue-exposure screens keep surfacing contract manufacturers as AI winners because their revenue line is exploding. This theme says that is a category error. Hon Hai runs a 6.12% gross margin and a 3.75% operating margin - up 59 basis points from 3.16% - on record AI-server revenue that passed 50% of total for the first time the evidence. There was never a large fixed opex block to spread, so the demand that gave NVIDIA 18.1 points of EBIT expansion gives the assembler basis points.

The mechanism. the mechanism shows the unit of sale inflating from a \$30,000 server to a \$2.8-6.5M rack. For a designer, the cost of selling that unit barely moved, which is the entire leverage. For the assembler, the cost of building it moved almost exactly in proportion, because it is materials - and it is hard to hold even a 10% margin on items that cost millions while the system vendor's role in the BOM is being reduced the structural sides.

The names. Hon Hai Precision (2317.TW) and Celestica (CLS) as the exposed-but-not-levered pair, with Fabrinet (FN) as the one manufacturing name that has actually shown EPS outgrowing revenue the peer table. The distinction matters: Fabrinet's leverage is real and small; Hon Hai's is real and smaller still.

What would have to be true. Rack ASP inflation continues while assembler operating margins stay below roughly 5%. Both halves are observable quarterly and neither requires a forecast.

What would kill it. Any of these names posting a 200bp-plus operating-margin expansion attributable to AI revenue. That would establish a fixed cost block that IS being spread, and the whole 'structurally excluded' claim collapses.

What is already priced. Partially, and unevenly. Hon Hai's low margin is well understood by anyone who has read it. What is less well priced is the DIRECTION of rack ASP inflation for these names - a larger unit at a fixed percentage margin raises working capital and inventory risk faster than it raises profit, which is a cost of the theme rather than a benefit.

Summary

So what: the unspent gap is the asset, and NVIDIA is not the furthest along the curve

FACTOR RESEARCH SUMMARY

The money in operating-leverage exhaustion sits with companies that still hold distance between today's opex ratio and the floor - Credo near 34% of revenue, Astera Labs in the mid-30s - and not with the names that have closed it. The finding that reframes the factor is that Broadcom's semiconductor segment already runs operating expense at 6% of segment revenue, BELOW NVIDIA's 9.5%, and has guided its first operating-margin step-down (to ~66% from a record 67.9%) while revenue is still accelerating. NVIDIA is a well-documented point on a curve a larger peer has travelled further along, not an outlier on it.

The landscape

COMPANY	TICKER	ROLE	DIRECTION	CONF.	WHY IT IS HERE
Broadcom	AVGO	incumbent	mixed	high	Segment opex already 6% of segment revenue; first margin step-down guided
Credo Technology	CRDO	emerging	benefits	high	~34% opex ratio on revenue that more than tripled to \$1.3B
Astera Labs	ALAB	emerging	benefits	high	Non-GAAP operating margin 36.2% to 39.1% to ~43% guided in three quarters
Arista Networks	ANET	incumbent	mixed	high	Opex flat at \$520M, but the gain absorbed gross margin 65% to 62%
Coherent	COHR	incumbent	benefits	high	Opex guided below its own 18% target model, leverage still ahead
Advanced Micro Devices	AMD	incumbent	benefits	medium	Room to spread, but on a 53% gross margin vs NVIDIA's 74.9%
Fabrinet	FN	adjacent	benefits	medium	FY2026 revenue +36%, non-GAAP EPS grew faster still
Celestica	CLS	adjacent	mixed	medium	CCS revenue +76% against a materials cost base
Hon Hai (Foxconn)	2317.TW	adjacent	mixed	high	6.12% gross margin, 3.75% operating margin on record AI revenue

COMPANY	TICKER	ROLE	DIRECTION	CONF.	WHY IT IS HERE
Alchip Technologies	3661.TW	emerging	mixed	medium	Revenue -58%, net income +1.6% - the reversal that did not bite
CoreWeave	CRWV	adjacent	hurt	medium	Costs in depreciation and interest, not spreadable opex
Cisco Systems	CSCO	adjacent	mixed	high	25% operating margin, no hypergrowth to spread against - the control

The ideas

1. **Own the spread, not the floor** (CRDO, ALAB) - the prize is the distance to the floor times the revenue it spreads over. Dies if the opex ratio stops falling for two quarters at unchanged revenue growth.
2. **The guided step-down is the marker** (AVGO, ANET) - a first margin step-down guided during acceleration dates the end of the spreading. Dies if Q1 prints back above 68% on the same mix.
3. **Mix is the only observed defence against the reversal** (3661.TW, FN) - counter-cyclical services mix is the only hedge this run found against reverse leverage. Dies if NRE and production revenue fall together in one quarter.
4. **The assembler is not the beneficiary** (2317.TW, CLS, FN) - rack ASPs to \$8.8M moved Hon Hai's operating margin 59bps. Dies if an EMS posts 200bps+ of AI-attributable operating-margin expansion.

What this run did NOT settle

The peer opex ratios are unevenly evidenced, and the unevenness is not random. Three of twelve rows - AMD, Celestica and Fabrinet - rest on structural inference rather than a disclosed operating-expense ratio, and all three are graded medium for that reason. Nobody publishes 'opex as a percentage of revenue' as a headline number, so every ratio in this bag was either stated by the company or computed from two figures in the same document. The corpus is deep on NVIDIA and on AI-supply-chain REVENUE and thin on peer COST disclosure; that asymmetry is the single biggest limitation on the table.

The confidence distribution is roughly balanced, but the one hurt row is not high. Seven rows are high and five medium. The mediums split two benefits, two mixed and one hurt, so the grades do not systematically flatter one side. The weakness is narrower: CoreWeave is the table's ONLY hurt row and it is medium, because the corpus chunk redacted its absolute loss figures. The squeezed side of this theme is therefore established by structure (Hon Hai, Celestica) rather than by a single well-quantified casualty, and a follow-up should fix that first.

The reversal case rests on two observations, one of which contradicts the other. NVIDIA's FY2023 (flat revenue, -55% GAAP EPS) is the theme's proof that reverse leverage bites. Alchip's Q1 2026 (-58% revenue, +1.6% net income) is a counter-example in which it did not, because mix moved the other way. Both are single

quarters. A run that had access to a proper panel - opex ratios and downturn outcomes across twenty names over ten years - could say which case is typical; this one cannot, and idea 3 is built on the tension rather than on a resolution of it.

Astera Labs' margin path is non-GAAP. The 36.2% to 43% progression is non-GAAP operating margin, and at a company that listed in March 2024 the gap to GAAP is stock compensation. The direction of travel is not in doubt; its size is overstated by the figures quoted.

The search budget shaped the fringe. This workbook was already at roughly 330 web searches against a soft budget of 150 when this run started, so the fringe search was deliberately limited to two targeted queries (Astera Labs, Credo) rather than a proper screen for pre-scale names. Three emerging rows came out of it. A dedicated screen for listed companies two years into an opex-spreading phase would likely find more, and that is the most valuable unfinished work in this bag.

Power and cooling names were dropped rather than guessed. Vertiv, nVent and Amphenol are plausibly exposed and the corpus contains articles about all three, but none carried an income statement, so no row was written for them. Their absence is a gap in the table, not a judgment that they are unexposed.